MIL-C-915A(SHIPS)
30 July 1952
SUPERSEDING
MIL-C-915(SHIPS)
15 November 1949

INTERIM MILITARY SPECIFICATION

CABLE, CORD AND WIRE, ELECTRICAL (SHIPBOARD USE)

1. SCOPE

1.1 Scope. - This specification covers electric cable for shipboard use. As distinguished by the type of insulation used, the general features of construction, and the service conditions involved, electric cables covered by this specification fall within the following general classifications:

Cables for nonflexing service.
Cables for repeated flexing service.
Cables, thin-walled.
Wires, miscellaneous.

1.2 Classification. -

1.2.1 Cables for nonflexing service. - Cables for nonflexing service shall be of the following types, as specified (see 6.2):

1.2.1.1 Heat and flame resistant types. -

Type SHFA - Single, heat and flame resistant, armored.
Type DHFA - Double, heat and flame resistant, armored.
Type THFA - Triple, heat and flame resistant, armored.
Type FHFA - Four, heat and flame resistant, armored.
Type SHFP - Single, heat and flame resistant, propulsion.
Type SHFR - Single, heat and flame resistant, radio.
Type DHFR - Double, heat and flame resistant, radio.
Type THFR - Triple, heat and flame resistant, radio.
Type SDGA - Single, degaussing, armored.
Type MDGA - Multiconductor, degaussing, armored.
Type MDGL - Multiconductor, degaussing, leaded.
Type MHFA - Multiconductor, heat and flame resistant, armored.
Type TTHFWA - Twisted pair, telephone, heat and flame resistant, armored.
Type PBJX - Pyrometer base lead wire.
Type PBTX - Pyrometer base lead wire.

1.2.1.2 Other types. -

Type DSS - Double conductor, special purpose, shielded.
Type TSS - Three conductor, special purpose, shielded.
Type FSS - Four conductor, special purpose, shielded.
Type MSS - Multiple conductor, special purpose, shielded.
Type TSP - Twisted pair, special purpose, plain.
Type MDGB - Multiconductor, degaussing, binnacle.
Type TTRSA - Twisted shielded pair, radio, armored.
Type MCSC - Multiple conductor, steel core.
Type DBSP - Double, small boat, shielded, plain.
Type TBSP - Triple, small boat, shielded plain.
Type FBSP - Four, small boat, shielded plain.
Type FCSF - Four, combination, special purpose, flexible.
Type MCGC - Multiple conductor, glass core.
Type SDU - Single, degaussing, unarmored.

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1.2.2 Cables for repeated flexing service. - Cables for repeated flexing service shall be of the following types, as specified (see 6.2):

1.2.2.1 Heat, flame, and oil resistant types. -

Type DCOP - Double conductor, oil resistant, portable.
Type TCOP - Triple conductor, oil resistant, portable.
Type MHFF - Multiconductor, heat and flame resistant, flexible.
Type TTOP - Twisted pair, telephone, oil resistant, portable.
Type MCOS - Multiple conductor, oil resistant, shielded.
Type MMOP - Multiconductor, microphone, oil resistant, portable.
Type SHOF - Single, heat and oil resistant, flexible.
Type DHOF - Double, heat and oil resistant, flexible.
Type THOF - Triple, heat and oil resistant, flexible.
Type FHOF - Four, heat and oil resistant, flexible.

1.2.2.2 Other types. -

Type TRF - Tough rubber jacket, flexible.
Type TRXF - Tough rubber jacket, extra flexible.
Type TTRS - Twisted shielded pair, radio, flexible.
Type CVSF - Combination, aircraft carrier, special purpose, flexible.

1.2.3 Thin-walled cables. - Thin-walled cables shall be of the following types and sizes, as specified (see 6.2):

1.2.3.1 Portable type. -

Type FCOTP-4 - Four conductor, oil resistant, thin walled, portable.

1.2.4 Miscellaneous wires. - Miscellaneous wires shall be of the following types, as specified (see 6.2):

Type SRI - Single resin insulated.
Type SRIB - Single resin insulated, braided.
Type SHFS - Single, heat and flame resistant, switch-board.
Type MRI - Multiconductor resin insulated.

1.3 Cable sizes. - The conductor size in thousands of circular mils, the number of conductors, or the number of twisted pairs in the wire or cable for each type shall be as indicated under detailed requirements by a numeral after the type letter designation.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications, standards, and publications, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL
NN-B-591 - Boxes, Fiberboard, Wood-Cleated (for Domestic Shipment).
NN-B-601 - Boxes, Wood-Cleated-Plywood, for Domestic Shipment.
NN-B-621 - Boxes, Wood, Nailed and Lock-Corner.
TT-A-468 - Aluminum-Pigment; Powder and Paste, for Paint.
LLL-B-631 - Boxes, Fiber, Corrugated (for Domestic Shipment).
LLL-B-636 - Boxes, Fiber, Solid (for Domestic Shipment).

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MILITARY

JAN-P-105 - Packaging and Packing for Overseas Shipment - Boxes, Wood, Cleated, Plywood.
JAN-P-106 - Packaging and Packing for Overseas Shipment - Boxes, Wood, Nailed.
MIL-B-107 - Boxes, Wood, Wirebound, (Overseas Type).
JAN-P-125 - Packaging and Packing for Overseas Shipment - Barrier-Materials, Waterproof, Flexible.
MIL-S-572 - Synthetic-Fiber; Organic: Cords, Yarns, and Mono-Filaments.
MIL-G-1140 - Glass-Fiber: Yarn, Cordage, Sleeving, Cloth, and Tape.
MIL-I-3053 - Insulation, Electrical, Asbestos-Fiber, Treated and Untreated.
MIL-I-3064 - Insulation, Electrical, Plastic-Sealer.
MIL-W-16970 - Wire and Ribbon, Resistance.

NAVY DEPARTMENT

General Specifications for Inspection of Material.
22C4 - Cable, Steel (Corrosion-Resisting), Flexible and Extra Flexible, Preformed Type (Aircraft Use).

STANDARDS

MILITARY

MIL-STD-104 - Limits for Electrical Insulation Color.
MIL-STD-129 - Marking of Shipments.

PUBLICATIONS

BUREAU OF SHIPS

N.B.S. 431 - Lubricating Oil; General Information, Requirements and Methods of Tests.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification. -

3.1.1 Electrical cable, except types TRF, TRXF, DBSP, TBSP, FBSP, CVSF, FCSF, MCGC, SRI, SRIB, MRI and SHFS, furnished under this specification shall be a product which has been tested and has passed the qualification tests specified in section 4 (see 6.3).

3.1.2 The types which are subject to qualification (see 3.1.1) are identified with the "approved sample" which was tested and on which the approval was based, and the manufacturer shall not change any of the component materials or the assembly of the components in completed cables offered for inspection under subsequent contracts or orders without first obtaining written approval for such modifications from the bureau or agency concerned.

3.2 Definitions of terms. - The following definitions shall apply to the technical terms wherever such terms appear in this specification:

3.2.1 Conductor. - A conductor is a slender rod or filament of drawn metal of circular cross section or a group of such rods or filaments not insulated from one another, suitable for transmitting an electric current.

3.2.2 Stranded conductor. - A stranded conductor is a conductor composed of more than one filament.

3.2.3 Strand. - A strand is one of the filaments, or one of the stranded groups, of a stranded conductor.

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3.2.4 Concentric lay. - A concentric lay conductor or cable is composed of a central core surrounded by one or more layers of helically wound strands or insulated conductors.

3.2.5 Bunched lay. - In a bunched lay conductor, the strands are twisted together in the same direction without regard to geometrical arrangement.

3.2.6 Rope lay. - In a rope lay conductor, the stranded members are twisted together with a concentric lay, and the stranded members themselves may have either a bunched, concentric, or rope lay.

3.2.7 Direction of lay. - The direction of lay is the lateral direction, either right-hand or left-hand, in which a strand or insulated conductor passes over the top as it recedes from an observer looking along the axis of the conductor or cable. In unilay or unidirectional lay constructions all components are laid in the same direction instead of reversing the direction of alternate layers.

3.2.8 Length of lay. - The lay of any helically wound strand or insulated conductor is the axial length of one turn of the helix, usually expressed in inches or as a multiple of the pitch diameter.

3.2.9 Pitch diameter. - The pitch diameter is the mean diameter of the helix described by the strands or insulated conductors in any layer.

3.2.10 Elongation. - Elongation is the extension or increase in length produced by a tension load in a section of a test specimen between bench marks placed on it, and is either expressed as a percentage of the original length between bench marks or indicated by specifying a minimum distance between bench marks.

3.2.11 Circular mil. - A circular mil is a unit of area equal to $\pi/4$ or 78.54 percent of a square mil. The cross-sectional area of a circle in circular mils is, therefore, equal to the square of its diameter in mils. A circular inch is equal to 1,000,000 circular mils.

3.2.12 Twisted pair. - A twisted pair is composed of two insulated conductors twisted together.

3.2.13 Shielded pair. - A shielded pair is a twisted pair over which a close braid of copper wire has been applied.

3.2.14 Completed wire or cable. - A completed wire or cable is one on which all manufacturing operations have been completed and which is ready for inspection.

3.2.15 Insulation resistance. - The insulation resistance of an insulated conductor is the electrical resistance offered by its insulation to an impressed direct current potential tending to produce a leakage of current through the same.

3.2.16 Shipping length. - A shipping length is defined as one continuous length of completed wire or cable, wound in a coil or on a reel, which is ready for packaging, packing and marking for shipment in accordance with section 5.

3.2.17 Nominal length. - The nominal length for wire or cable on a reel is the unit ordering length, for each type and size, as shown in table XX. The nominal length or unit ordering length, for wire or cable in coils (see 5.1.1.1) is 500 feet.

3.2.18 Standard length. - A standard length is defined as any length between 110 and 90 percent of the nominal length

3.2.19 Random length. - A random length is defined as any length at least 30 and less than 90 percent of the nominal length.

3.2.20 Remnant length. - A remnant length is defined as any length at least 10 and less than 30 percent of the nominal length.

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3.2.21 Scrap length. - A scrap length is defined as any length less than 10 percent of the nominal length.

3.3 Material. - All material used in the construction of electrical wires and cables shall be of the quality and form best suited for the purpose intended.

3.4 Minimum standards. - The requirements specified hereinafter for each component material and completed wire or cable constitute the minimum acceptable standards of quality and performance under tests. The detail requirements for cables shall be as specified in the individual specification sheets (see 3.12 to 3.49, inclusive); in the event of any conflict between the general requirements of this specification and the detail requirements for the particular type of cable the latter shall govern.

3.5 Standardized construction. - The necessity for standardization of the fittings, fixtures, et cetera (etc.), with which the cable is supported, terminated, or served, particularly as regards the watertight integrity of the vessel, requires similar uniformity as to the type, size and construction, as well as electrical performance, of the cable itself. The general construction details specified hereinafter for each type and size of cable, shall be strictly complied with so as to assure the necessary uniformity and interchangeability without regard to the source of manufacture or date of procurement.

3.6 Watertight construction. - All sizes of cable types required to be of watertight construction (see 3.12 to 3.25, inclusive, 3.27, 3.28 and 3.34) shall contain sealing compounds which have been approved as a result of qualification tests (see 3.1). Leakage in excess of three times the value specified in 70, column 1 of Appendix I is indicative of inadequate processing control, and the cable represented by such samples shall be inspected carefully for the presence of watertight sealing compounds Failure to include the approved watertight sealing compounds in all void spaces within the copper conductors and the core of the cable shall be sufficient cause for rejection.

3.7 Dimensional tolerances. -

3.7.1 Overall dimensions. - Unless otherwise specified herein, the overall diameter of each completed wire or cable specified hereinafter shall be considered the maximum value allowable. No plus tolerance will be permissible. Where the minimum overall diameter of a completed wire or cable is specified hereinafter no value below this minimum will be permissible. Where the minus tolerance is not specified definitely, the minimum permissible diameter shall be not less than 92-1/2 percent of the specified overall diameter of the completed wire or cable.

3.7.2 Nominal dimensions. - Where the diameter over successive component layers of insulation, braid, sheath, armor, etc., is specified hereinafter, these diameters shall be considered as nominal values, and the nominal thickness of any layer shall be calculated as one half of the diameter increase of that layer. The average thickness throughout the length shall be within either $\pm$0.003 inch or $\pm$15 percent of the calculated nominal thickness, whichever allows the greater tolerance, provided that the overall diameter of the completed cable is within the specified limits.

3.7.3 Minimum thicknesses. - Where the thicknesses of the conductor insulation or cable sheath are specified as a minimum value, as in cables for repeated flexing service, the average thickness throughout the length shall be not less than the specified minimum. The minimum thickness measured at any cross section shall be not less than 90 percent of the specified minimum thickness.

3.7.4 Centering and circularity. -

3.7.4.1 Insulated conductors. - The several types of insulating walls for the various classes of electric cable shall be applied concentrically about the conductor and so cured, processed or maintained, as to provide for accurate centering of the conductor or core within the finished wall and retention of a circular cross section. The insulation on the individual conductors shall be reasonably uniform in diameter throughout the conductor length, and at any cross section the maximum wall thickness shall not exceed the minimum by more than 25 percent for specified thicknesses greater than 0.025 inch, nor by more than 40 percent for specified thicknesses of 0.025 inch and less.

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3.7.4.2 Outer sheaths. - The outer sheath shall be applied concentrically to the cable core in order to maintain circularity in the completed cable. The maximum wall thickness of the sheath at any cross section shall not exceed the minimum by more than 66 percent.

3.8 Manufacturer's identification. -

3.8.1 General. - Except where otherwise specifically required or approved, all wires and cables shall be provided with an easy and indelible means of identifying the manufacturer. The means employed shall be such that it cannot be confused with the requirements for circuit identification specified in 3.9. The use of a printed marker tape or colored marker threads, as applicable to the particular type or types of cable in question, shall be considered standard; any departures therefrom shall be subject to specific approval of the bureau or agency concerned.

3.8.2 Marker tape. - Marker tape shall consist of a thin tape, approximately 1/8 inch wide, of cotton or cellulose, laid directly under the outer sheath, or other location as specifically approved. There shall appear on one side of the tape, at intervals of approximately 1 foot, the following printed information:

Name of manufacturer.
Name or location of plant.
Date of manufacture (year).
Specification MIL-C-915.
Serial number.

3.8.3 Marker threads. - Marker threads shall be used only when satisfactory to the bureau or agency concerned, and shall be limited to the miscellaneous wires (see 1.2.4) of 0.300-inch overall diameter and smaller, and to the small flexible cables (see 1.2.2.1) of 0.300-inch overall diameter and smaller or where it is not practicable to employ marker tape. Marker threads shall be of cotton, rayon, or artificial silk. The number of marker threads and the color of each thread used by the manufacturer shall be satisfactory to the bureau or agency concerned.

3.8.4 Molded marking. - Except for the specific requirements of 3.30 and 3.34, molded marking will be permitted at the manufacturer's option on cables for repeated flexing service (see 1.2.2); however, when molded marking is used, it shall be in addition to and not a substitute for standard marker tape or threads. The marking shall show the manufacturer's name or his trademark or trade-name and the year of manufacture. This marking shall be molded into the sheath during processing and it shall be either by indentation of or by raised lettering on the sheath itself, or shown on thin slips, colored if desired, which are themselves molded into the sheath and do not project above the surface of the sheath.

3.9 Circuit identification. -

3.9.1 General. - Means shall be provided for ready identification of each individual insulated conductor in respect to the electrical distribution system of which it is to form a part. This shall be so clear and distinctive as to permit its connection into a standard arrangement (phase sequence, circuit arrangement, load tracing, etc.) in the system; and also to provide a rapid and accurate means for its identification and reconnection into the system after a serious casualty. The standard means for circuit identification shall be: (1) colored insulation, (2) colored thread wind, (3) colored tape, and (4) color-coded braid, as specifically required or permitted by 3.12 to 3.49, inclusive. By any of these methods, each color shall be readily distinguishable from every other color and, in the completed cable, the shade of colors shall conform closely to those specified in 3.9.6.

3.9.2 Colored insulation. - Where circuit identification is obtained through the use of colored insulation, the following sequence of solid colors shall be employed: (1) black, (2) white, (3) red, (4) green, (5) orange, (6) blue, (7) brown, (8) gray, (9) yellow, (10) purple, (11) tan, and (12) pink. Where a wall of saturated asbestos or glass fiber insulation is employed, on individual conductors in combination with other insulation, circuit identification may be obtained by coloring this saturated wall.

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3.9.3 Colored thread wind. - Where circuit identification is obtained through the use of color-coded thread winds, the winds shall employ the sequence of solid colors specified in 3.9.2. Such winds shall completely cover the outer periphery of the insulation and shall be suitably treated to prevent fraying. Means shall be provided for obtaining satisfactory adherence of the wind so that circuit identification is maintained when the individual conductors are separated from the cable core. Unless otherwise specified herein, the thread for the winds shall be of cotton, rayon, cellulose acetate, or glass, at the manufacturer's option.

3.9.4 Colored tape. - In general, colored tape is to be limited to those cable types wherein a tape or separator is specified for the accomplishment of other purposes, and coloring simply permits the added purpose of circuit identification to be served. This paragraph shall not be considered as authority for the addition of a tape not already specified; such additions shall be satisfactory to the bureau or agency concerned. The tape colors shall conform to the sequence of solid colors specified in 3.9.2.

3.9.5 Color-coded braid. - In general, color-coded braid shall be limited to those cable types wherein an inner braid (see 3.11.16) is specified over the individual conductor insulation of a multiple conductor or twisted pair telephone cable. This paragraph shall not be considered as authority for the addition of a braid not already specified; such addition shall be satisfactory to the bureau or agency concerned. Where a combination of two or more colors is specified for the braid, the "base" color shall be the background, with the "tracer" color (or colors) providing the particular combination required. The "tracer" colors shall be on 3 adjacent carriers for each color.

3.9.6 Color shades. - The shades of color used shall be fast to light and to heat, and shall fall within the light and dark limits of the color chips specified in Standard MIL-STD-104. In the case of synthetic rubber insulation (see 3.11.3 and 3.11.5), polychloroprene insulation (see 3.11.6), or compound-filled tapes (see 3.11.15), for circuit identification, somewhat wider limits will be permitted in color shades provided all colors in the cable are easily distinguishable from each other.

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3.9.7 Color code. -

3.9.7.1 Multiple conductor cable. - Except where otherwise specifically required hereinafter, the color code shown in table I shall apply to all multiple conductor (not twisted pair) cables having from 2 to 44 individually insulated conductors within a common protective sheath. For example, all single conductor cables would be "1 Black"; all 2-conductor cables would consist of "1 Black" and "2 White"; all 3-conductor cables would consist of "1 Black", "2 White" and "3 Red" etc., up to a 44-conductor cable, where all the color combinations listed would be included. In the case of cables having more than one layer of conductors, the numbering shown above shall be from the innermost to the outermost that is (i.e.), the number (No.) 1 conductor shall be the center conductor (or one of the center conductors where two or more are used as a center) of the concentric lay.

Table I - Color code of multiple conductor cable.

Conductor No.	Base color	Tracer color	Tracer color
1	Black	.....	.....
2	White	.....	.....
3	Red	.....	.....
4	Green	.....	.....
5	Orange	.....	.....
6	Blue	.....	.....
7	White	Black	.....
8	Red	Black	.....
9	Green	Black	.....
10	Orange	Black	.....
11	Blue	Black	.....
12	Black	White	.....
13	Red	White	.....
14	Green	White	.....
15	Blue	White	.....
16	Black	Red	.....
17	White	Red	.....
18	Orange	Red	.....
19	Blue	Red	.....
20	Red	Green	.....
21	Orange	Green	.....
22	Black	White	Red
23	White	Black	Red
24	Red	Black	White
25	Green	Black	White
26	Orange	Black	White
27	Blue	Black	White
28	Black	Red	Green
29	White	Red	Green
30	Red	Black	Green
31	Green	Black	Orange
32	Orange	Black	Green
33	Blue	White	Orange
34	Black	White	Orange
35	White	Red	Orange
36	Orange	White	Blue
37	White	Red	Blue
38	Brown	.....	.....
39	Brown	Black	.....
40	Brown	White	.....
41	Brown	Red	.....
42	Brown	Green	.....
43	Brown	Orange	.....
44	Brown	Blue	.....

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3.9.7.2 Twisted pair telephone cable. - Individual conductors and pairs in twisted pair telephone cables shall be color coded by pairing the solid colors in sequence (see 3.9.2) first, 1 (black) with 2 to 12 inclusive in the order listed, then 2 (white) with 3 to 12 inclusive in the order listed, then 3 (red) with 4 to 12 inclusive in the order listed, etc.

3.10 Standard copper conductors. -

3.10.1 General. - Conductors shall be of soft or annealed round copper wire having the properties specified in 3.11.1.

3.10.2 Area. - The cross sectional area of each copper conductor in a completed wire or cable shall be not less than the minimum area specified in tables II and III. Nominal areas are based on the exact or calculated wire diameters specified in table IV. The nominal strand diameters are specified in tables II, III and IV for convenience.

Table II - Conductor data - concentric strand.

Standard copper conductor size	Number of strands	Strand diameter	Diameter over conductor	Cross sectional area of conductor		Maximum conductor resistance (d.c.) per 1000 feet at 25°C.		Weight per 1000 feet
	Minimum	Nominal	Nominal	Nominal	Minimum	Bare	Coated	Approx.
		Inch	Inches	Cir. mils	Cir. mils	Ohms	Ohms	Pounds
1(7)	7	0.013	0.038	1,119	1,096	9.89	10.3	3.4
2(7)	7	.016	.048	1,779	1,743	6.25	6.50	5.5
3(7)	7	.020	.060	2,828	2,771	3.92	4.07	8.7
3(19)	19	.013	.063	3,036	2,975	3.64	3.80	9.3
4(7)	7	.025	.076	4,497	4,407	2.46	2.52	14
6(7)	7	.031	.092	6,512	6,382	1.69	1.73	20
9(7)	7	.036	.108	9,016	8,836	1.23	1.25	28
14(7)	7	.045	.136	14,340	14,050	0.770	0.784	44
23(7)	7	.057	.171	22,800	22,340	.486	.493	70
30(19)	19	.040	.202	30,860	30,240	.358	.365	95
40(19)	19	.045	.226	38,910	38,130	.284	.288	120
50(19)	19	.051	.254	49,080	48,100	.225	.228	150
60(37)	37	.040	.282	60,090	58,890	.185	.189	190
75(37)	37	.045	.317	75,780	74,260	.146	.149	230
100(61)	61	.040	.363	99,060	97,080	.112	.115	310
125(61)	61	.045	.407	124,900	122,400	.0888	.0904	390
150(61)	61	.051	.457	157,600	154,400	.0704	.0716	490
200(61)	61	.057	.514	198,700	194,700	.0560	.0570	610
250(61)	61	.064	.577	250,500	245,500	.0444	.0453	770
300(91)	91	.057	.628	296,400	290,500	.0375	.0382	910
350(91)	91	.062	.682	349,800	342,800	.0316	.0321	1,100
400(127)	127	.057	.742	413,600	405,400	.0268	.0273	1,300
500(127)	127	.064	.832	521,600	511,100	.0214	.0217	1,600
650(127)	127	.072	.936	657,600	644,500	.0169	.0172	2,000
800(127)	127	.081	1.050	829,300	812,700	.0134	.0136	2,600
1000(127)	127	.091	1.180	1,046,000	1,025,000	.0106	.0108	3,200
1300(127)	127	.102	1.325	1,318,000	1,292,000	.00843	.00851	4,100
1600(127)	127	.114	1.485	1,662,000	1,629,000	.00668	.00676	5,100
2000(127)	127	.128	1.670	2,097,000	2,055,000	.00530	.00536	6,300

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Table III - Conductor data-bunch and rope strand.

Standard copper conductor size	Number of strands	Strand diameter	Diameter over conductor	Cross sectional area of conductor		Maximum conductor resistance (d. c.) per 1,000 ft. at 25°C.		Weight per 1,000 ft.
	Minimum	Nominal	Nominal	Nominal	Minimum	Bare	Coated	Approximate
		Inch	Inches	Cir. Mils	Cir. Mils	Ohms	Ohms	Pounds
Bunch:								
1/2(21)	21	0.005	0.028	525	515	21.1	22.2	1.6
3/4(7)	7	.010	.030	703	689	15.7	16.4	2.2
1(10)	10	.010	.038	1,005	985	11.0	11.5	3.1
1(26)	26	.006	.042	1,034	1,014	10.7	11.2	3.2
1-1/2(16)	16	.010	.049	1,608	1,576	6.88	7.17	4.9
1-1/2(41)	41	.006	.049	1,630	1,597	6.78	7.15	5.0
2-1/2(26)	26	.010	.061	2,613	2,561	4.23	4.48	8.0
2-1/2(65)	65	.006	.061	2,594	2,542	4.26	4.51	8.0
4(41)	41	.010	.077	4,121	4,038	2.68	2.81	13
6(65)	65	.010	.097	6,533	6,402	1.69	1.78	20
9(90)	90	.010	.120	9,045	8,864	1.22	1.28	28
14(140)	140	.010	.145	14,070	13,790	.786	.823	43
Rope:								
20(49)	7x7	.020	.180	19,800	19,400	.562	.582	61
23(228)	19x12	.010	.190	22,190	22,910	.499	.523	73
26(49)	7x7	.023	.210	26,250	25,730	.428	.440	82
30(304)	19x16	.010	.220	30,550	29,940	.374	.392	97
33(336)	7x48	.010	.235	33,370	33,090	.344	.360	106
42(209)	19x11	.014	.260	42,110	41,280	.272	.284	130
42(49)	7x7	.029	.260	41,740	40,910	.268	.275	130
53(532)	19x28	.010	.304	53,470	52,400	.213	.218	169
60(304)	19x16	.014	.310	61,260	60,040	.187	.196	190
66(133)	19x7	.022	.330	66,370	65,040	.171	.175	210
84(2107)	2107	[1]	.410	83,690	82,020	.138	----	270
83(418)	19x22	.014	.380	84,230	82,560	.136	.142	270
105(2646)	2646	[1]	.460	105,500	103,400	.108	----	340
105(259)	37x7	[1]	.410	105,500	103,400	.108	.113	330
133(3325)	3325	[1]	.520	133,100	130,400	.0876	----	430
133(259)	37x7	[1]	.460	133,100	130,400	.0856	.0892	420
133(684)	19x36	.014	.480	137,800	135,100	.0830	.0867	440
150(760)	19x40	.014	.510	153,100	150,100	.0747	.0780	490
168(427)	61x7	[1]	.520	167,800	164,400	.0681	.0710	530
200(988)	19x52	.014	.580	199,100	195,100	.0575	.0600	630
220(259)	37x7	.029	.610	220,700	216,300	.0494	.0507	718
250(1254)	19x66	.014	.680	252,700	247,700	.0453	.0472	800
400(2052)	19x108	.014	.850	413,500	405,300	.0277	.0289	1,300
800(4033)	37x109	.014	1.150	812,700	796,500	.0141	.0148	2,600

[1]These are conductors for flexible cables, types TRF and TRXF, for which the minimum number of strands and minimum area are specified. The individual strand diameter is determined from the required area and the actual number of strands used.

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Table IV - Standard copper wires.

Standard copper wire sizes			Maximum resistance (d.c.) per 1000 feet at 25°C.		Approx. weight per 1000 feet
A.W.G. No.	Strand diameter		Cross sectional area		
	Nominal	Calculated	Nominal	Bare	Coated	
	Inch	Inch	Cir. mils	Ohms	Ohms	Pounds
36	0.005	0.005000	25.00	449.	473	0.0757
34	.006	.006305	39.75	280.	296	.120
30	.010	.01003	100.5	108.	114	.304
28	.013	.01264	159.8	68.2	71.2	.484
27	.014	.01420	201.5	54.1	56.5	.610
26	.016	.01594	254.1	42.9	44.7	.769
24	.020	.02010	404.0	27.0	27.5	1.22
...	.022	.02234	499.1	21.9	22.4	1.55
...	.023	.02315	535.9	20.3	20.7	1.70
22	.025	.02535	642.4	16.9	17.2	1.94
...	.029	.02919	852.0	12.8	13.1	2.69
...	.031	.03050	930.3	11.7	11.9	2.82
19	.036	.03589	1,288.	8.46	8.64	3.90
18	.040	.04030	1,624.	6.71	6.85	4.92
17	.045	.04526	2,048.	5.31	5.42	6.20
16	.051	.05082	2,583.	4.21	4.29	7.82
15	.057	.05707	3,257.	3.35	3.42	9.86
...	.062	.06200	3,844.	2.82	2.88	11.6
14	.064	.06408	4,107.	2.66	2.72	12.4
13	.072	.07196	5,178.	2.10	2.14	15.7
12	.081	.08081	6,530.	1.67	1.69	19.8
11	.091	.09074	8,234.	1.32	1.34	24.9
10	.102	.1019	10,380.	1.05	1.06	31.4
9	.114	.1144	13,090.	0.832	.844	39.6
8	.128	.1285	16,510.	.660	.668	50.0

3.10.3 Coating, metallic. - Except where otherwise specified on the individual specification sheets, all conductor strands shall be bare, uncoated copper wires. When a metallic coating is specified on the individual specification sheets, the wire shall be thoroughly and evenly coated with a thin adherent layer of pure tin which, prior to stranding or braiding, shall meet the tests specified in 4.8.22. An alloy coating may be employed only when specifically approved by the bureau or agency concerned as an exception limited to a specific contract or order.

3.10.4 Types of stranding. - Stranded conductors shall be furnished with the following types of stranding as specified hereinafter and in tables II and III.

3.10.4.1 Concentric stranding. - Except where otherwise specified herein, all conductors shall be concentrically stranded. One strand shall form the central core, and be surrounded by one or more layers of helically laid wires, the direction of lay being alternated for the successive layers. The pitch of lay or any layer of wires shall not exceed the values specified in table V, except that for conductors composed of 37 wires or more, this requirement shall apply only to the two outer layers. The number of strands shall be as specified in table II. The diameter of all strands shall be substantially equal.

3.10.4.2 Bunch stranding. - Where specifically required under detailed requirements stranded conductors shall be bunch stranded. A collection of approximately the specified number of strands shall be twisted together in the same direction without regard to geometrical arrangement. The pitch or lay shall not exceed the values specified in table V. The diameter of all strands shall be substantially equal.

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3.10.4.3 Rope stranding. - Where specifically required, stranded conductors shall be rope stranded. The required number of stranded members shall be laid up concentrically to form the completed conductor. The lay of wires or strands within each stranded member, whether concentric, bunch or rope lay, and the lay of the two outer layers of stranded members shall not exceed the values specified in table V. In the case of the extra flexible rope stranded cable, type TRXF, the direction of lay of the component members may all be in the same direction in lieu of the alternately reversed direction specified in 3.10.4.1.

Table V - Lay of wire stranding.

Concentric stranding			Bunch stranding	
Wires or strands per conductor	Number of layer	Lay expressed in diameters of individual wires or strands	Nominal area of bunch	Maximum lay
		Maximum	Cir. mils	Inches
			0 to 1000	0.63
7	First	48	1001 to 2500	.80
19	Second	80	2501 to 4000	1.00
37	Third	112	4001 to 6000	1.30
61	Fourth	144	6001 to 8000	1.60
91	Fifth	176	8001 to 12000	2.10
127	Sixth	208	12001 to 17000	2.75
			17001 to 24000	3.50

3.10.5 Equivalent to standard conductor. - An alternate or equivalent to the specified standard bunch or rope stranded conductor may be used provided this equivalent shall be a conductor having a total cross-sectional area not less than, and be made up with strands not fewer in number than, that specified for the standard copper conductor.

3.10.6 Joints. - Necessary joints in solid wires and rods shall be made prior to final drawing in accordance with the best commercial practice. The joint shall be made in a neat and workmanlike manner so as not to increase the diameter of the strand nor lessen its breaking strength by more than 5 percent. No splice shall be made in a stranded conductor as a whole. Not more than one of the solid wires in a concentric conductor of 19 wires or less, nor more than 5 percent of the solid wires in a concentric conductor of more than 19 wires, shall be spliced in any one foot of the conductor. In rope stranded conductors, the component members forming the rope shall be considered the equivalent of a solid wire in a concentric conductor, and may be spliced as a unit under the same limitations.

3.10.7 Electrical resistance. -

3.10.7.1 The resistance of completed single conductor cable, and of insulated single conductors intended for use in multiple conductor cable when measured in the preliminary stage, shall not exceed the maximum value specified in table II or III. The resistance of the conductors in completed multiple conductor cable shall not exceed the maximum value shown on the individual specification sheet (see 3.12 to 3.49, inclusive).

3.10.7.2 When the resistance measurements are made at a temperature other than 25° Centigrade (C.), suitable correction shall be made in accordance with 3.11.1.2.2.

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3.11 Principal components. - The following requirements cover the several principal components used in the manufacture of electrical wires and cables.

3.11.1 Copper wire. -

3.11.1.1 Physical properties. -

3.11.1.1.1 Each solid conductor and each wire previous to stranding shall be so drawn and annealed that it shall conform to the values specified in table VI. Each wire shall be uniform in cross section, free from flaws, scales, and other imperfections.

3.11.1.1.2 After stranding, but before insulating, the maximum tensile strength permitted shall be increased 5 percent, and the minimum elongation permitted shall be reduced 5 percent in numerical value (for example from 30 to 25 percent) from the values specified in table VI.

Table VI - Physical properties of round copper wires.

Diameter of copper wire	Tensile strength (max.)		Elongation in 10 inches (min.)	
	Bare	Coated	Bare	Coated
Inch	Pounds per square inch	Pounds per square inch	Percent	Percent
0.0030 to 0.0116	40,000	40,000	20	10
.0117 to .0207	40,000	39,000	20	15
.0208 to .105	38,500	38,500	25	20
.106 to .298	37,000	37,000	30	25

3.11.1.2 Electrical resistance. -

3.11.1.2.1 The resistance of the copper wires prior to stranding or insulating shall not exceed the maximum values specified in table IV, and shall be determined only when specifically required.

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3.11.1.2.2 Copper resistance shall be referred to a temperature of 25°C. by multiplying the measured value by a conversion factor corresponding to the temperature of the measurement. These conversion factors are shown in table VII. For temperatures between those tabulated, the conversion factor may be obtained by interpolation or calculated from the formula:

$$\text{Conversion factor} = \frac{259.5}{234.5 + t}$$

Where:

t = Test temperature, °C.

Table VII - Copper resistance temperature conversion factor.

Temperature of copper		Conversion factor	Temperature of copper		Conversion factor	Temperature of copper		Conversion factor
°C.	°F.		°C.	°F.		°C.	°F.	
0	32.0	1.107	20	68.0	1.020	40	104.0	0.945
2	35.6	1.097	22	71.6	1.012	42	107.6	.939
4	39.2	1.088	24	75.2	1.004	44	111.2	.932
6	42.8	1.079	26	78.8	0.996	46	114.8	.925
8	46.4	1.070	28	82.4	.989	48	118.4	.919
10	50.0	1.061	30	86.0	.981	50	122.0	.912
12	53.6	1.052	32	89.6	.974	52	125.6	.906
14	57.2	1.044	34	93.2	.967	54	129.2	.900
16	60.8	1.036	36	96.8	.959	56	132.8	.893
18	64.4	1.028	38	100.4	.952	58	136.4	.887

3.11.2 Insulation, synthetic resin. -

3.11.2.1 General. - Specimens of insulation removed from the insulated conductor of a complete wire or cable shall meet the following requirements when tested as specified in 4.8.1.

3.11.2.1.1 Tensile strength. - The tensile strength shall be not less than 1800 pounds per square inch.

3.11.2.1.2 Elongation. - Specimens shall stretch until the 2-inch bench marks are not less than 4-1/2 inches apart before rupture.

3.11.2.1.3 Set. - The set, after elongating specimens until the 2-inch bench marks are 4 inches apart, shall be not more than 30/64 inch when measured 2 minutes after release.

3.11.2.1.4 Air oven aging. - After 96 hours aging at 100°C. in the air oven (see 4.8.1.3), the ultimate elongation shall be not less than 80 percent of the value measured before aging.

3.11.3 Insulation, synthetic rubber. -

3.11.3.1 General. - Specimens of insulation removed from the insulated conductor of a completed wire or cable shall meet the following requirements when tested as specified in 4.8.1.

3.11.3.1.1 Specimens taken from conductor sizes 14 and larger. -

3.11.3.1.1.1 Tensile strength. - The tensile strength shall be not less than 600 pounds per square inch.

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3.11.3.1.1.2 Elongation. - Specimens shall stretch until the 2-inch bench marks are not less than 6-1/2 inches apart before rupture.

3.11.3.1.1.3 Set. - The set, after elongating the specimen until the 2-inch bench marks are 6 inches apart, shall not exceed 30/64 inch.

3.11.3.1.1.4 Air pressure heat aging. - After 20 hours aging at 126.7°C. in the air pressure heat bomb (see 4.8.1.3), the tensile strength and elongation shall be not less than 50 percent of the values measured before aging.

3.11.3.1.1.5 Oxygen pressure aging. - After 168 hours aging at 80°C. in the oxygen pressure bomb (see 4.8.1.3), the tensile strength and elongation shall be not less than 50 percent of the value measured before aging.

3.11.3.1.2 Specimens taken from conductor sizes 9 and smaller. -

3.11.3.1.2.1 Tensile strength. - The tensile strength shall be not less than 600 pounds per square inch.

3.11.3.1.2.2 Elongation. - Specimens shall stretch until the 2-inch bench marks are not less than 6-1/2 inches apart before rupture.

3.11.3.1.2.3 Oxygen pressure aging. - After 96 hours aging at 70°C. in the oxygen pressure bomb (see 4.8.1.3), the elongation shall be not less than 70 percent of the value measured before aging.

3.11.4 Insulation, polythene. -

3.11.4.1 General. - Specimens of the insulation removed from the insulated conductor of the completed cable shall conform to the following requirements when tested as specified in 4.8.1:

3.11.4.1.1 Tensile strength. - The tensile strength shall be not less than 1400 pounds per square inch.

3.11.4.1.2 Elongation. - Specimens shall stretch until the 2-inch bench marks are not less than 8 inches apart before rupture.

3.11.4.1.3 Set. - The set, after elongating specimens until the 2-inch bench marks are 4 inches apart, shall not exceed 35/64 inch.

3.11.4.1.4 Air oven aging. - After 96 hours aging at 100°C. in the air oven (see 4.8.1.3), the ultimate elongation shall be not less than 60 percent of the value measured before aging.

3.11.5 Insulation, butyl rubber. -

3.11.5.1 General. - Specimens of the insulation removed from the insulated conductor of the completed cable shall conform to the following requirements when tested as specified in 4.8.1:

3.11.5.1.1 Tensile strength. - The tensile strength shall be not less than 600 pounds per square inch.

3.11.5.1.2 Elongation. - Specimens shall stretch until the 2-inch bench marks are not less than 11 inches apart before rupture.

3.11.5.1.3 Set. - The set, after elongating specimens until the 2-inch bench marks are 6 inches apart, shall not exceed 30/64 inch.

3.11.5.1.4 Air pressure heat aging. - After 20 hours aging at 126.7°C. in the air pressure heat bomb (see 4.8.1.3), the tensile strength shall be not less than 70 percent and the elongation shall be not less than 75 percent of the values measured before aging.

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3.11.6 Insulation, polychloroprene. -

3.11.6.1 General. - Samples of the insulation removed from the insulated conductor of a completed wire or cable shall meet the following requirements when tested as specified in 4.8.1:

3.11.6.1.1 Tensile strength. - The tensile strength shall be not less than 1200 pounds per square inch.

3.11.6.1.2 Elongation. - Specimens shall stretch until the 2-inch bench marks are not less than 7 inches apart before rupture.

3.11.6.1.3 Set: - The set, after elongating specimens until the 2-inch bench marks are 6 inches apart, shall not exceed 3/8 inch.

3.11.6.1.4 Oxygen pressure aging. - After 96 hours aging at 70°C. in the oxygen pressure bomb (see 4.8.1.3), the tensile strength shall be not less than 1000 pounds per square inch and the elongation not less than 2 to 6 inches.

3.11.6.1.5 Hot oil aging. - After subjection to the hot oil test (see 4.8.17), the tensile strength and elongation shall be not less than 60 percent of the values measured before aging.

3.11.7 Insulation, varnished-cambric. -

3.11.7.1 Insulation. - The wall of varnished cambric insulation shall consist of multiple layers of high grade varnished fabric as follows, with a lubricating compound conforming to 3.11.8. Samples of tape removed from the completed cable shall pass the tests specified in 4.8.3.

3.11.7.2 Fabric. - The fabric shall be a closely woven cotton cloth having at least 60 and 55 threads per inch for warp and filler, respectively.

3.11.7.3 Varnish coating. - The highest grade of electrical insulating varnish, shall be applied to the fabric, preferably in two or three successive coats, each coat being dried slowly at a low baking temperature. The finished product shall be of uniform thickness, free from wrinkles, blisters, and other imperfections. The varnish coating shall be pliable and shall not crack when the varnished cloth is doubled upon itself.

3.11.7.4 Thickness of varnished cloth. - The finished varnish cloth shall have a uniform average thickness of not less than 0.005 inch nor more than 0.013 inch. The allowable variation of individual tapes shall be plus or minus 20 percent but shall not exceed 0.002 inch.

3.11.7.5 Tensile strength. - The tensile strength of the varnished tapes (measured lengthwise) (see 4.8.3.2), at room temperature, shall be not less than 2,400 pounds per square inch cross section, upon samples removed from the completed cable.

3.11.7.6 Varnished-cambric tape. - Used as a tape, the varnished cambric shall be of a suitable width, and so applied that samples taken from the completed cable shall meet the physical and electrical tests specified in 4.8.3. The nominal thickness of tape used shall be not less than 0.005 inch nor more than 0.013 inch, as may be most suitable for the type and size of cable in which used. The tension used in wrapping successive layers of tape around the conductor shall be such that all layers fit snugly and that there is no crinkling of the inner layers attributable to failures in applying or maintaining correct tension at all times.

3.11.7.7 Lap. - In general, and except where otherwise specifically required, the amount of lap used in applying the varnished cambric tape, and whether positive or negative, will be left to the discretion of the manufacturer.

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3.11.7.8 Thin varnished cambric walls. - For thin walls of varnished cambric, particularly those of 0.048 inch thickness or less, and/or where used in combination with asbestos, the thickness of varnished cambric tape used shall be given special consideration. For such walls it is required that there be a multiplicity of tape layers, with the best possible distribution of joints, and hence improved structure as a dielectric. Unless otherwise specified or approved for a particular type and size of cable (see 3.12 to 3.49, incl.), the tape thickness shall be such that at no point in the insulation shall there be less than 3 thicknesses of varnished cambric.

3.11.7.9 Alternates for varnished cambric. - Where varnished cambric is specified as a component of electric cables the use of the suitable alternate material will be permitted when satisfactory to the bureau or agency concerned. The suitability of such alternates shall be determined on the basis of tests specified in 3.1.1.

3.11.7.10 Effect of hot petrolatum. - After aging 15 minutes at 150°C. in hot petrolatum (see 4.8.3.3) and cooling to room temperature, the varnish coating on the tapes shall not be materially softer or more tacky than before being placed therein.

3.11.7.11 Dielectric strength. - The average dielectric strength of the varnished-cambric tapes (see 4.8.3.4), as removed from the completed cable, in terms of volts per mil of thickness at room temperature, shall be not less than 750 for single conductor cables and not less than 600 for multiple conductor cables.

3.11.7.12 Aging. - After 150 hours aging at 125°C. in an air oven (see 4.8.3.5) the varnish film on the tape shall show no evidence of cracking or peeling visible to the unaided eye, when bent around a mandrel 1/8 inch in diameter.

3.11.8 Lubricating compound for varnished-cambric insulation. -

3.11.8.1 Use. - A suitable compound shall be used for excluding air and moisture and for the lubrication of the varnished cambric insulation. It shall assure freedom from tearing or rupturing of the insulation when the completed cables are subjected to the bending test (see 4.8.9).

3.11.8.2 General properties. - The compound shall be a mineral base, acid-free compound, which will not dry out, oxidize, or combine with the varnish on the cambric. The compound shall have no deleterious effect upon any other component material in the cable construction.

3.11.8.3 Amount used. - The amount of compound used shall be such that the completed cable will pass the drip test specified in 4.8.8. The outer layers of varnished cambric, in the larger size conductors especially, may be applied without added compound, provided the tape itself is sufficiently lubricated to allow free motion between the layers in bending of the cable.

3.11.9 Asbestos. -

3.11.9.1 General. - All asbestos used in the construction of electric cables shall be of a form, type, and class suitable for the particular purpose, and shall conform to the applicable requirements of Specification MIL-I-3053. The composition and structure shall be such as will enable the completed cable to meet all specified test requirements.

3.11.9.2 Asbestos electrical insulating walls. - Asbestos electrical insulating walls shall consist of plain or reinforced asbestos applied either in treated or untreated form. The asbestos shall be suitably applied and compressed about the conductor so as to provide a dense felted wall of circular cross sections, in which the conductor is well centered. Such insulating walls are normally to be employed in combination with walls of insulating material of high dielectric strength, such as synthetic resin or varnished cambric. Asbestos for this application shall conform to either class 2 or class 4 of Specification MIL-I-3053.

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3.11.9.3 Asbestos heat and flame resistant walls. - The asbestos for the wall or beltings applied about two or more individually insulated conductors, which have been cabled together, shall consist of plain or reinforced asbestos, applied either as treated or untreated. The asbestos shall be suitably applied and compressed about the cable core so as to provide a dense felted wall of circular cross section in which the cable core is well centered. Asbestos for this application shall conform to either class 4 or class 6 of Specification MIL-I-3053. The thickness of this felted wall shall be such as to permit strict adherence of the completed cable with the specified overall dimensions.

3.11.9.4 Impregnation. - The asbestos over the individually insulated conductor (or conductors) and over the cable core shall be thoroughly impregnated with a moisture and flame retardant insulating compound in such a manner as to fill all interstices and make the insulation one homogeneous structure. The compound used to saturate the asbestos shall consist of suitable high grade ingredient and shall retain its initial qualities during the normal service life of the cable. The compound shall develop no injurious chemical action within itself or with any other of the component parts of the completed cable. The compound shall not drip from the completed cable when tested as specified in 4.8.8.

3.11.9.5 Asbestos fillers. - Fillers shall meet the requirements of 3.11.11.

3.11.10 Glass fiber: -

3.11.10.1 General. - All glass fibers used in the construction of electric cables shall be of a class and form suitable for the particular purpose, and shall conform to the applicable requirements of Specification MIL-G-1140.

3.11.10.2 Admixtures. - Where glass fiber insulation is specified as such, it shall not be used in a direct combination or mixture with any organic fabric or textile material, i.e., glass fiber rovings, yard, tape, cloth, braid, cord, bonded mat, or other form of glass fibrous material, and shall not be part cotton, part silk, part linen, etc., unless otherwise specifically permitted or required in 3.12 to 3.49, inclusive.

3.11.10.3 Glass fiber servings. - Servings shall be of multiple wound continuous fiber yarn, size 900-2/2 or the approximate equivalent, except where otherwise specified or approved.

3.11.10.4 Glass fiber braids. - The yarn for glass fiber braids shall be the appropriate size of either staple or continuous fiber yarn conforming to Specification MIL-G-1140. The braid angle, weave and thickness shall be as specified in 3.11.6.

3.11.10.5 Glass fiber electrical insulating walls. - Glass fiber electrical insulating walls shall consist of staple, fiber roving or condensed silver suitably applied and compressed about the conductor so as to provide a dense wall of circular cross section in which the conductor is well centered. Such electrical insulating walls are normally to be employed in combination with walls of insulating materials of high dielectric strength, such as synthetic resin or varnished cambric.

3.11.10.6 Glass fiber heat and flame resistant walls. - The glass fiber for walls or beltings applied about two or more individually insulated conductors, which have been cabled together, shall consist of staple fiber roving or condensed silver. The glass fiber shall be suitably applied and compressed about the cable core so as to provide a dense wall of circular cross section in which the cable core is well centered. The thickness of the felted wall shall be such as to permit strict adherence of the completed cable with the specified overall dimensions.

3.11.10.7 Impregnation. - The glass fiber over the individually insulated conductor (or conductors), including braids, and over the cable core shall be thoroughly impregnated with a moisture and flame retardant insulating compound in such a manner as to fill all interstices and make the insulation one homogeneous structure. The compound used to saturate the glass fibers shall consist of suitable high grade ingredients and shall retain its initial qualities during the normal service life of the cable. The compound shall develop no injurious chemical action within itself or with any other of the component parts of the completed cable. The compound shall not drip from the completed cable under the test conditions specified in 4.8.8.

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3.11.10.8 Glass fiber fillers. - Fillers shall meet the requirements of 3.11.11.

3.11.10.9 Glass fiber impervious sheath reinforcement. - Where glass fiber is used as a heat and flame resistant textile reinforcement in the impervious sheath construction for cables (see 3.11.18), it shall be of either staple or continuous fiber yarn.

3.11.11 Fillers. -

3.11.11.1 General. - Except where otherwise specified, filler material shall be cotton, jute, synthetic fibers, glass fibers, or asbestos. Filler material shall be of long fiber and free from defects, dirt, knots and lumps. Dry or untreated fillers shall not be used.

3.11.11.2 For heat and flame resistant cable types. -

3.11.11.2.1 General. - Asbestos or glass fibers, pre-impregnated with suitable moisture and flame retardent insulating compounds, shall be used in conjunction with the approved watertight core sealing compounds as fillers for heat and flame resistant cables (see 1.2.1.1). Fibrous fillers may be omitted, at manufacturer's option, only from cables having three or more conductors of standard size 3(7) or smaller, provided the required firmness of core is retained with the sealing compound alone. For the larger size cables, the asbestos or glass fibers shall occupy not less than 50 percent of the total valley space so as to provide a firm, well-rounded assembly as specified on the individual specification sheets.

3.11.11.2.2 Asbestos. - Where asbestos is employed as heat and flame resistant fillers, it shall conform to the applicable requirements for class 4 or class 6 of Specification MIL-I-3053.

3.11.11.2.3 Glass fiber. - Where glass fiber is employed as heat and flame resistant fillers it shall be of staple rovings and condensed silver and shall conform to the applicable requirements of Specification MIL-G-1140.

3.11.11.2.4 Filler impregnation. - The moisture and flame retardant impregnating compound shall consist of high grade insulating ingredients which retain their initial qualities during the normal service life of the cable. The compound shall develop no injurious chemical action within itself or with any other of the component parts of the completed cable. Except in flexible cables, the fillers shall be compacted and thoroughly impregnated in such a manner as to completely fill all interstices of the core and make the fillers one homogeneous structure. The amount of compound shall be not less than 10 percent of the weight of the saturated filler. In no case shall the compound from the fillers drip from the completed cable under the test conditions specified in 4.8.8.

3.11.11.3 For other cable types. - Fillers used in all multiple conductor cables (see 1.2.1.2, 1.2.2 and 1.2.3.1), except heat and flame resistant types (see 1.2.1.1), shall be of cotton, jute, glass or asbestos treated with a moisture resistant compound. The amount of compound shall be not less than 10 percent of the weight of the saturated fillers.

3.11.12 Cotton. -

3.11.12.1 General. - All cotton used shall be clean, dry and free from all foreign matter deleterious to the insulating qualities.

3.11.12.2 Separator. - Where a serving of cotton is used as a separator around a copper conductor, a minimum No. 60 single-ply cotton yarn shall be applied with full coverage.

3.11.12.3 Reinforcement. - Cotton used for reinforcing the sheath of cables intended for repeated flexing service, shall be multiple ends of seine twine, hawser cord, or heavy cabled cotton yarn of a size most suitable for the particular construction in which used. Alternate reinforcement may be used only when satisfactory to the bureau or agency concerned.

3.11.12.4 Tie cord. - Cotton tie cord shall be of a braided construction, having a breaking strength of not less than 30 pounds (see 4.8.28.2), and impregnated with a waterproofing flexible compound to a diameter of not more than 0.065 inch.

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3.11.12.5 Fillers and braids. - When used as fillers and braids, the requirements of 3.11.11 and 3.11.16 shall also be met.

3.11.13 Rayon and other synthetic fiber. - Rayon and other synthetic fiber used in the construction of electrical cable and wire shall be of the best quality and in the best form suited for the purpose intended. It shall be clean, dry, and free from foreign matter deleterious to its insulating qualities and shall conform to Specification MIL-S-572.

3.11.14 Separators. -

3.11.14.1 Where a separator is specified or permitted, the following materials, when properly employed, will be considered suitable. Other materials shall be used only when satisfactory to the bureau or agency concerned.

3.11.14.1.1 For heat and flame resistant types. - A wind of glass fibers, or other suitable, heat-resistant synthetic fiber or a thin tape of moisture-resistant cellophane, cellulose acetate, or flame-resistant synthetic resin will be suitable. The wall thickness of the applied separator shall not exceed 0.004 inch.

3.11.14.1.2 For other types. - A wind or tape of cotton, paper, rayon or cellulose acetate will be suitable. This type of separator shall not be used for circuit identification.

3.11.15 Tape, compound-filled. -

3.11.15.1 General. - The tape shall be made of cotton sheeting, evenly and firmly woven from good cotton, free from unsightly defects, dirt, knots, lumps and irregularities of twist. It shall be thoroughly impregnated and evenly coated on both sides or one side when specifically approved, with an insulating compound. The compound shall adhere firmly to the fabric. The compound shall develop no injurious chemical action within itself or with any other of the component parts of the complete cable.

3.11.15.2 Thickness. - The thickness of the tape shall be not less than 0.008 inch.

3.11.15.3 Application. - The tape shall be applied in a manner to insure a smooth surface and a circular section. The tape shall be applied with not less than 1/8-inch lap for diameters under 0.2-inch, and not less than 1/4-inch lap for larger diameters.

3.11.16 Braids, textile. -

3.11.16.1 General. - Braids shall be of cotton (see 3.11.12), rayon (see 3.11.13), or glass fiber (see 3.11.10) as specified for the particular type and size of cable.

3.11.16.2 Inner (marker) braids. -

3.11.16.2.1 General. - Except where otherwise specified or approved, inner braids shall be closely woven of 2-ply cotton, or an approved weave of rayon or glass fibers at the manufacturer's option. These braids shall be color-coded for circuit identification, as specified in 3.9. The braids shall be suitably treated so that they will not fray or loosen at the cut ends after reasonable handling incident to use.

3.11.16.2.2 Thread size. - Cotton marker braids shall be woven from yarn not smaller than that specified in table VIII.

Table VIII - Thread size.

Diameter under braid Inches	Minimum thread size Size/ply
0.050 to 0.400	40/2
.401 to .800	30/2
.801 to 3.000	20/2

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3.11.16.2.3 Braid angle. - The tangent of the angle between the cotton thread and the axis of the wire shall be not less than that specified in table IX.

Table IX - Braid angle.

Diameter under braid	Value of tangent	Corresponding braid angle
Inches		Degrees
0.050 to 1.000	0.700	35
1.001 to 1.500	.839	40
1.501 to 3.000	1.000	45

3.11.16.3 Outer cotton braids. -

3.11.16.3.1 General. - Outer cotton braids shall be closely woven from cotton thread having not less than 2 plies. The braid shall be thoroughly saturated with a flexible, hard, black (except where otherwise specified) moisture and flame retardant, insulating compound, which shall not ooze out or drip off at a temperature of 95°C.

3.11.16.3.2 Braid angle. - The tangent of the angle between the cotton thread and the axis of the wire shall be not less than that specified in table IX.

3.11.16.3.3 Braid thickness. - The thickness of outer cotton braids shall be not less than that specified in table X.

Table X - Braid thickness.

Diameter under braid	Thickness of braid minimum	Corresponding size and ply of cotton yarn minimum
Inches	Inch	Size/ply
0.050 to 0.100	0.014	40/2
.101 to .200	.016	30/2
.201 to .350	.017	26/2
.351 to .800	.020	20/2
.801 to 1.500	.026	12/2
1.501 to 3.000	.032	8/2

3.11.16.4 Glass braids. - Glass braids shall conform to 3.11.10. and 3.11.16.1 to 3.11.16.2, inclusive, as far as practicable.

3.11.16.5 Rayon braids. - Rayon braids shall conform to 3.11.13, and 3.11.16.1 to 3.11.16.2, inclusive, as far as practicable.

3.11.16.6 Copper shielding braid. - Shielding braids shall be of bare, uncoated copper if applied over synthetic resin insulation (see 3.11.2) or polythene insulation (see 3.11.4). Tinned copper shall be employed over rubber insulation (see 3.11.3 and 3.11.5). The percent of coverage shall conform to 3.11.20.6. The braid angle shall not be less than 30 degrees.

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3.11.17 Sheath, polychloroprene. - Polychloroprene sheath shall fit closely to the cable core and, except in types TRF and SHOF, shall be readily removable without damage to the conductor insulation. Samples of the sheath, removed from completed cable shall meet the following requirements when tested as specified in 4.8.1.

3.11.17.1 Tensile strength. - The tensile strength shall be not less than 1800 pounds per square inch.

3.11.17.2 Elongation. - Specimens shall stretch until the 2-inch bench marks are not less than 8 inches apart before rupture.

3.11.17.3 Set. - The set, after elongating specimens until the 2-inch bench marks are 6 inches apart, shall not exceed 3/8 inch.

3.11.17.4 Air pressure heat aging. - After 20 hours aging at a temperature of 126.7°C. in the air pressure heat bomb (see 4.8.1.3), the tensile strength and elongation shall be not less than 65 percent of the values measured before aging.

3.11.17.5 Hot oil immersion. - After subjection to the hot oil test (see 4.8.17) the tensile strength and elongation shall be not less than 65 percent of the values measured before aging.

3.11.18 Sheath, impervious. -

3.11.18.1 General. - Where specified in 3.12 to 3.49, inclusive, cable shall be provided with an impervious sheath as approved on the basis of qualification tests (see 3.1). The sheath shall be applied to the cable core as a close-fitting extruded tube to provide a concentric wall of uniform texture throughout. For cables with braided metal armor, application of the sheath in laminated form, or the inclusion of textile reinforcement within the sheath wall or to the outer surface of the sheath, with the object of subsequently molding such laminations or several component materials together to form an impervious sheath, will not be acceptable. Cables for repeated flexing service may have a suitable reinforcement under or within the body of the impervious sheath, when so employed in the approved sample (see 3.1). The sheath compound shall be black, unless a colored sheath is specified.

3.11.18.2 Imperviousness. - The sheath shall be of such composition and construction as to render the completed cable impervious to water and air under pressure when tested as specified in 4.8.13.

3.11.18.3 Specific characteristics. -

3.11.18.3.1 General. - The composition and construction of the impervious sheath shall provide a combination of the following basic properties that is properly proportioned for the particular purpose intended:

(a) Resistance to flame.
(b) Stability under long continued exposure to elevated temperatures.
(c) Absence of noticeable odor at cable operating temperatures.
(d) Deformability at low temperature (i.e. cold bending qualities).
(e) Stability under exposure to oil, sulfuric acid, and sea water.
(f) Stability under exposure to sunlight.

3.11.18.3.2 Nonflexing service. - Particular emphasis shall be placed on the flame resistance and the stability under heating. The remaining characteristics shall be maintained at as high a level as can be made compatible with this objective.

3.11.18.3.3 Repeated flexing service. - Particular emphasis shall be placed on the flexing endurance characteristic. The remaining characteristics shall be maintained at as high a level as can be made compatible with this objective.

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3.11.19 Sheath, lead. -

3.11.19.1 General. - The lead sheath shall consist of commercially pure lead (not less than 99.85 percent pure) without flaws, and be tightly formed about the cable core.

3.11.19.2 Thickness. - Unless otherwise specified for the particular type and size of cable involved, lead sheaths shall have an average thickness not less than that specified in table XI. In no case shall the sheath thickness be less, at any point than 90 percent of the required average thickness.

Table XI - Average thickness of lead sheath.

Diameter over cable core	Thickness of lead sheath
Inches	Inch
0.701 to 1.050	0.078
1.051 to 1.500	.094
1.501 to 2.000	.109
2.001 to 3.000	.125

3.11.20 Armor, braided metal. -

3.11.20.1 General. - Armoring wire shall be of 0.0126 ± 0.0005 inch diameter aluminum alloy, except where otherwise specified. The wire shall be of uniform diameter free from cracks, splints, or other flaws. Splices in the wire shall be infrequent, staggered, and inconspicuous, so as not to increase the normal diameter of the cable or result in rough spots. The weave shall be such that the wires of a strand are laid closely together, flat and parallel, tightly and firmly binding the sheathed cable. The braid shall be applied with sufficient tension to prevent loosening and creeping, but without so much tension that excessive broken ends occur. It shall remain snug to the cable when cut and not spring away from the sheath. It shall withstand the tests specified in 4.8.7. It shall provide reasonable mechanical protection to the sheath during installation and handling, and shall retard deformation of the cable under compression from stuffing tubes and cable hangers.

3.11.20.2 Steel armor. -

3.11.20.2.1 Steel armor shall be of soft annealed wire and shall be thoroughly and evenly galvanized.

3.11.20.2.2 Galvanized steel wire 0.0095 ± 0.0005 inch in diameter shall have a tensile strength of not less than 50,000 pounds per square inch, and the elongation in 8 inches shall be not less than 12 percent, as tested before application to the cable.

3.11.20.2.3 Galvanized steel wire 0.0126 ± 0.0005 inch in diameter shall have a tensile strength of not less than 50,000 pounds per square inch, and the elongation in 8 inches shall be not less than 12 percent, as tested before application to the cable.

3.11.20.3 Bronze armor. - Bronze armor shall be of silicon, phosphor, or cadmium bronze wire and of a suitable composition to resist sea water corrosion. The wire shall be 0.0126 ± 0.0005 inch in diameter.

3.11.20.4 Aluminum armor. -

3.11.20.4.1 Aluminum armor shall be of an aluminum alloy wire containing not more than 0.2 percent copper and not less than 92.5 percent aluminum. The wire shall be 0.0126 ± 0.0005 inch in diameter

3.11.20.4.2 Aluminum wire shall have a tensile strength of not less than 50,000 pounds per square inch, and the elongation in 10 inches shall be not less than 2 percent, as tested before application to the cable.

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3.11.20.5 Weave. - The weave shall be of either the "one over-one under" or the "two over-two under" type. The selection of the number of ends per carrier and the number of carriers per braider shall be such as to produce a basket weave with a braid angle within the limits shown in table XII. The number of ends per carrier shall be not more than the values shown in table XIII.

Table XII - Braid angle.

Diameter over sheath	Braid angle	
	Minimum	Maximum
Inches	Degrees	Degrees
0.100 to 0.600	30	60
.601 to 1.000	35	60
1.001 to 1.500	40	70
1.501 to 2.000	45	70
2.001 to 3.000	50	80

Table XIII - Ends per carrier.

Cable diameter under braid	Maximum number of ends per carrier	
	One over-one under	Two over-two under
Inches		
0.100 to 0.400	8	5
.401 to .800	12	10
.801 to 3.000	16	10

3.11.20.6 Coverage. - The coverage for all types and sizes of cable shall be no less than 88 percent, as determined by the following formula:

$$K = (2F - F^2) \times 100.$$

Where: K = Percent coverage.
a = Angle of braid with axis of cable.
$\tan a = 2\pi DP/C$.
d = Diameter of individual braid wire in inches.
C = Number of carriers.
D = Diameter of cable under armor in inches.
F = NPd/Sin a.
N = Number of wires per carrier.
P = Picks per inch of cable length.

3.11.21 Paint. -

3.11.21.1 General. - Where specified (see 3.12 to 3.49, inclusive), braided metal armored cables shall be suitably coated with aluminum paint.

3.11.21.2 Aluminum paint. - Aluminum paint shall consist of aluminum pigment (paste form) conforming to Specification TT-A-468, in a suitable vehicle of the synthetic resinous (phenol-formaldehyde or glyceride) type designed for good penetration, high bonding power, high resistance to water and to oil, and with a high degree of flexibility at temperatures up to 100°C.

3.11.21.3 Application. - Painting shall be done as soon as practicable after completion of prior manufacturing operations so as to prevent accumulation of dirt, moisture and foreign matter on the braid or in the interstices. The paint shall be properly dried and shall not be tacky. The painting shall be done preferably by the immersion method so as to fill in all interstices in the outer cable surface. The

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paint finish shall provide a relatively smooth overall coating that will facilitate making watertight glands with Navy standard stuffing tubes and approved packing. It shall be flexible and adhere firmly so that it will not crack, peel, or flake off under the cold bend (see 4.8.9) and the heat endurance (see 4.8.12) tests.

3.11.22 End seals. -

3.11.22.1 General. - All types and sizes of cable specified herein, except miscellaneous wires (see 1.2.4), shall have both ends of every shipping length provided with a seal to prevent the entrance of moisture. The end seals shall be constructed as shown on figure 1, unless specific approval is given to the manufacturer for the use of an alternate method. Approval of alternate methods shall be granted only after tests made at a Government laboratory (see 3.1), and when the design is considered to be such that uniformly satisfactory sealing will be obtained in routine factory applications. Having once fixed upon a given design of end seal and having received qualification the manufacturer shall adhere strictly under subsequent deliveries of cable to the construction so tested and approved.

3.11.22.2 Watertightness. - The end seal shall be of such composition and shall be applied in such manner as to prevent the entrance of water into the cable under an externally applied pressure. A sample of the completed cable with end seal in place shall withstand without leakage, the water and air leakage test specified in 4.8.13.

3.11.23 Natural rubber. - Unless specifically authorized, the use of natural rubber either as a component of electric cables or as a component of end seals, will not be permitted.

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Individual Specification Sheet dated 30 July 1952

3.12 Type SHFA cable.

CONSTRUCTION

3.12.1 Types SHFA-3 to 9, inclusive:

First. A standard conductor of bare, uncoated copper (see 3.10).
Second. A wall of synthetic resin insulation (see 3.11.2).
Third. A wall of felted asbestos (see 3.11.9) or glass fiber (see 3.11.10).
Fourth. With or without separator (see 3.11.14) at manufacturer's option.
Fifth. An impervious sheath (see 3.11.18).
Sixth. Braided metal armor (see 3.11.20).
Seventh. Paint (see 3.11.21).

3.12.2 Types SHFA-14 to 800, inclusive:

First. A standard conductor of bare, uncoated copper (see 3.10).
Second. With or without separator (see 3.11.14) at manufacturer's option.
Third. A wall of felted asbestos (see 3.11.9) or glass fiber (see 3.11.10).
Fourth. A separator (see 3.11.14).
Fifth. A wall of varnished cambric insulation (see 3.11.7)
Sixth. A separator (see 3.11.14).
Seventh. A wall of felted asbestos (see 3.11.9) or glass fiber (see 3.11.10).
Eighth. With or without separator (see 3.11.14) at manufacturer's option.
Ninth. An impervious sheath (see 3.11.18).
Tenth. Braided metal armor (see 3.11.20).
Eleventh. Paint (see 3.11.21).

TEST REQUIREMENTS

Drip (see 4.8.8).
Cold bend (see 4.8.9).
Flammability (see 4.8.10).
Current overload (see 4.8.11).
Water and air leakage (see 4.8.13).
Electrical tests (see following table).
Watertightness (see Appendix I).

Type	Standard copper conductor	Diameters (inches)			Dielectric strength (see 4.8.19)		Conductor resistance (d.c.) per 1000 feet at 25°C. (see 4.8.24)	Insulation resistance 1000 feet (see 4.8.25)	Cold bend mandrel diameter (see 4.8.9)
		Over-synthetic resin		Overall	After flame test	Full reel	Bare		
					Minimum	Minimum	Maximum	Minimum	
		Minimum			Volts	Volts	Ohms	Megohms	Inches
SHFA-3	3(7)	0.092		0.355	1,000	2,000	3.92	100	5
SHFA-4	4(7)	.115		.500	1,000	2,000	2.46	100	6
SHFA-9	9(7)	.163		.656	1,000	2,000	1.23	100	8
		Over first asbestos or glass	Over varnished cambric						
SHFA-14	14(7)	0.226	0.334	.684	1,500	2,000	.784	225	9
SHFA-23	23(7)	.261	.369	.719	1,500	2,000	.493	200	9
SHFA-30	30(19)	.291	.399	.749	1,500	2,000	.365	180	10
SHFA-40	40(19)	.316	.424	.774	1,500	2,000	.288	170	10
SHFA-50	50(19)	.344	.452	.802	1,500	2,000	.228	160	11
SHFA-75	75(37)	.407	.515	.865	1,500	2,000	.149	135	11
SHFA-100	100(61)	.453	.561	.911	1,500	2,000	.115	125	12
SHFA-150	150(61)	.547	.655	1.005	1,500	3,000	.0716	100	13
SHFA-200	200(61)	.604	.712	1.062	1,500	3,000	.0570	90	13
SHFA-300	300(91)	.718	.826	1.176	1,500	3,000	.0382	80	15
SHFA-400	400(127)	.832	.988	1.338	1,500	3,000	.0273	70	17
SHFA-500	500(127)	.923	1.079	1.429	1,500	3,000	.0217	65	18
SHFA-650	650(127)	1.026	1.182	1.532	1,500	3,000	.0172	60	20
SHFA-800	800(127)	1.141	1.297	1.647	1,500	3,000	.0136	50	21

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Individual Specification Sheet dated 30 July 1952

3.13 Type DHFA cable.

CONSTRUCTION

3.13.1 Types DHFA-3 to 9 inclusive:

First. A standard conductor of bare, uncoated copper (see 3.10).
Second. A wall of synthetic resin insulation (see 3.11.2).
Third. A wall of felted asbestos (see 3.11.9) or glass fiber (see 3.11.10).
Fourth. Two conductors shall be twisted together with a lay not greater than 24 times the pitch diameter. Fillers (see 3.11.11) shall be employed to obtain a firm, well rounded assembly.
Fifth. A wall of felted asbestos (see 3.11.9) or glass fiber (see 3.11.10).
Sixth. With or without separator (see 3.11.14) at manufacturer's option.
Seventh. An impervious sheath (see 3.11.18).
Eighth. Braided metal armor (see 3.11.20).
Ninth. Paint (see 3.11.21).

3.13.2 Types DHFA-14 to 400, inclusive:

First. A standard conductor of bare, uncoated copper (see 3.10).
Second. With or without separator (see 3.11.14) at manufacturer's option.
Third. A wall of felted asbestos (see 3.11.9) or glass fiber (see 3.11.10).
Fourth. A separator (see 3.11.14).
Fifth. A wall of varnished cambric insulation (see 3.11.7).
Sixth. A separator (see 3.11.14).
Seventh. A wall of felted asbestos (see 3.11.9) or glass fiber (see 3.11.10).
Eighth. Two conductors shall be twisted together with a lay not greater than 24 times the pitch diameter. Fillers (see 3.11.11) shall be employed to obtain a firm, well rounded assembly.
Ninth. A wall of felted asbestos (see 3.11.9) or glass fiber (see 3.11.10).
Tenth. With or without separator (see 3.11.14) at manufacturer's option.
Eleventh. An impervious sheath (see 3.11.18).
Twelfth. Braided metal armor (see 3.11.20).
Thirteenth. Paint (see 3.11.21).

TEST REQUIREMENTS

Drip (see 4.8.8).
Cold bend (see 4.8.9).
Flammability (see 4.8.10).
Current overload (see 4.8.11).
Water and air leakage (see 4.8.13).
Electrical tests (see following table).
Watertightness (see Appendix I).

Type	Standard copper conductor	Diameters (inches): Over synthetic resin	Over first asbestos or glass	Over varnished cambric	Over second asbestos or glass	Overall	Dielectric strength (see 4.8.19): After flame test, Minimum	Full reel, Minimum	Conductor resistance (d.c.) per 1000 feet at 25°C. (see 4.8.24): Bare, Maximum	Insulation resistance 1,000 feet (see 4.8.25): Minimum	Cold bend mandrel diameter (see 4.8.9)
		Minimum					Volts	Volts	Ohms	Megohms	Inches
DHFA-3	3(7)	0.092	0.140			0.530	1,000	2,000	4.00	100	7
DHFA-4	4(7)	.115	.214			.778	1,000	2,000	2.50	100	10
DHFA-9	9(7)	.163	.246			.842	1,000	2,000	1.26	100	11
		Over first asbestos or glass	Over varnished cambric	Over second asbestos or glass							
DHFA-14	14(7)	0.166	0.226	0.286		.922	1,500	2,000	.800	275	12
DHFA-23	23(7)	.201	.261	.321		.992	1,500	2,000	.506	250	13
DHFA-30	30(19)	.241	.313	.373		1.108	1,500	2,000	.362	225	14
DHFA-40	40(19)	.266	.338	.398		1.168	1,500	2,000	.294	215	15
DHFA-50	50(19)	.294	.366	.426		1.224	1,500	2,000	.233	200	16
DHFA-60	60(37)	.332	.404	.484		1.340	1,500	2,000	.193	190	17
DHFA-75	75(37)	.367	.457	.537		1.474	1,500	2,000	.152	175	19
DHFA-100	100(61)	.423	.513	.593		1.586	1,500	2,000	.117	160	20
DHFA-125	125(61)	.467	.557	.637		1.674	1,500	2,000	.0922	150	22
DHFA-150	150(61)	.537	.627	.717		1.854	1,500	3,000	.0730	135	24
DHFA-200	200(61)	.594	.684	.774		1.968	1,500	3,000	.0581	125	25
DHFA-250	250(61)	.669	.757	.847		2.142	1,500	3,000	.0462	120	27
DHFA-300	300(91)	.718	.326	.916		2.280	1,500	3,000	.0390	110	30
DHFA-400	400(127)	.832	.940	1.030		2.508	1,500	3,000	.0278	100	32

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3.14 Type THFA cable.

CONSTRUCTION

3.14.1 Types THFA-3 to 9, inclusive:

First. A standard conductor of bare, uncoated copper (see 3.10).
Second. A wall of synthetic resin insulation (see 3.11.2).
Third. A wall of felted asbestos (see 3.11.9) or glass fiber (see 3.11.10).
Fourth. Three conductors shall be twisted together with a lay not greater than 24 times the pitch diameter. Fillers (see 3.11.11) shall be employed to obtain a firm, well rounded assembly.
Fifth. A wall of felted asbestos (see 3.11.9) or glass fiber (see 3.11.10).
Sixth. With or without separator (see 3.11.14) at manufacturer's option.
Seventh. An impervious sheath (see 3.11.18).
Eighth. Braided metal armor (see 3.11.20).
Ninth. Paint (see 3.11.21).

3.14.2 Types THFA-14 to 400, inclusive:

First. A standard conductor of bare, uncoated copper (see 3.10).
Second. With or without separator (see 3.11.14) at manufacturer's option.
Third. A wall of felted asbestos (see 3.11.9) or glass fiber (see 3.11.10).
Fourth. A separator (see 3.11.14).
Fifth. A wall of varnished cambric insulation (see 3.11.7).
Sixth. A separator (see 3.11.14).
Seventh. A wall of felted asbestos (see 3.11.9) or glass fiber (see 3.11.10).
Eighth. Three conductors shall be twisted together with a lay not greater than 24 times the pitch diameter. Fillers (see 3.11.11) shall be employed to obtain a firm, well rounded assembly.
Ninth. A wall of felted asbestos (see 3.11.9) or glass fiber (see 3.11.10).
Tenth. With or without separator (see 3.11.14) at manufacturer's option.
Eleventh. An impervious sheath (see 3.11.18).
Twelfth. Braided metal armor (see 3.11.20).
Thirteenth. Paint (see 3.11.21).

TEST REQUIREMENTS

Drip (see 4.8.8).
Cold bend (see 4.8.9).
Flammability (see 4.8.10).
Current overload (see 4.8.11).
Water and air leakage (see 4.8.13).
Electrical tests (see following table).
Watertightness (see Appendix I).

Type	Standard copper conductor	Diameters (inches): Over synthetic resin	Over first asbestos or glass	Over var-nished cambric	Over second asbestos or glass	Over-all	Dielectric strength (see 4.8.19): After flame test, Minimum	Full reel, Minimum	Conductor resistance (d.c.) per 1000 feet at 25°C. (see 4.8.24): Bare, Maximum	Insulation resistance 1,000 feet: (see 4.8.25), Minimum	Cold bend mandrel diameter (see 4.8.9)
		Minimum					Volts	Volts	Ohms	Megohms	Inches
THFA-3	3(7)	0.092	0.140			0.560	1,000	2,000	4.00	100	7
THFA-4	4(7)	.115	.214			.812	1,000	2,000	2.50	100	10
THFA-9	9(7)	.163	.246			.881	1,000	2,000	1.28	100	11
		Over first asbestos or glass		Over var-nished cambric	Over second asbestos or glass						
THFA-14	14(7)	0.166	0.226	0.286	.968		1,500	2,000	.800	275	12
THFA-23	23(7)	.201	.261	.321	1.043		1,500	2,000	.505	250	13
THFA-30	30(19)	.241	.313	.373	1.168		1,500	2,000	.362	225	15
THFA-40	40(19)	.266	.338	.398	1.232		1,500	2,000	.294	215	16
THFA-50	50(19)	.294	.366	.426	1.292		1,500	2,000	.233	200	17
THFA-60	60(37)	.332	.404	.484	1.417		1,500	2,000	.193	190	18
THFA-75	75(37)	.367	.457	.537	1.560		1,500	2,000	.152	175	20
THFA-100	100(61)	.423	.513	.593	1.681		1,500	2,000	.117	160	22
THFA-125	125(61)	.467	.557	.637	1.776		1,500	2,000	.0922	150	23
THFA-150	150(61)	.537	.627	.717	1.969		1,500	3,000	.0730	135	25
THFA-200	200(61)	.594	.684	.774	2.092		1,500	3,000	.0581	125	27
THFA-250	250(61)	.667	.757	.847	2.278		1,500	3,000	.0462	120	29
THFA-300	300(91)	.718	.826	.916	2.427		1,500	3,000	.0390	110	31
THFA-350	350(91)	.772	.880	.970	2.545		1,500	3,000	.0328	105	33
THFA-400	400(127)	.832	.940	1.030	2.673		1,500	3,000	.0278	100	34

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3.15 Type FHFA cable.

CONSTRUCTION

3.15.1 Types FHFA-3 to 9, inclusive:

First. A standard conductor of bare, uncoated copper (see 3.10).
Second. A wall of synthetic resin insulation (see 3.11.2).
Third. A wall of felted asbestos (see 3.11.9) or glass fiber (see 3.11.10).
Fourth. Four conductors shall be twisted together with a lay not greater than 24 times the pitch diameter. Fillers (see 3.11.11) shall be employed to obtain a firm, well-rounded assembly.
Fifth. A wall of felted asbestos (see 3.11.9) or glass fiber (see 3.11.10).
Sixth. With or without separator (see 3.11.14) at manufacturer's option.
Seventh. An impervious sheath (see 3.11.18).
Eighth. Braided metal armor (see 3.11.20).
Ninth. Paint (see 3.11.21).

3.15.2 Types FHFA-23 and 50:

First. A standard conductor of bare, uncoated copper (see 3.10).
Second. With or without separator (see 3.11.14) at manufacturer's option.
Third. A wall of felted asbestos (see 3.11.9) or glass fiber (see 3.11.10).
Fourth. A separator (see 3.11.14).
Fifth. A wall of varnished cambric insulation (see 3.11.7).
Sixth. A separator (see 3.11.14).
Seventh. A wall of felted asbestos (see 3.11.9) or glass fiber (see 3.11.10).
Eighth. Four conductors shall be twisted together with a lay not greater than 24 times the pitch diameter. Fillers (see 3.11.11) shall be employed to obtain a firm, well-rounded assembly.
Ninth. A wall of felted asbestos (see 3.11.9) or glass fiber (see 3.11.10).
Tenth. With or without separator (see 3.11.14) at manufacturer's option.
Eleventh. An impervious sheath (see 3.11.18).
Twelfth. Braided metal armor (see 3.11.20).
Thirteenth. Paint (see 3.11.21).

TEST REQUIREMENTS

Drip (see 4.8.8).
Cold bend (see 4.8.9).
Flammability (see 4.8.10).
Current overload (see 4.8.11).
Water and air leakage (see 4.8.13).
Electrical tests (see following table).
Watertightness (see Appendix I).

Type	Standard copper conductor	Diameter (inches): Over synthetic resin, Minimum	Diameter (inches): Over first asbestos or glass	Diameter (inches): Over varnished cambric	Diameter (inches): Over second asbestos or glass	Diameter (inches): Overall	Dielectric strength (see 4.8.19): After flame test	Dielectric strength (see 4.8.19): Full reel	Conductor resistance (d.c.) per 1,000 feet at 25°C. (see 4.8.24): Bare, Maximum	Insulation resistance (see 4.8.25) 1,000 ft.: Minimum	Cold bend mandrel diameter (see 4.8.9)
							Volts	Volts	Ohms	Megohms	Inches
FHFA-3	3(7)	0.092	0.140	-----	------	0.610	1,000	2,000	4.00	100	8
FHFA-4	4(7)	.115	.214	-----	------	.865	1,000	2,000	2.50	100	11
FHFA-9	9(7)	.163	.246	-----	------	.920	1,000	2,000	1.26	100	12
FHFA-23	23(7)	-----	.201	0.261	0.321	1.130	1,500	2,000	.505	250	14
FHFA-50	50(19)	-----	.294	.366	.426	1.410	1,500	2,000	.233	200	18

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3.16 Type SHFP cable (For 5000 volt service).

CONSTRUCTION

First. A standard conductor of bare, uncoated copper (see 3.10).
Second. With or without separator (see 3.11.14) at manufacturer's option.
Third. A wall of felted asbestos (see 3.11.9) or glass fiber (see 3.11.10).
Fourth. A separator (see 3.11.14).
Fifth. A wall of varnished cambric insulation (see 3.11.7).
Sixth. A separator (see 3.11.14).
Seventh. A wall of felted asbestos (see 3.11.9) or glass fiber (see 3.11.10).
Eighth. With or without separator (see 3.11.14) at manufacturer's option.
Ninth. An impervious sheath (see 3.11.18).
Tenth. Braided bronze armor (see 3.11.20).

TEST REQUIREMENTS

Drip (see 4.8.8).
Cold bend (see 4.8.9).
Flammability (see 4.8.10).
Water and air leakage (see 4.8.13).
Electrical tests (see following table).
Watertightness (see Appendix I).

Type	Diameters: Standard copper conductor	Over first asbestos or glass	Over varnished cambric	Over second asbestos or glass	Over-all	Dielectric strength (see 4.8.19): After flame test: Minimum	Full reel: Minimum	Conductor resistance (d.c.) per 1000 feet at 25°C. (see 4.8.24): Bare Maximum	Insulation resistance 1,000 feet (see 4.8.25): Minimum	Cold bend mandrel diameter (see 4.8.9)
	Inches	Inches	Inches	Inches		Volts	Volts	Ohms	Megohms	Inches
SHFP-200	200(61)	0.586	0.838	0.940	1.260	3,000	12,000	0.0570	225	16
SHFP-300	300(91)	.700	.952	1.054	1.374	3,000	12,000	.0382	200	18
SHFP-400	400(127)	.814	1.066	1.168	1.488	3,000	12,000	.0273	175	20
SHFP-500	500(127)	.905	1.157	1.259	1.579	3,000	12,000	.0217	160	21
SHFP-650	650(127)	1.008	1.260	1.362	1.682	3,000	12,000	.0172	140	22
SHFP-800	800(127)	1.123	1.375	1.477	1.797	3,000	12,000	.0136	125	23

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3.17 Type SHFR cable (For 3,000-volt service).

CONSTRUCTION

First. A standard conductor of bare, uncoated copper (see 3.10).
Second. A wall of synthetic resin insulation (see 3.11.2).
Third. With or without separator (see 3.11.14) at manufacturer's option.
Fourth. A wall of varnished-cambric insulation (see 3.11.7).
Fifth. A separator (see 3.11.14).
Sixth. A wall of felted asbestos (see 3.11.9) or glass fiber (see 3.11.10).
Seventh. With or without separator (see 3.11.14) at manufacturer's option.
Eighth. An impervious sheath (see 3.11.18).
Ninth. Braided metal armor (see 3.11.20).
Tenth. Paint (see 3.11.21).

TEST REQUIREMENTS

Cold bend (see 4.8.9).
Flammability (see 4.8.10).
Water and air leakage (see 4.8.13).
Electrical tests (see following table).
Watertightness (see Appendix I).

Type	Standard copper conductor	Diameters: Over synthetic resin insulation, Minimum	Diameters: Over varnished cambric	Diameters: Overall	Dielectric strength (see 4.8.19): After flame test, Minimum	Dielectric strength (see 4.8.19): Full reel, Minimum	Conductor resistance (d.c.) per 1000 feet at 25°C. (see 4.8.24): Bare, Maximum	Insulation resistance 1,000 feet (see 4.8.25): Minimum	Cold bend mandrel diameter (see 4.8.9)
		Inch	Inch	Inch	Volts	Volts	Ohms	Megohms	Inches
SHFR-4	4(7)	0.115	0.185	0.540	3,000	8,000	2.46	300	7

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3.18 Type DHFR cable (For 3,000 volt service).

CONSTRUCTION

First. A standard conductor of bare, uncoated copper (see 3.10).
Second. A wall of synthetic resin insulation (see 3.11.2).
Third. With or without separator (see 3.11.14) at manufacturer's option.
Fourth. A wall of varnished-cambric insulation (see 3.11.7).
Fifth. A separator (see 3.11.14).
Sixth. A wall of felted asbestos (see 3.11.9) or glass fiber (see 3.11.10).
Seventh. Two conductors shall be twisted together with a lay not greater than 24 times the pitch diameter. Fillers (see 3.11.11 shall be employed to obtain a firm, well rounded assembly.
Eighth. A wall of felted asbestos (see 3.11.9) or glass fiber (see 3.11.10).
Ninth. With or without separator (see 3.11.14) at manufacturer's option.
Tenth. An impervious sheath (see 3.11.18).
Eleventh. Braided metal armor (see 3.11.20).
Twelfth. Paint (see 3.11.21).

TEST REQUIREMENTS

Cold bend (see 4.8.9).
Flammability (see 4.8.10)
Water and air leakage (see 4.8.13).
Electrical tests (see following table).
Watertightness (see Appendix I).

Type	Diameters: Standard copper conductor	Diameters: Over synthetic resin Minimum	Diameters: Over varnished cambric	Diameters: Over first asbestos or glass	Diameters: Over-all	Dielectric strength (see 4.8.19): After flame test Minimum Volts	Dielectric strength (see 4.8.19): Full reel Minimum Volts	Conductor resistance (d.c.) per 1000 feet at 25°C. (see 4.8.24): Bare Maximum Ohms	Insulation resistance 1,000 feet (see 4.8.25): Minimum Megohms	Cold bend mandrel diameter (see 4.8.9): Inches
		Inch	Inch	Inch	Inch	Volts	Volts	Ohms	Megohms	Inches
DHFR-4	4(7)	0.115	0.185	0.247	0.844	3,000	8,000	2.50	400	10

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3.19 Type THFR cable (For 3,000 volt service).

CONSTRUCTION

First. A standard conductor of bare, uncoated copper (see 3.10).
Second. A wall of synthetic resin insulation (see 3.11.2).
Third. With or without separator (see 3.11.14) at manufacturer's option.
Fourth. A wall of varnished-cambric insulation (see 3.11.7).
Fifth. A separator (see 3.11.14).
Sixth. A wall of felted asbestos (see 3.11.9) or glass fiber (see 3.11.10).
Seventh. Three conductors shall be twisted together with a lay not greater than 24 times the pitch diameter. Fillers (see 3.11.11) shall be employed to obtain a firm, well rounded assembly.
Eighth. A wall of felted asbestos (see 3.11.9) or glass fiber (see 3.11.10).
Ninth. With or without separator (see 3.11.14) at manufacturer's option.
Tenth. An impervious sheath (see 3.11.18).
Eleventh. Braided metal armor (see 3.11.20).
Twelfth. Paint (see 3.11.21).

TEST REQUIREMENTS

Cold bend (see 4.8.9).
Flammability (see 4.8.10).
Water and air leakage (see 4.8.13).
Electrical tests (see following table).
Watertightness (see Appendix I).

Type	Diameters: Standard copper conductor	Diameters: Over synthetic resin Minimum	Diameters: Over varnished cambric	Diameters: Over first asbestos or glass	Diameters: Over-all	Dielectric strength (see 4.8.19): After flame test Minimum	Dielectric strength (see 4.8.19): Full reel Minimum	Conductor resistance (d.c.) per 1000 feet at 25°C. (see 4.8.24): Bare Maximum	Insulation resistance 1,000 feet (see 4.8.25): Minimum	Cold bend mandrel diameter (see 4.8.9)
		Inch	Inch	Inch	Inch	Volts	Volts	Ohms	Megohms	Inches
THFR-4	4(7)	0.115	0.185	0.247	0.883	3,000	8,000	2.50	400	11

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3.20 Type SDGA cable.

CONSTRUCTION

First. A standard conductor of bare uncoated copper (see 3.10).
Second. With or without separator (see 3.11.14) at manufacturer's option.
Third. A wall of felted asbestos (see 3.11.9) or glass fiber (see 3.11.10).
Fourth. A separator (see 3.11.14).
Fifth. A wall of varnished-cambric insulation (see 3.11.7), except that less than 3 thicknesses of varnished cambric shall be permitted, provided the minimum thickness at any point is not less than one-half of the calculated nominal thickness.
Sixth. With or without separator (see 3.11.14) at manufacturer's option.
Seventh. A wall of felted asbestos (see 3.11.9) or glass fiber (see 3.11.10). This wall may be omitted at manufacturer's option provided the thickness of the other components (asbestos wall "Third", varnished cambric "Fifth" and/or sheath "Ninth") is increased so as to permit strict adherence of the completed cable to the specified overall dimensions.
Eighth. With or without separator (see 3.11.14) at manufacturer's option.
Ninth. An impervious sheath (see 3.11.18).
Tenth. Braided metal armor (see 3.11.20).
Eleventh. Paint (see 3.11.21).

TEST REQUIREMENTS

Drip (see 4.8.8).
Cold bend (see 4.8.9).
Flammability (see 4.8.10).
Current overload (see 4.8.11).
Water and air leakage (see 4.8.13).
Electrical tests (see following table).
Watertightness (see Appendix I).

Type	Standard copper conductor	Diameters: Over first asbestos or glass	Diameters: Over varnished cambric	Diameters: Overall	Dielectric strength (see 4.8.19): After flame test: Minimum	Dielectric strength (see 4.8.19): Full reel: Minimum	Conductor resistance (d.c.) per 1000 feet at 25°C. (see 4.8.24): Bare: Maximum	Insulation resistance 1,000 feet (see 4.8.25): Minimum	Cold bend mandrel diameter (see 4.8.9)
		Inches	Inches	Inches	Volts	Volts	Ohms	Megohms	Inches
SDGA-400	400(127)	0.792	0.854	1.270	500	1,000	0.0273	25	20
SDGA-500	500(127)	.883	.945	1.360	500	1,000	.0217	20	20
SDGA-650	650(127)	.986	1.048	1.460	500	1,000	.0172	20	22
SDGA-800	800(127)	1.101	1.163	1.580	500	1,000	.0136	15	24
SDGA-1000	1000(127)	1.237	1.299	1.710	500	1,000	.0108	15	24
SDGA-1300	1300(127)	1.386	1.466	1.880	500	1,000	.00851	10	26
SDGA-1600	1600(127)	1.540	1.620	2.030	500	1,000	.00676	10	26
SDGA-2000	2000(127)	1.750	1.830	2.240	500	1,000	.00536	10	28

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3.21 Type MDGA cable.

CONSTRUCTION

First. A standard conductor of bare, uncoated copper (see 3.10).
Second. With or without separator (see 3.11.14) at manufacturer's option.
Third. A wall of felted asbestos (see 3.11.9) or glass fiber (see 3.11.10).
Fourth. A separator (see 3.11.14).
Fifth. A wall of varnished cambric insulation (see 3.11.7).
Sixth. With or without separator (see 3.11.14) at manufacturer's option.
Seventh. An inner braid (see 3.11.16) of rayon impregnated with a clear moisture and flame retardant compound.
Eighth. Nineteen conductors shall be cabled together with a lay not greater than 24 times the pitch diameter of the layer. Except for size 19(6), fillers (see 3.11.11) shall be used to obtain a firm, well rounded assembly.
Ninth. A suitable binder over the conductor assembly.
Tenth. An impervious sheath (see 3.11.18).
Eleventh. Braided metal armor (see 3.11.20).
Twelfth. Paint (see 3.11.21).

TEST REQUIREMENTS.

Drip (see 4.8.8).
Cold bend (see 4.8.9).
Flammability (see 4.8.10).
Current overload (see 4.8.11).
Water and air leakage (see 4.8.13).
Electrical tests (see following table).
Watertightness (see Appendix I).

Type	Standard copper conductor	Diameters: Over asbestos or glass	Diameters: Over varnished cambric	Diameters: Overall	Dielectric strength (see 4.8.19): After flame test, Minimum	Dielectric strength (see 4.8.19): Full reel, Minimum	Conductor resistance (d.c.) per 1000 feet at 25°C. (see 4.8.24): Bare, Maximum	Insulation resistance 1,000 feet (see 4.8.25): Minimum	Cold bend mandrel diameter (see 4.8.9)
		Inch	Inch	Inches	Volts	Volts	Ohms	Megohms	Inches
MDGA-19(6)	6(7)	0.122	0.170	1.285	500	1,000	1.78	50	16
MDGA-19(14)	14(7)	.166	.226	1.520	500	1,000	.300	50	19
MDGA-19(23)	23(7)	.211	.283	1.820	500	1,000	.505	50	21
MDGA-19(40)	40(19)	.266	.338	2.100	500	1,000	.294	50	25

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3.22 Type MDGL cable.

CONSTRUCTION

First. A standard conductor of bare, uncoated copper (see 3.10).
Second. With or without separator (see 3.11.14) at manufacturer's option.
Third. A wall of felted asbestos (see 3.11.9) or a glass fiber (see 3.11.10).
Fourth. A separator (see 3.11.14).
Fifth. A wall of varnished-cambric insulation (see 3.11.7).
Sixth. With or without separator (see 3.11.14) at manufacturer's option.
Seventh. An inner braid (see 3.11.16) of rayon impregnated with a clear moisture and flame retardant compound.
Eighth. Nineteen conductors shall be cabled together with a lay not greater than 24 times the pitch diameter of the layer. Fillers (see 3.11.11) shall be used in size 19(14) to obtain a firm, well rounded assembly.
Ninth. A suitable binder over the conductor assembly.
Tenth. A lead sheath (see 3.11.19).
Eleventh. An outer protective covering of duck tape, or a tape (see 3.11.15) and an outer cotton braid (see 3.11.16). The lead sheathed cable shall be coated with hot asphalt or tar compound immediately before the application of tape. Duck tape shall be cut from high grade 8-ounce duck which has been presaturated with asphalt or tar compound. The average thickness of the duck tape shall be not less than 0.020 inch. The cut edges of the duck tape shall not fray nor disclose any unsaturated fibers. Duck tape shall be applied helically with an overlap of not less than 25 percent of its width. A suitable asphalt or pitch compound shall be applied over the duck tape to provide a smooth, hard, dry surface. The cable shall be finished with a suitable material which will prevent sticking of adjacent turns on a reel.

TEST REQUIREMENTS

Drip (see 4.8.8).
Cold bend (see 4.8.9).
Electrical tests (see following table).
Watertightness (see Appendix I).

Type	Standard copper conductor	Diameters: Over asbestos or glass	Diameters: Over varnished cambric	Diameters: Overall	Dielectric strength (see 4.8.19)	Conductor resistance (d.c.) per 1,000 feet at 25°C. (see 4.8.24)	Insulation resistance 1,000 feet (see 4.8.25)	Cold bend mandrel diameter (see 4.8.9)
					Minimum	Bare: Maximum	Minimum	
		Inch	Inch	Inches	Volts	Ohms	Megohms	Inches
MDGL-19(6)	6(7)	0.122	0.170	1.285	1,000	1.78	50	16
MDGL-19(14)	14(7)	.166	.226	1.550	1,000	.800	50	18

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3.23 Type MHFA cable.

CONSTRUCTION

First. The standard conductor size 3(7) of bare, uncoated copper (see 3.10).
Second. A wall of synthetic resin insulation (see 3.11.2).
Third. A wall of felted asbestos (see 3.11.9) or glass fiber (see 3.11.10).
Fourth. An inner braid (see 3.11.16) of rayon impregnated with a clear moisture and flame retardant compound.
Fifth. The specified number of conductors shall be cabled together with a lay not greater than 16 times the pitch diameter of the layer. Fillers (see 3.11.11) shall be employed, where necessary, to obtain a firm, well rounded assembly.
Sixth. An approved binder over the conductor assembly.
Seventh. A wall of felted asbestos (see 3.11.9) or glass fiber (see 3.11.10).
Eighth. With or without separator (see 3.11.14) at manufacturer's option.
Ninth. An impervious sheath (see 3.11.18).
Tenth. Braided metal armor (see 3.11.20).
Eleventh. Paint (see 3.11.21).

TEST REQUIREMENTS

Cold bend (see 4.8.9).
Flammability (see 4.8.10).
Water and air leakage (see 4.8.13).
Electrical tests (see following table).
Watertightness (see Appendix I).

Type	Number of conductors	Diameters: Over synthetic resin, Minimum	Diameters: Over asbestos or glass	Diameters: Overall	Dielectric strength (see 4.8.19): After flame test, Minimum Volts	Dielectric strength (see 4.8.19): Full reel, Minimum Volts	Conductor resistance (d.c.) per 1000 feet at 25°C. (see 4.8.24): Bare, Maximum Ohms	Insulation resistance 1,000 feet (see 4.8.25): Minimum Megohms	Cold bend mandrel diameter (see 4.8.9): Individual conductors Inch	Cold bend mandrel diameter (see 4.8.9): Completed cable Inches
MHFA-7	7	0.092	0.145	0.859	1,000	2,000	4.03	100	0.50	11
MHFA-10	10	.092	.145	1.037	1,000	2,000		100		13
MHFA-14	14	.092	.145	1.118	1,000	2,000		100		15
MHFA-19	19	.092	.145	1.209	1,000	2,000		100		16
MHFA-24	24	.092	.145	1.375	1,000	2,000		100		17
MHFA-30	30	.092	.145	1.456	1,000	2,000		100		19
MHFA-37	37	.092	.145	1.569	1,000	2,000		100		20
MHFA-44	44	.092	.145	1.734	1,000	2,000		100		22

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3.24 Type TTHFWA cable.

CONSTRUCTION

First. The standard conductor size 3/5(7) of bare, uncoated copper (see 3.10).
Second. A wall of synthetic resin insulation (see 3.11.2).
Third. Two (see note) conductors shall be twisted together with a lay not more than 2-1/4 inches.
Fourth. The specified number of twisted pairs shall be cabled together with a lay not greater than 10 inches. Fibrous fillers (see 3.11.11) shall not be employed.
Fifth. An approved binder over the assembled pairs.
Sixth. A wall of felted asbestos (see 3.11.9) or glass fiber (see 3.11.10).
Seventh. With or without separator (see 3.11.14) at manufacturer's option.
Eighth. An impervious sheath (see 3.11.18)
Ninth. Braided metal armor (see 3.11.20).
Tenth. Paint (see 3.11.21).

TEST REQUIREMENTS

Cold bend (see 4.8.9).
Flammability (see 4.8.10).
Water and air leakage (see 4.8.13).
Electrical tests (see following table).
Watertightness (see Appendix I).

Type	Number of pairs	Diameters: Over insulated conductor, Minimum	Diameters: Overall	Dielectric strength (see 4.8.19): Between conductors, Minimum	Dielectric strength (see 4.8.19): Conductors to ground, Minimum	Conductor resistance (d.c.) per 1,000 feet at 25°C. (see 4.8.24), Bare, Maximum	Insulation resistance: 1,000 feet (see 4.8.25), Minimum	Cold bend mandrel diameter (see 4.8.9)
		Inch	Inches	Volts	Volts	Ohms	Megohms	Inches
TTHFWA-1-1/2	1-1/2	0.062	0.380	1,000	2,000	16.20	100	5
TTHFWA-3	3	.062	.500	1,000	2,000		100	7
TTHFWA-5	5	.062	.590	1,000	2,000		100	8
TTHFWA-10	10	.062	.690	1,000	2,000		100	9
TTHFWA-15	15	.062	.800	1,000	2,000		100	10
TTHFWA-20	20	.062	.880	1,000	2,000		100	11
TTHFWA-30	30	.062	1.030	1,000	2,000		100	13
TTHFWA-40	40	.062	1.130	1,000	2,000		100	15
TTHFWA-50	50	.062	1.265	1,000	2,000		100	16
TTHFWA-60	60	.062	1.350	1,000	2,000		100	17

NOTE: Type TTHFWA-1-1/2 comprises 1-1/2 pairs, or three, instead of two, conductors twisted together.

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3.25 Types PBJX and PBTX cable.

CONSTRUCTION

First. One conductor of iron and one conductor of constantan in type PBJX, one conductor of copper and one conductor of constantan in type PBTX, each conductor a solid wire 0.064-inch in diameter. (Type letter designations for other conductor metals have not been adopted.)

Second. A wall of synthetic resin insulation (see 3.11.2). Circuit identification as specified under "Fourth".

Third. A braid (see 3.11.16) of glass fiber filled with a flexible, hard, moisture and flame retardant compound which shall not ooze out or drip off at a temperature of 95°C. Circuit identification as specified under "Fourth".

Fourth. The two conductors shall be twisted together with a lay not greater than 16 times the pitch diameter. When iron and constantan are used, color coding (see 3.9.2 or 3.9.5) shall be employed, except the color over the iron shall be gray; the constantan, red. In the case of other metals, chromel shall be yellow, copper shall be blue, noble metal shall be black, for positive conductors; the negative conductor of any combination shall be red. Fillers (see 3.11.11) shall be used to obtain a firm, well rounded assembly.

Fifth. An approved binder over the conductor assembly.

Sixth. An impervious sheath (see 3.11.18).

Seventh. Braided metal armor (see 3.11.20).

Eighth. Paint (see 3.11.21).

TEST REQUIREMENTS

Cold bend (see 4.8.9).
Flammability (see 4.8.10).
Electrical tests (see following table).
Watertightness (see Appendix I).

Type	Diameters			Dielectric strength (see 4.8.19)	Insulation resistance 1,000 ft. (see 4.8.25)	Cold bend mandrel diameter (see 4.8.9)
	Over synthetic resin	Overall				
	Minimum	Minimum	Maximum	Minimum	Minimum	
	Inch	Inch	Inch	Volts	Megohms	Inches
PBJX-4	0.095	0.450	0.480	500	100	6
PBTX-4	.095	.450	.480	500	100	6

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3.26 Types DSS, TSS, FSS and MSS.

CONSTRUCTION

First - A standard copper conductor (see 3.10) with metallic coating (see 3.10.3) and watertight compounds (see 3.6).
Second - A wall of synthetic insulation (see 3.11.3 or 3.11.5).
Third - (a) Types DSS, TSS and FSS shall have 2, 3, and 4 conductors, respectively, with plain colored insulation (see 3.9.2).
(b) Type MSS-6 shall have six insulated conductors comprising two shielded pairs and two singles. Each shielded pair shall comprise two insulated conductors twisted together with a braided shield of 0.0063 inch diameter tinned copper wire over the pair (see 3.11.16,6). The colors shall be black and white in one pair, blue and red in the other pair, one yellow single and one green single obtained by colored insulation (see 3.9.2), thread wind (see 3.9.3), or braid (see 3.9.5) at manufacturer's option.
Fourth - The required number of conductors and shielded pairs shall be twisted together with a lay not exceeding that specified in the following table. Fillers of soft synthetic compound shall be employed with a minimum amount of fibrous material to obtain a firm, well-rounded cross section.
Fifth - An approved binder over the conductor assembly.
Sixth - A close braid of 0.0063-inch tinned copper wire providing at least 88 percent coverage on types DSS, TSS and FSS.
Seventh - A polychloroprene sheath (see 3.11.17) having a minimum thickness (see 3.7.3) of 0.050-inch. The assembled components shall be molded together compactly to enable the completed cable to meet the hydrostatic test. The individual conductors shall be readily separable from fillers and binder without damage to the insulation.

TEST REQUIREMENTS

Electrical tests (see following table).
Flammability (see 4.8.10).
Flexing endurance (see 4.8.14), except 1000 cycles of bending only on 4-inch diameter mandrel at room temperature.
Hydrostatic test (see 4.8.31 and Appendix I): permissible leakage - 2 cubic inches without price reduction.

Type	Standard copper conductor	No of cdr	Insulation thickness (Min)	Lay of twist (Max)	Overall diameter (Min)	Overall diameter (Max)	Dielectric strength: Cdr to cdr (Min)	Dielectric strength: Shield to grd (Min)	Copper resist. (dc) per 1000 ft. (Max)	Insulation resist. 1000 ft. (Min)
			inch	Inches	Inch	Inch	Volts	Volts	Ohms	Megohms
DSS-2	2(7)	2	0.025	2.0	0.375	0.390	2000	1500	6.64	250
DSS-3	3(7)	2	.031	2.5	.425	.445	2000	1500	4.15	250
TSS-4	4(7)	3	.031	2.5	.480	.500	2000	1500	2.57	200
FSS-2	2(7)	4	.031	2.5	.450	.465	2000	1500	6.64	250
MSS-6	1(7)	6	.015	3.0	.470	.490	1000	1500	10.5	150

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3.27 Type TSP cable.

CONSTRUCTION

First. A standard conductor of bare, uncoated, copper, size 3/5(7) (see 3.6 and 3.10).
Second. A wall of synthetic resin insulation (see 3.11.2).
Third. Two conductors twisted together with a maximum lay of 2-1/2 inches to form a twisted pair with color coded insulation (see 3.9.7.2).
Fourth. Eleven twisted pairs twisted together with a maximum lay of 4 inches in the outer layer. Watertight sealing compounds but no fibrous fillers shall be employed.
Fifth. A suitable binder and/or separator may be applied over the assembled pairs.
Sixth. An impervious sheath (see 3.11.18) of the thermoplastic type.

LENGTH REQUIREMENT

Shipping length (see 5.1.1.1) - 500 foot coils.

TEST REQUIREMENTS

Flammability (see 4.8.10) - same as specified for type TTOP.
Electrical tests (see following tabulation).
Watertightness (see Appendix I).

Type	Number of		Thickness of		Overall	Dielectric strength	Insulation resistance	Conductor resistance
	Conductors	Pairs	Insulation	Sheath	diameter	conductor to ground	conductor to ground	per 1000 ft. at 25°C.
			Min.	Min.	Max.	Min.	Min.	Max.
			Inch	Inch	Inch	Volts	Megohms	Ohms.
TSP - 11	22	11	0.015	0.080	0.735	2,500	100	16.6

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3.28 Type MDGB cable.

CONSTRUCTION

First. A standard conductor of bare, uncoated copper (see 3.10).
Second. A wall of synthetic resin insulation (see 3.11.2). Color coded (see 3.9.2).
Third. The twelve insulated conductors shall be cabled together with a lay.[1] Fibrous fillers shall not be used.
Fourth. With or without a binder at manufacturer's option (see 3.11.15).
Fifth. A gray impervious sheath (see 3.11.18).

[1]Not exceeding 16 times the pitch diameter of the layer.

TEST REQUIREMENTS

Flammability (see 4.8.10).
Cold bend (see 4.8.9).
Electrical tests (see following table).
Watertightness (see Appendix I).

Type	Standard copper conductor	Thickness		Overall diameter	Dielectric strength (see 4.8.19)	Conductor resistance (d.c.) per 1000 feet at 25°C. (see 4.8.24)	Insulation resistance 1000 feet (see 4.8.25)	Cold bend mandrel diameter (see 4.8.9)
		Insulation	Sheath			Bare		
		Min.	Min.		Min.	Max.	Min.	
		Inch	Inch	Inch	Volts	Ohms	Megohms	Inches
MDGB-12(1-1/2)	2(7)	0.015	0.062	0.590	1,000	6.37	100	7

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3.29 Type TTRSA cable.

CONSTRUCTION

First.	A standard conductor of bare, uncoated copper (see 3.10) size 1(7). With or without an approved separator (see 3.11.14), at manufacturer's option. Circuit identification by means of not less than 4 colored threads among the strands of the conductor, as satisfactory to the bureau or agency concerned, unless accomplished under "Second". [1]
Second.	A wall of polythene (see 3.11.4), to the diameter and within the dimensional tolerances required to obtain the specified degree of performance as limited by the maximum overall diameter of the finished cable. Circuit identification (see 3.9.2), unless accomplished under "first". [1]
Third.	Two such insulated conductors shall be combined to form a pair. They shall be twisted together with any lay necessary to obtain the specified mechanical and electrical characteristics. One conductor shall be color-coded white, and the other shall be color-coded black. [1]
Fourth.	A braided shield of either 0.005 or 0.0063 inch diameter copper wire (see 3.11.16.6).
Fifth.	An inner braid (see 3.11.16.2) of cotton (see 3.11.12) or rayon (see 3.11.13). The braid shall be lightly impregnated with a synthetic resin material to afford satisfactory abrasion resistance, and shall be color-coded (see 3.9.5) for identification of each shielded pair.
Sixth.	The number of shielded pairs indicated in the following table shall be cabled together with a lay not to exceed 14 inches. Impregnated asbestos or glass fiber fillers (see 3.11.11) shall be used if necessary to obtain circularity of cross section. The grouped pairs shall be bound together by a thin single faced friction tape, cellophane, or other material, satisfactory to the bureau or agency concerned for the purpose. Manufacturer's identification (see 3.8.3), if transparent binder tape is used, unless accomplished under "Seventh".
Seventh.	A wall of felted asbestos (see 3.11.9) or glass fiber (see 3.11.10) compressed and impregnated with a neutral insulating cement to form a continuous and homogeneous layer. The thickness of the felted layer shall be such as to permit adherence of the finished cable to the specified overall dimensions. Manufacturer's identification tape (see 3.8.2) unless accomplished under "Sixth".
Eighth.	An impervious sheath (see 3.11.18).
Ninth.	Braided metal armor (see 3.11.20).
Tenth.	Paint (see 3.11.21).

[1] In order to facilitate installation, it is desired that color coding of individually insulated conductors be obtained wherever practicable by means of colored insulating compounds (see 3.9.2). If color insulation is used, it is desired that one conductor of each pair be colored white (natural) and the other conductor distinctive, preferably dark (green, blue), color if a black insulation cannot be obtained.

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TEST REQUIREMENTS

Characteristic impedance (surge impedance) of a pair (see 4.8.27). - 71.5 ohms (min.), 85.0 ohms (max.) as near 76 ohms as practicable measured at 25°C. and 1 megacycle per second.
Capacitance of a pair (see 4.8.27.3). - 25.0 (max.) micromicrofarads per foot at 25°C. at 1 megacycle per second.
Dielectric strength (see 4.8.19). - 2000 volts applied for 1 minute for white (natural) conductor of each pair to other conductor and shield, and for both conductors of each pair to shield, shield being grounded.
Cold bend (see 4.8.9 and the following table).
Flammability (see 4.8.10).
Water and air leakage (see 4.8.13).
Conductor resistance (d.c.) per 1000 feet at 25°C., (see 4.8.24) - 10.3 ohms maximum.
Insulation resistance - 1000 megohms per 1000 feet (see 4.8.25).

Type	Number of shielded pairs	Diameter over metal armoring	Cold bend mandrel diameter (see 4.8.9)
TTRSA-2	2	0.740	9.60
TTRSA-4	4	.800	10.00
TTRSA-6	6	.940	12.00
TTRSA-8	8	1.050	13.00
TTRSA-10	10	1.140	15.00
TTRSA-12	12	1.160	15.00
TTRSA-16	16	1.250	16.00

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3.30 Type MCSC cable.

CONSTRUCTION

First. A flexible strength member employed as a core, comprising a 5/32-inch nominal diameter steel cable rolled and compacted to a nominal diameter of 0.140 inch, insulated with a 0.031-inch minimum wall (see 3.7.3) of synthetic rubber (see 3.11.3) to a nominal overall diameter of 0.215 inch. The steel stranding shall conform to class B of Specification 22C4.

Second. A standard conductor of bare, uncoated copper (see 3.10), size 1-1/2(41), with a separator (see 3.11.14). A wall of synthetic rubber insulation (see 3.11.3) colored for circuit identification (see 3.9.2).

Third. Suitable soft synthetic rubber fillers having a nominal diameter of 0.095 inch.

Fourth. Five insulated conductors and five fillers, alternating in position so that no two conductors or fillers are adjacent, shall be cabled helically about the insulated strength member with a lay not greater than 3 inches.

Fifth. A polychloroprene sheath (see 3.11.17) over the assembly specified in "Fourth" the assembled components so molded together that the completed cable shall be one compact mass from which the individual conductors shall be readily separable without damage to the insulation. The minimum sheath thickness shall be not less than 0.085 inch. Manufacturer's identification (see 3.8) shall include marking (see 3.8.4) at intervals not greater than 25 feet, showing the cable type letter designation, the manufacturer's name or trade mark and the year of manufacture.

LENGTH REQUIREMENT

Type MCSC cable shall be delivered in nominal lengths of not less than 1,800 feet and not more than 1,850 feet. Not more than 30 percent of the total footage of each lot offered for inspection may be accepted in random lengths of not less than 600 feet and not more than 630 feet. Lengths between 630 and 1,800 feet, and lengths of less than 600 feet, are not suitable for the purpose intended.

TEST REQUIREMENTS

Flammability (see 4.8.10).
Water air leakage (see 4.8.13), except that the sheath shall be tested on the completed cable at 550 pounds per square inch water pressure.
Flexing endurance (see 4.8.14), except 1,000 cycles of bending only on a 5-inch diameter mandrel at room temperature.
Breaking strength (see 4.8.28): 2,500 pounds minimum.
Electrical tests (see following table).

Type	Number of conductors	Insulation			Dielectric strength conductors to strength member and ground (see 4.8.19)	Conductor resistance (d.c.) per 1,000 feet at 25°C. (see 4.8.24)	Insulation resistance 1,000 feet to strength member and ground (see 4.8.25)
		Thickness	Diameter	Overall diameter			
		Minimum	Minimum		Minimum	Maximum	Minimum
		Inch	Inch	Inch	Volts	Ohms	Megohms
MCSC-5	5	0.015	0.082	0.595	2,000	7.30	200

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3.31 Types DBSP, TBSP and FBSP cable.

CONSTRUCTION

First. A standard conductor of bare, uncoated copper (see 3.10), a separator (see 3.11.14) shall be used when synthetic insulation (see 3.11.3, 3.11.5, or 3.11.6) is employed.

Second. A wall of extruded synthetic insulation.

Third. The specified number of conductors shall be twisted together with a lay not greater than 20 times the pitch diameter. Fillers may be employed only where necessary to obtain a smooth, uniform overall diameter on the completed cable.

Fourth. A thin copper tape not more than 0.003 inch thick, applied helically over the conductor assembly with a positive lap.

Fifth. A gray impervious sheath.

TEST REQUIREMENTS

Electrical tests (see following table).

Flammability (see 4.8.10).

Heat shock (see 4.8.30) - No cracking of insulation or sheath.

Accelerated aging. - Specimens of the completed cable shall be aged 20 hours at 126.7°C. in the air pressure heat bomb (see 4.8.1.3) and, after completion of subsequent bending and dielectric tests, shall exhibit no cracks in the insulation or sheath as determined by dissection. After cooling and bending in U-shape on a radius not less than 3 times the diameter of the specimen, the specimen shall withstand the dielectric strength tests shown in the following table and the sheath compound shall have an elongation of not less than 200 percent.

Hot oil. - After subjection to the hot oil test (see 4.8.17), the elongation of the sheath compound shall be not less than 200 percent.

Cold bend (see 4.8.9) - No cracking of sheath when bent over a mandrel having a diameter twice that of the cable specimen.

Size	Standard copper conductor	Insulation thickness, Minimum, Inch	Sheath thickness, Minimum, Inch	Maximum over-all diameter, Type DBSP, Inch	Maximum over-all diameter, Type TBSP, Inch	Maximum over-all diameter, Type FBSP, Inch	Dielectric strength (see 4.8.18), Shield to ground, Minimum, Volts	Dielectric strength (see 4.8.18), Conductors to shield, Minimum, Volts	Conductor resistance (d.c.) 1,000 feet at 25°C. (see 4.8.24), Bare, Maximum, Ohms	Conductor resistance (d.c.) 1,000 feet at 25°C. (see 4.8.24), Tinned, Maximum, Ohms	Insulation resistance 1,000 feet (see 4.8.25), Minimum, Megohms
3/5	3/5(7)	0.015	0.030	0.206	0.216	0.232	500	1,000	16.0	16.7	100
1	1(7)	.015	.030	.224	.234	.253	500	1,000	10.1	10.5	100
2	2(7)	.015	.030	.245	.257	.278	500	1,000	6.37	6.64	100
3	3(7)	.020	.035	.303	.319	.346	500	1,500	4.00	4.15	100
4	4(7)	.025	.035	.359	.378	.413	500	1,500	2.51	2.57	100
6	6(7)	.030	.040	.426	.450	---	500	1,500	1.72	1.76	100
9	9(7)	.030	.045	.471	.498	---	500	1,500	1.25	1.27	100
14	14(7)	.035	.045	.552	.586	---	500	2,000	.785	.800	100
23	23(7)	.035	.050	.639	.677	---	500	2,000	.495	.503	100

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3.32 Type FCSF cable.

CONSTRUCTION

First - Standard conductors of bare, uncoated copper (see 3.10 and following tabulation).
Second - A separator (see 3.11.14).
Third - A wall of butyl rubber (see 3.11.5).
Fourth - A compound filled tape (see 3.11.15) or an inner braid (see 3.11.16.2) on each conductor is permitted but not required. Color coding, one black and one white on each conductor size is required.
Fifth - The four conductors cabled together with fillers of synthetic insulating compound or polychloroprene to obtain a firm, well rounded solid core, with a minimum amount of fibrous material. Outer fillers may be integral with the sheath compound or may be laid in separately.
Sixth - A suitable binder over the conductor assembly is permitted but not required.
Seventh - A polychloroprene sheath (see 3.11.17). Fillers and sheath material in the completed cable shall be readily separable from, and without damage to, the conductor insulation. Reinforcement (see 3.11.12) shall be applied under the sheath, or under at least half the sheath thickness.

LENGTH REQUIREMENT

Shipping lengths (feet) - FCSF-66 - 1590 min., 1650 max.
FCSF-220 - 1590 min., 1650 max.

TEST REQUIREMENTS

Flammability (see 4.8.10).
Water and air leakage (see 4.8.13).
Cold bend (see 4.8.9). No cracking of sheath when bent two turns around mandrel with diameter twice that of specimen at -30°C.
Electrical tests (see following tabulation).

Type	Standard copper conductor	Number of conductors	Insulation thickness	Lay of conductors	Overall diameter		Dielectric strength (see 4.8.19)	Conductor resistance (d.c.) per 1000 ft. at 25°C. (see 4.8.24)	Insulation resistance per 1000 ft. (see 4.8.25)
		Total	Minimum	Maximum	Minimum	Maximum	Minimum	Maximum	Minimum
			Inch	Inches	Inches	Inches	Volts	Ohms	Megohms
FCSF-66	2 of 20(49) and 2 of 66(133)	4	0.062 .062	12	1.300	1.370	2500	0.573 .174	300
FCSF-220	2 of 20(49) and 2 of 220(259)	4	.062 .078	20	1.920	2.000	3500	.373 .0504	300

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3.33 Type MCGC cable.

CONSTRUCTION

First. A standard conductor of bare, uncoated copper (see 3.10), size 1-1/2(16), with separator (see 3.11.14).

Second. A wall of synthetic rubber (see 3.11.3) having a minimum wall thickness (see 3.7.3) of 0.031 inch.

Third. An inner braid (see 3.11.16).

Fourth. Six conductors shall be twisted with a maximum lay of 3 inches around a central core of continuous filament glass fibers (see 3.11.11.2.3). Fibrous fillers shall not be used in the outer interstices and no binder shall be applied over the assembly.

Fifth. A polychloroprene sheath (see 3.11.17) to an overall diameter of not less than 0.565 inch nor more than 0.600 inch. The sheath shall be extruded with a tight fit so that the sheath material fills all interstices over the conductor assembly. The sheath shall be readily removable from the completed cable without damage to the insulation.

LENGTH REQUIREMENT

Shipping lengths shall be not less than 1000 nor more than 1050 feet.

TEST REQUIREMENTS

Water and air leakage (see 4.8.13).
Flexing endurance (see 4.8.14).
1000 twisting and bending cycles at +75°C.
500 twisting and bending cycles at -30°C.
Electrical tests.
Dielectric strength (see 4.8.19) - 2000 volts.
Insulation resistance (see 4.8.25) - 300 megohms per length.
Conductor resistance (d.c.) per 1000 feet at 25°C. (see 4.8.24) - 7.22 ohms maximum.

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3.34 Type SDU cable.

CONSTRUCTION

First. A standard conductor of bare, uncoated copper (see 3.10), except that all of the 36 strands comprising the outer layer of the stranded conductor shall be metallically coated (see 3.10.3). All interstices of the conductor shall be filled with watertight sealing compound (see 3.6).

Second. A wall of butyl rubber insulation (see 3.11.5).

Third. With or without a nonmetallic reinforcement at manufacturer's option. When reinforcement is employed, it may be applied under or over the insulation but not over the sheath.

Fourth. A polychloroprene sheath (see 3.11.17). Manufacturer's identification shall consist of a molded marking (see 3.8.4) at intervals not greater than 25 feet, showing the cable type letter designation, the manufacturer's name or trade-mark, and the year of manufacture.

TEST REQUIREMENTS

Flammability (see 4.8.10).
Impact resistance (see 4.8.15).
Abrasion resistance (see 4.8.16).
Electrical tests (see following table).
Hydrostatic test (see 4.8.31 and Appendix I): Permissible leakage - 1 cubic inch without price reduction:

Type and size	Standard copper conductor	Insulation thickness	Sheath thickness	Overall diameter	Dielectric strength (see 4.8.19)	Conductor resistance (d.c.) per 1000 feet at 25°C. (see 4.8.24)	Insulation resistance per 1000 feet at 15.6°C. (see 4.8.25)
		Minimum	Minimum	Maximum	Minimum	Maximum	Minimum
		Inch	Inch	Inches	Volts	Ohms	Megohms
SDU-500	500(127)	0.050	0.062	1.210	2000	0.0216	2000
SDU-800	800(127)	.066	.062	1.465	2000	.0136	2000

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3.35 Types DCOP and TCOP cord.

CONSTRUCTION

First - A standard conductor of bare, uncoated copper (see 3.10).
Second - (a) A separator (see 3.11.14) when synthetic rubber insulation (see 3.11.3) is used, or
(b) with or without separator (see 3.11.14), at manufacturer's option when synthetic resin insulation (see 3.11.2) is used.
Third - A wall of synthetic insulation (see 3.11.2 or 3.11.3).
Fourth - (a) A thread wind (see 3.9) will be permitted over the insulation on size DCOP-1.
(b) Plain colored insulation (see 3.9) is required on size DCOP - 1-1/2.[1]
(c) An inner braid (see 3.11.16) will be permitted on sizes DCOP-2 and TCOP-2.
Fifth - The specified number of conductors shall be twisted together with either a right-hand or left-hand lay, with the lay not greater than that specified in the following table. Cotton tie cord (see 3.11.12) or approved alternate simulating a third or dummy conductor shall be laid straight as a central core in size DCOP - 1-1/2. Fillers (see 3.11.11) shall be employed where necessary to obtain a firm, well rounded assembly.
Sixth - (a) A braid of minimum No. 60/2 ply cotton or approved alternate shall be used over the conductor assembly of size DCOP-1 and is permitted but not required on size DCOP - 1-1/2.
(b) An approved binder shall be used over the conductor assembly of sizes DCOP-2 and TCOP-2.
Seventh - A sheath of polychloroprene (see 3.11.17) or an impervious sheath (see 3.11.18).

Because of peculiar lay-up of type DCOP - 1-1/2 cable, the centering and circularity requirements (see 3.7.4) shall not apply, providing the minimum thickness requirements (see 3.7.3) are met.

TEST REQUIREMENTS

Water and air leakage (see 4.8.13).
Flexing endurance (see 4.8.14) 1000 twisting and bending cycles at +75°C., 500 twisting and bending cycles at -30°C.
Impact resistance (see 4.8.15).
Abrasion resistance (see 4.8.16).
Electrical tests (see following table).

Type	Standard copper conductor	Insulation thickness	Lay of conductors	Overall diameter		Dielectric strength (see 4.8.19)	Conductor resistance (d.c.) per 1000 feet at 25°C. (see 4.8.24)	Insulation resistance 1000 feet (see 4.8.25)
		Min.	Max.	Min.	Max.	Min.	Max.	Min.
		Inch	Inches	Inch	Inch	Volts	Ohms	Megohms
DCOP-1	1(10)	0.015	1	0.240	0.250	1000	11.60	100
DCOP-1-1/2	1-1/2(16)	.015	3/8	.300	.310	1500	13.10	100
DCOP-2	1-1/2(16)	.015	2	.315	.330	1500	7.22	100
TCOP-2	1-1/2(16)	.015	2	.330	.345	1500	7.22	100

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3.36 Type MHFF cable.

CONSTRUCTION

First. The standard conductor size 2-1/2(26) of bare, uncoated copper (see 3.10).

Second. A wall of synthetic resin insulation (see 3.11.2) to a diameter of 0.092 inch minimum.

Third. A wall of felted asbestos (see 3.11.9) or glass fiber (see 3.11.10) to a diameter of 0.125 maximum.

Fourth. An inner braid (see 3.11.16) of rayon or glass fiber yarn treated suitably with a flame-retardant compound to prevent fraying and to provide adequate resistance to abrasion due to flexing of the cable. Diameter shall be not less than 0.130 inch nor more than 0.140 inch.

Fifth. The specified number of conductors shall be cabled together in one or more layers, all in a left-hand direction, with a minimum amount of asbestos or glass fiber fillers (see 3.11.11) in the interstices between conductors. The length of lay of the first and successive layers shall be a suitable function of the pitch diameter of that layer, designed to meet the flexing endurance test on completed cable. The conductors may be laid around a central core of asbestos or glass fiber (see 3.11.11). A single conductor shall not be used as a center for the cable core.

Sixth. An approved binder over the conductor assembly.

Seventh. An impervious sheath (see 3.11.18) of the thermoplastic type. The minimum sheath thickness shall be not less than 0.060-inch.

TEST REQUIREMENTS

Cold bend (see 4.8.9).
Flammability (see 4.8.10).
Water and air leakage (see 4.8.13).
Flexing endurance (see 4.8.14):
1000 twisting and bending cycles at +75°C.
500 twisting and bending cycles at -30°C.
Impact resistance (see 4.8.15).
Abrasion resistance (see 4.8.16).
Electrical tests (see following table).

Type	Number of conductors	Over-all diameter	Dielectric strength (see 4.8.19) After flame test	Dielectric strength (see 4.8.19) Full reel	Conductor resistance (d.c.) per 1000 feet at 25°C. (see 4.8.24)	Insulation resistance 1,000 feet (see 4.8.25)	Cold bend mandrel diameter (see 4.8.9) Individual conductors	Cold bend mandrel diameter (see 4.8.9) Completed cable
			Minimum	Minimum	Bare Maximum	Minimum		
		Inches	Volts	Volts	Ohms	Megohms	Inch	Inches
MHFF-2	2	0.460	750	2,000		100		6.00
MHFF-4	4	.520	750	2,000		100		6.70
MHFF-7	7	.627	750	2,000		100		8.00
MHFF-10	10	.795	750	2,000		100		10.00
MHFF-14	14	.844	750	2,000	4.37	100	0.50	11.00
MHFF-19	19	.995	750	2,000		100		13.00
MHFF-24	24	1.120	750	2,000		100		14.00
MHFF-30	30	1.194	750	2,000		100		15.00
MHFF-37	37	1.290	750	2,000		100		16.00
MHFF-44	44	1.420	750	2,000		100		18.00

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3.37 Type TTOP cable.

CONSTRUCTION

First. A standard conductor of bare, uncoated copper (see 3.10), size 1(10).

Second. A wall of synthetic resin insulation (see 3.11.2). Color coding (see 3.9.2) unless accomplished under "Third".

Third. With or without an inner braid (see 3.9.5), at manufacturer's option suitably treated to prevent fraying and to provide adequate resistance to abrasion due to flexing of the cable. Color coding of this covering is required unless accomplished under "Second".

Fourth. Two conductors shall be twisted together with a suitable lay to form a twisted pair.

Fifth. The specified number of twisted pairs shall be cabled together in one or more layers all in a left-hand direction. The length of lay of the first and successive layers shall be a suitable function of the pitch diameter of that layer, designed to meet the flexing endurance test on completed cable. Circularity to cross section shall be obtained by the use of a minimum amount of asbestos or glass fiber fillers (see 3.11.11) lightly impregnated. A Single pair shall not be used as a center for the cable core.

Sixth. An approved binder over the assembled pairs.

Seventh. An impervious sheath (see 3.11.18) of the thermoplastic type. The minimum sheath thickness shall be not less than 0.060 inch.

TEST REQUIREMENTS

Flammability (see 4.8.10).
Water and air leakage (see 4.8.13).
Flexing endurance (see 4.8.14):
1000 twisting and bending cycles at +75°C.
500 twisting and bending cycles at -30°C.

Impact resistance (see 4.8.15).
Abrasion resistance (see 4.8.16).
Electrical tests (see following table).

Type	Number of pairs	Over-all diameter	Dielectric strength (see 4.8.19)		Conductor resistance (d.c.) per 1000 feet at 25°C. (see 4.8.24)	Insulation resistance 1000 feet (see 4.8.25)
			Between conductors	Conductors to ground	Bare	
			Minimum	Minimum	Maximum	Minimum
		Inch	Volts	Volts		Megohms
TTOP-3	3	0.480	1,000	2,000	11.4	100
TTOP-5	5	.590	1,000	2,000		100
TTOP-10	10	.700	1,000	2,000		100
TTOP-15	15	.830	1,000	2,000		100

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3.38 Type MCOS cable.

CONSTRUCTION

First. A standard conductor of bare, uncoated copper (see 3.10).

Second. A wall of synthetic resin insulation (see 3.11.2).

Third. (a) Sizes 2, 4 and 7 shall have 2, 4 and 7 conductors, respectively, with plain colored insulation (see 3.9.2).

(b) Size 5 shall have five insulated conductors comprising one shielded pair and three singles. The shielded pair shall comprise two insulated conductors twisted together with a braided shield of either 0.005 or 0.0063 inch diameter copper wire (see 3.11.16.6) over the pair. The colors shall be black and white in the pair; singles shall be one red, one green, and one blue; obtained by colored insulation (see 3.9.2).

(c) Size 6 shall have six insulated conductors comprising two shielded pairs and two singles. Each shielded pair shall comprise two insulated conductors twisted together with a braided shield of either 0.005 or 0.0063-inch diameter copper wire (see 3.11.16.6) over the pair. The colors shall be black and white in one pair; blue and red in the other pair; one yellow single, and one green single; obtained by colored insulation (see 3.9.2), thread wind (see 3.9.3) or braid (see 3.9.5) at manufacturer's option.

Fourth. The required single conductors and shielded pairs shall be cabled together all in a left-hand direction with a minimum amount of filler (see 3.11.11) in the interstices. The length of lay shall be a suitable function of the pitch diameter, designed to meet the flexing endurance (see 4.8.14) on completed cable and, in addition for sizes 5 and 6 only, the electrical requirements of 4.8.29. The conductors of size 7 shall be laid around a central core of cotton, asbestos, or glass fiber (see 3.11.11). A single conductor shall not be used as a center for the cable core.

Fifth. (a) A braided shield of either 0.005 or 0.0063-inch diameter copper wire (see 3.11.16.6) plus a separator (see 3.11.14) over the assembly of sizes 2, 4 and 7. At the manufacturer's option a separator (see 3.11.14) or a tape (see 3.11.15) may be applied under the copper braid.

(b) An approved binder over the assembly of sizes 5 and 6.

Sixth. (a) For sizes 2, 4, 6 and 7, an impervious sheath (see 3.11.18) of the thermoplastic type, having a minimum thickness of 0.060 inch.

(b) For size 5, a polychloroprene sheath (see 3.11.17) or an impervious sheath (see 3.11.18), having a minimum thickness of 0.030 inch.

TEST REQUIREMENTS

Flammability (see 4.8.10).
Water and air leakage (see 4.8.13).
Flexing endurance (see 4.8.14):
1000 twisting and bending cycles at 75°C.
500 twisting and bending cycles at -30°C.

Impact resistance (see 4.8.15).
Abrasion resistance (see 4.8.16).
Induced voltage (see 4.8.29): Five microvolts.
Electrical tests (see following table).

Type	Number of conductors	Standard copper conductors	Insulation thickness	Overall diameter		Dielectric strength (see 4.8.19)		Insulation resistance, 1,000 feet (see 4.8.25)	Conductor resistance (d.c.) per 1,000 feet at 25°C. (see 4.8.24)
						Between Conductors	Shield to ground		
			Minimum	Minimum	Maximum	Minimum	Minimum	Minimum	Bare Maximum
			Inch	Inch	Inch	Volts	Volts	Megohms	Ohms
MCOS-2	2	1-1/2(16)	0.025	0.440	0.460	2,000	1,500	200	7.08
MCOS-4	4	1-1/2(16)	.025	.490	.510	2,000	1,500	200	7.08
MCOS-5	5	1(26)	.015	.375	.390	1,000	500	100	10.90
MCOS-6	6	1(10)	.015	.450	.465	1,000	1,500	100	11.40
MCOS-7	7	1-1/2(16)	.025	.575	.595	2,000	1,500	200	7.08

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3.39 Type MMOP cable.

CONSTRUCTION

First. A standard conductor of bare, uncoated copper (see 3.10).

Second. A wall of synthetic resin insulation (see 3.11.2). Color coding (see 3.9.2).

Third. Five such color-coded conductors shall be twisted together with a maximum lay not greater than 2 inches. Glass fibers tie cord (see 3.11.10) shall be cabled with the 5 conductors laid straight as a central core. Fillers (see 3.11.11) shall be employed where necessary to obtain a firm, well rounded assembly.

Fourth. A braid of a minimum No. 60/2 ply cotton or approved alternate over the conductor assembly.

Fifth. A sheath of polychloroprene (see 3.11.17) or an impervious sheath (see 3.11.18).

TEST REQUIREMENTS

Flammability (see 4.8.10).
Water and air leakage (see 4.8.13).
Flexing endurance (see 4.8.14).
1000 twisting and bending cycles at +75°C.
500 twisting and bending cycles at -30°C.
Impact resistance (see 4.8.15).
Abrasion resistance (see 4.8.16).
Electrical tests (see following table).

Type	Standard copper conductor	Number of conductors	Diameters		Dielectric strength (see 4.8.19)	Conductor resistance (d.c.) per 1000 feet at 25°C. (see 4.8.24)	Insulation resistance 1,000 feet (see 4.8.25)
			Over insulated conductor	Overall		Bare	
			Minimum	Maximum	Minimum	Maximum	Minimum
			Inch	Inch	Volts	Ohms	Megohms
MMOP-5	1/2(21)	5	0.065	0.305	1000	21.7	100

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3.40 Types SHOF, DHOF, THOF and FHOF cable.

CONSTRUCTION

First. A standard conductor of bare, uncoated copper (see 3.10).

Second. A separator (see 3.11.14).

Third. A wall of butyl rubber insulation (see 3.11.5).

Fourth. (a) An inner braid (see 3.11.16) will be permitted on sizes 3 and larger.
(b) A tape (see 3.11.15) or an inner braid (see 3.11.16) will be permitted on sizes 23 and larger.

Fifth. (a) Reinforcement (see 3.11.12) on type SHOF 23 and larger only.
(b) The required number of conductors twisted together with a left-hand lay not greater than specified in the following table. Fillers (see 3.11.11) shall be employed to obtain a firm, well rounded assembly. An approved binder over the conductor assembly.

Sixth. A polychloroprene sheath (see 3.11.17) or impervious sheath (see 3.11.18) on sizes 42 and smaller. A polychloroprene sheath (see 3.11.17) on sizes larger than 42.

TEST REQUIREMENTS

Flammability (see 4.8.10).
Cold bend (see 4.8.9) (SHOF only): No cracking of sheath when bent two turns around mandrel with diameter twice that of specimen at -30°C.
Water and air leakage (see 4.8.13).
Flexing endurance (see 4.8.14) (types DHOF, THOF and FHOF only):
1000 twisting and bending cycles at +75°C.
500 twisting and bending cycles at -30°C.
Impact resistance (see 4.8.15).
Abrasion resistance (see 4.8.16).
Electrical tests (see following table).

Type	Standard copper conductor	Insulation thickness	Lay of conductors	Overall diameter		Dielectric strength (see 4.8.19)	Conductor resistance (d.c.) per 1000 feet at 25°C. (see 4.8.24)	Insulation resistance 1000 feet (see 4.8.25)
							Bare	
		Minimum	Maximum	Minimum	Maximum	Minimum	Maximum	Minimum
		Inch	Inches	Inches	Inches	Volts	Ohms	Megohms
SHOF-3	2-1/2(65)	0.031	---	0.195	0.210	2,000	4.26	300
SHOF-23	23(228)	.040	---	.440	.460	2,500	.499	300
SHOF-60	60(304)	.050	---	.570	.600	2,500	.187	250
SHOF-150	150(760)	.070	---	.830	.870	2,500	.0747	200
SHOF-200	200(988)	.070	---	.940	.980	2,500	.0575	200
SHOF-250	250(1254)	.070	---	1.035	1.085	2,500	.0453	150
SHOF-800	800(4033)	.100	---	1.600	1.670	2,500	.0141	100
DHOF-3	2-1/2(26)	.031	2	.405	.425	2,000	4.43	300
DHOF-4	4(41)	.031	2	.440	.460	2,000	2.81	250
DHOF-6	6(65)	.031	2-1/2	.490	.510	2,000	1.76	200
DHOF-9	9(90)	.031	3	.540	.570	2,000	1.27	200
DHOF-14	14(140)	.040	4	.675	.705	2,500	0.810	200
DHOF-23	23(228)	.040	5-1/2	.820	.860	2,500	.513	150
DHOF-30	30(304)	.050	7-1/2	.920	.960	2,500	.382	150
DHOF-83	83(418)	.070	16	1.390	1.450	2,500	.139	150
DHOF-250	250(1254)	.070	24	2.000	2.100	2,500	.0462	100
DHOF-400	400(2052)	.090	30	2.400	2.500	2,500	.0283	100
THOF-3	2-1/2(26)	.031	2	.430	.450	2,000	4.43	300
THOF-4	4(41)	.031	2-1/2	.460	.480	2,000	2.81	250
THOF-6	6(65)	.031	3	.520	.550	2,000	1.76	200
THOF-9	9(90)	.031	3-1/2	.570	.600	2,000	1.27	200
THOF-14	14(140)	.040	4-1/2	.720	.750	2,500	.810	200
THOF-23	23(228)	.040	6	.860	.900	2,500	.513	150
THOF-42	42(209)	.070	12	1.200	1.250	2,500	.278	150
THOF-150	150(760)	.070	18	1.740	1.820	2,500	.0762	125
THOF-250	250(1254)	.070	24	2.140	2.240	2,500	.0462	100
THOF-400	400(2052)	.090	30	2.680	2.800	2,500	.0283	100
FHOF-3	2-1/2(26)	.031	2-1/2	.460	.480	2,000	4.43	300
FHOF-4	4(41)	.031	3	.520	.550	2,000	2.81	250
FHOF-9	9(90)	.031	4	.630	.660	2,000	1.27	200
FHOF-133	133(674)	.070	20	1.900	2.000	2,500	0.0846	125

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3.41 Type TRF cable.

CONSTRUCTION

First. A standard conductor of bare, uncoated copper (see 3.10).
Second. A separator (see 3.11.14).
Third. A wall of synthetic insulation (see 3.11.3.1.1 or 3.11.5), with or without tape (see 3.11.15) at manufacturer's option.
Fourth. Cotton reinforcement (see 3.11.12).
Fifth. A sheath of polychloroprene (see 3.11.17) bonded to the underlying insulation.

TEST REQUIREMENTS

Flammability (see 4.8.10).
Cold bend (see 4.8.9): No cracking of sheath when bent 2 turns around mandrel with diameter twice that of specimen at -30°C.
Impact resistance (see 4.8.15).
Abrasion resistance (see 4.8.16).
Electrical tests (see following table).

Type	Standard copper conductor	Diameters		Dielectric strength (see 4.8.19)	Conductor resistance (d.c.) per 1000 feet at 25°C. (see 4.8.24)
		Over insulation	Overall		Bare
				Minimum	Maximum
		Inch	Inch	Volts	Ohms.
TRF-105	105(259)	0.58	0.76	1,200	0.108
TRF-133	133(259)	.61	.81	1,200	.0856
TRF-168	168(427)	.67	.86	1,200	.0681

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3.42 Type TRXF cable.

CONSTRUCTION

First. A standard conductor of bare, uncoated copper (see 3.10).
Second. A separator (see 3.11.14).
Third. A sheath of polychloroprene (see 3.11.17).

TEST REQUIREMENTS

Flammability (see 4.8.10).
Cold bend (see 4.8.9): No cracking of sheath when bent 2 turns around mandrel with diameter twice that of specimen at -30°C.
Impact resistance (see 4.8.15).
Abrasion resistance (see 4.8.16).
Electrical tests (see following table).

Type	Standard copper conductor	Overall diameter	Dielectric strength (see 4.8.19)	Conductor resistance (d.c.) per 1000 feet at 25°C. (see 4.8.24)
				Bare
			Minimum	Maximum
		Inch	Volts	Ohms
TRXF-84	84(2107)	0.60	1,200	0.138
TRXF-105	105(2646)	.68	1,200	.108
TRXF-133	133(3325)	.75	1,200	.0876

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3.4.3 Type TTRS cable.

CONSTRUCTION

First. A standard conductor of bare, uncoated copper (see 3.10), size 1(7). With or without an approved separator (see 3.11.14), at manufacturer's option. Circuit identification by means of not less than 4 colored threads among the strands of the conductor, as satisfactory to the bureau or agency concerned, unless accomplished under "Second".[1]

Second. A wall of polythene (see 3.11.4) to the diameter, and within the dimensional tolerances required to obtain the specified degree of performance as limited by the maximum overall diameter of the finished cable. Circuit identification (see 3.9.2) unless accomplished under "First".[1]

Third. Two such insulated conductors shall be combined to form a pair. They shall be twisted together with any lay necessary to obtain the specified mechanical and electrical characteristics. One conductor shall be color-coded white, and the other shall be color-coded black.[1]

Fourth. A braided shield of 0.005-inch diameter copper wire (see 3.11.16.6).

Fifth. An inner braid (see 3.11.16.2) of cotton (see 3.11.12) or rayon (see 3.11.13). The braid shall be lightly impregnated with a synthetic resin material to afford satisfactory abrasion resistance, and shall be color-coded (see 3.9.5) for identification of each shielded pair.

Sixth. The number of shielded pairs indicated in the following table shall be cabled together with a left-hand lay. The length of lay shall be a suitable function of the pitch diameter of that layer, designed to meet the flexing endurance test on completed cable. Wherever practicable, the component pairs shall be cabled in a single layer, using, if necessary, a soft center core of asbestos (see 3.11.9), glass (see 3.11.10), or synthetic fibers. A single pair shall not be used as a core for the cabled construction. Asbestos, glass, or synthetic fiber fillers (see 3.11.11) shall be used if necessary to obtain circularity of cross section.

Seventh. The cabled pairs shall be bound by a loose braid or fabric binder in such a manner as to permit sliding between the pairs themselves and between the pairs and the overall braid or binder when the cable is sharply and repeatedly bent and twisted. Manufacturer's identification tape (see 3.8.2).

Eighth. An impervious sheath (see 3.11.18 and the following table), of the thermoplastic type.

[1]In order to facilitate installation, it is desired that color coding of individually insulated conductors be obtained wherever practicable by means of colored insulating compounds (see 3.9.2). If color insulation is used, it is desired that one conductor of each pair be colored white (natural) and the other conductor a distinctive, preferably dark (green, blue), color if a black insulation cannot be obtained.

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TEST REQUIREMENTS

Lot acceptance tests (see 4.5). Lot acceptance tests shall apply to type TTRS cable except that the second sample (see 4.8.14.1.2) shall not be tested unless a flexing endurance failure occurs in the first. If the second sample passes the flexing endurance test, the lot shall be accepted provided the first sample was satisfactory in all other respects. In the event of failure of both samples, the lot shall be rejected.

Characteristic impedance (surge impedance) of a pair (see 4.8.27). - 71.5 ohms (min.), 85.0 ohms (max.) (as near to 76 ohms as practicable) measured at 25°C. and 1 megacycle per second.

Capacitance of a pair (see 4.8.27.3). - 25.0 (max.) micromicrofarads per foot at 25°C. at 1 megacycle per second.

Dielectric strength. - (see 4.8.19) 2000 volts applied for 1 minute for white (natural) conductor of each pair to other conductor and shield, and for both conductors of each pair to shield, shield being grounded.

Insulation resistance. - 1000 megohms per 1000 feet at 15.5°C. (see 4.8.25).

Flammability (see 4.8.10).

Water and air leakage (see 4.8.13).

Conductor resistance (d.c.) per 1000 feet at 25°C., (see 4.8.24) - 10.3 ohms maximum.

Flexing endurance (see 4.8.14).

1000 twisting and bending cycles at +75°C.

500 twisting and bending cycles at -30°C.

Type	Number of shielded pairs	Diameter over sheath
		Inches
TTRS-2	2	0.680
TTRS-4	4	.740
TTRS-6	6	.880
TTRS-8	8	.990
TTRS-10	10	1.080
TTRS-12	12	1.100
TTRS-16	16	1.190

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3.44 Type CVSF cable.

CONSTRUCTION

First. Three uncoated copper conductors (see 3.10), size 53(532) (see table III), with separator (see 3.11.14) insulated with synthetic insulation (see 3.11.3.2 and 3.11.5) having a minimum wall thickness (see 3.7.3) of 0.070 inch. Color-coding (see 3.9.2 or 3.9.4) is required.

Second. One uncoated copper conductor (see 3.10), size 33(336) (see table III), with no covering.

Third. One pair, comprising two uncoated copper conductors (see 3.10) size 2-1/2(26) (see table III) with separator (see 3.11.14) each insulated with color-coded (see 3.9.2) synthetic rubber (see 3.11.3) having a minimum wall thickness (see 3.7.3) of 0.031 inch, the two conductors twisted with fillers (see 3.11.11) having a lay of not more than 2 inches and held with a suitable open thread binder.

Fourth. The three large insulated conductors cabled with a lay of not more than 14 inches, placing the uninsulated conductor in one outer valley and the twisted pair in another outer valley, employing fillers (see 3.11.11) to obtain a well-rounded flexible core, and bound with a compound-filled tape (see 3.11.15) or braid (see 3.11.16).

Fifth. A polychloroprene sheath (see 3.11.17) having a minimum thickness (see 3.7.3) of 0.125 inch to an overall diameter not exceeding 1.45 inches for the completed cable.

TEST REQUIREMENTS

Dielectric strength (see 4.8.19) - 2500 volts.
Insulation resistance (see 4.8.25) - 150 megohms.
Flammability (see 4.8.10).
Water and air leakage (see 4.8.13).
Flexing endurance (see 4.8.14) - 500 bending cycles only at -30°C.
Conductor resistance (d.c.) per 1000 feet at 25°C. (see 4.8.24).
0.217 ohms maximum for size 53(532),
0.351 ohms maximum for size 33(336),
4.35 ohms maximum for size 2-1/2(26).

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Individual Specification Sheet dated 30 July 1952

3.45 Type FCOTP-4 cable.

CONSTRUCTION

First. A standard conductor of bare, uncoated copper (see 3.10).
Second. Synthetic resin insulation (see 3.11.2).
Third. Four such insulated conductors cabled together with a lay not to exceed 2 inches. Circularity of cross section shall be obtained by means of cotton, asbestos, or glass fillers (see 3.11.11). A binder thread or open-weave braid over-cabled conductors, at manufacturer's option.
Fourth. An impervious sheath (see 3.11.18) or polychloroprene sheath (see 3.11.17), colored an orange color (see 3.9.6).

TEST REQUIREMENTS

Flammability (see 4.8.10).
Water and air leakage (see 4.8.13).
Flexing endurance (see 4.8.14):
1000 twisting and bending cycles at +75°C.
500 twisting and bending cycles at -30°C.
Impact resistance (see 4.8.15).
Abrasion resistance (see 4.8.16).
Electrical tests (see following table).

Type	Standard copper conductor	Diameter over synthetic resin		Lay of conductor	Overall diameter		Dielectric strength (see 4.8.19)	Conductor resistance (d.c.) per 1000 feet at 25°C. (see 4.8.24)	Insulation resistance 1000 feet (see 4.8.25)	Cold bend mandrel diameter (see 4.8.9)
		Minimum	Maximum	Maximum	Minimum	Maximum	Minimum	Bare Maximum	Minimum	
		Inch	Inch	Inches	Inch	Inch	Volts	Ohms	Megohms	Inches
FCOTP-4	4(41)	0.105	0.115	2	0.400	0.440	2,000	2.80	200	6

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Individual Specification Sheet dated 30 July 1952

3.46 Type SRI wire.

CONSTRUCTION

First. A standard conductor of bare, uncoated copper (see 3.10).
Second. A wall of synthetic resin insulation (see 3.11.2).

TEST REQUIREMENTS

Cold bend (see 4.8.9).
Electrical tests (see following table).
Flammability (see 4.8.10).

Type	Standard copper conductor	Overall diameter		Dielectric strength (see 4.8.19)				Conductor resistance (d.c.) per 1000 feet at 25°C. (see 4.8.24)	Insulation resistance 1,000 feet (see 4.8.25)	Cold bend mandrel diameter (see 4.8.9)
				Spark	Full reel	Short length	After cold bend			
		Minimum	Maximum	Minimum	Minimum	Minimum	Minimum	Bare Maximum	Minimum	
		Inch	Inch	Volts	Volts	Volts	Volts	Ohms	Megohms	Inches
SRI-1(10)	1(10)	0.068	0.075	2,500	1,000	6,000	3,000	11.5	100	0.30
SRI-1-1/2(41)	1-1/2(41)	.080	.090	5,000	1,000	6,000	3,000	7.15	100	.40
SRI-2-1/2(26)	2-1/2(26)	.092	.102	5,000	1,000	6,000	3,000	4.43	100	.40
SRI-3	3(19)	.095	.105	5,000	1,000	6,000	3,000	3.80	100	.40
SRI-4(7)	4(7)	.115	.125	7,500	1,500	6,000	3,000	2.52	100	.50
SRI-4(41)	4(41)	.115	.125	7,500	1,500	6,000	3,000	2.81	100	.50
SRI-6(7)	6(7)	.135	.150	7,500	1,500	6,000	3,000	1.73	100	.60
SRI-6(65)	6(65)	.145	.160	7,500	1,500	6,000	3,000	1.78	100	.60
SRI-9(7)	9(7)	.163	.178	7,500	1,500	6,000	3,000	1.25	100	.70
SRI-9(90)	9(90)	.175	.190	7,500	1,500	6,000	3,000	1.28	100	.70
SRI-14	14(140)	.200	.215	7,500	1,500	6,000	3,000	.823	100	.85
SRI-26	26(49)	.265	.285	7,500	1,500	6,000	3,000	.440	100	1.10
SRI-42	42(49)	.315	.340	7,500	1,500	6,000	3,000	.275	100	1.40
SRI-66	66(133)	.395	.420	7,500	1,500	6,000	3,000	.175	100	1.70
SRI-100	100(61)	.428	.453	7,500	1,500	6,000	3,000	.115	100	1.80

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Individual Specification Sheet dated 30 July 1952

3.47 Type SRIB wire.

CONSTRUCTION

First. A standard conductor of bare, uncoated copper (see 3.10).
Second. A wall of synthetic resin insulation (see 3.11.2).
Third. An outer cotton braid (see 3.11.16), unless a rayon braid is specifically required. The braid shall be saturated or filled, and finished with moisture and flame retardant, insulating compounds providing a hard, smooth, dull finish, without gloss. The color of finish shall be black (see 3.9.6) unless otherwise specified in the contract or order.

TEST REQUIREMENTS

Cold bend (see 4.8.9).
Flammability (see 4.8.10).
Creepage (see 4.8.20).
Electrical tests (see following table).

Type	Standard copper conductor	Diameter over synthetic resin	Overall diameter		Dielectric strength (see 4.8.19)			Conductor resistance (d.c.) per 1000 feet at 25°C. (see 4.8.24)	Cold bend mandrel diameter (see 4.8.9)
					Spark	Short length	After cold bend		
		Minimum	Minimum	Maximum	Minimum	Minimum	Minimum	Bare Maximum	
		Inch	Inch	Inch	Volts	Volts	Volts	Ohms	Inches
SRIB-1(10)	1(10)	0.068	0.093	0.110	2,500	6,000	3,000	11.5	0.40
SRIB-1-1/2(41)	1-1/2(41)	.080	.105	.125	5,000	6,000	3,000	7.15	.50
SRIB-2-1/2(26)	2-1/2(26)	.092	.117	.137	5,000	6,000	3,000	4.43	.55
SRIB-3	3(19)	.095	.120	.140	5,000	6,000	3,000	3.80	.55
SRIB-4	4(7)	.115	.145	.165	7,500	6,000	3,000	2.52	.65
SRIB-6	6(7)	.135	.165	.190	7,500	6,000	3,000	1.73	.75
SRIB-9	9(7)	.163	.193	.218	7,500	6,000	3,000	1.25	.90
SRIB-14	14(140)	.200	.230	.255	7,500	6,000	3,000	.823	1.00
SRIB-26	26(49)	.265	.295	.325	7,500	6,000	3,000	.440	1.30
SRIB-42	42(49)	.315	.345	.380	7,500	6,000	3,000	.275	1.50
SRIB-66	66(133)	.395	.430	.465	7,500	6,000	3,000	.175	1.85

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Individual Specification Sheet dated 30 July 1952

3.48 Type SHFS wire.

CONSTRUCTION

First. A standard conductor of bare, uncoated copper (see 3.10).

Second. A wall of synthetic resin insulation (see 3.11.2).

Third. A wall of felted asbestos (see 3.11.9) or glass fiber (see 3.11.10).

Fourth. (a) A rayon braid (see 3.11.16) on sizes 1 to 3, inclusive.

(b) An outer cotton braid (see 3.11.16) on sizes 4 to 100, inclusive.

Fifth. The braid shall be filled and finished with moisture and flame retardant compounds, providing a hard, smooth dull finish, without gloss. The color of the finish shall be gray (see 3.9.6) unless otherwise specified in the contract or order.

TEST REQUIREMENTS

Cold bend (see 4.8.9).
Flammability (see 4.8.10).
Creepage (see 4.8.20).
Electrical tests (see following table).

Type	Standard copper conductor	Diameter over synthetic resin	Diameter over glass or asbestos	Overall diameter		Dielectric strength (see 4.8.19)			Conductor resistance (d.c.) per 1000 feet at 25°C. (see 4.8.24): Bare	Cold bend mandrel diameter (see 4.8.9)
						Spark	Short length	After cold bend		
		Minimum		Minimum	Maximum	Minimum	Minimum	Minimum	Maximum	
		Inch	Inch	Inch	Inch	Volts	Volts	Volts	Ohms	Inches
SHFS-1	1(10)	0.068	0.100	0.115	0.125	2,500	6,000	3,000	11.5	0.50
SHFS-1-1/2	1-1/2(41)	.080	.115	.130	.140	5,000	6,000	3,000	7.15	.50
SHFS-2-1/2	2-1/2(26)	.092	.122	.135	.145	5,000	6,000	3,000	4.43	.60
SHFS-3	3(19)	.095	.135	.150	.160	5,000	6,000	3,000	3.80	.60
SHFS-4	4(7)	.115	.155	.185	.195	7,500	6,000	3,000	2.52	.80
SHFS-6	6(7)	.135	.180	.205	.220	7,500	6,000	3,000	1.73	.90
SHFS-9	9(7)	.163	.208	.233	.248	7,500	6,000	3,000	1.25	1.00
SHFS-14	14(140)	.200	.245	.270	.285	7,500	6,000	3,000	.823	1.15
SHFS-26	26(49)	.265	.325	.345	.365	7,500	6,000	3,000	.440	1.50
SHFS-42	42(49)	.315	.380	.405	.425	7,500	6,000	3,000	.275	1.70
SHFS-66	66(133)	.395	.460	.485	.505	7,500	6,000	3,000	.175	2.00
SHFS-100	100(61)	.428	.493	.523	.543	7,500	6,000	3,000	.115	2.15

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Individual Specification Sheet dated 30 July 1952

3.49 Type MRI wire.

CONSTRUCTION

First. A standard conductor of bare, uncoated copper (see 3.10).
Second. A wall of synthetic resin insulation (see 3.11.2).
Third. The specified number of conductors shall be twisted together with a lay not greater than that specified in the following table.

TEST REQUIREMENTS

Flammability (applies to single conductors after untwisting) (see 4.8.10).
Electrical tests (see following table).

Type	Number of conductors	Standard copper conductor	Diameter over synthetic resin		Lay of conductors	Dielectric strength (see 4.8.19)			Conductor resistance (d.c.) per 1000 feet at 25°C. (see 4.8.24)	Insulation resistance 1,000 feet (see 4.8.25)
			Minimum	Maximum	Maximum	Spark	Full reel	Short length	Bare Maximum	Minimum
			Inch	Inch	Inches	Volts	Volts	Volts	Ohms	Megohms
MRI-D-1	2	1(10)	0.068	0.075	2	2,500	1,000	6,000	11.7	100
MRI-D-2-1/2	2	2-1/2(26)	.092	.102	3	5,000	1,500	6,000	4.52	100
MRI-T-2-1/2	3	2-1/2(26)	.092	.102	3	5,000	1,500	6,000	4.52	100

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3.50 Workmanship. - The workmanship shall be first class in every respect and uniform throughout each lot of cable offered for delivery.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Inspection procedures. - For Naval purchases, the general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

4.2 Qualification tests at a Government laboratory. - Qualification tests shall be conducted at a Government laboratory designated by the Bureau of Ships. These tests shall consist of the tests specified in 4.7.

4.3 Sampling. -

4.3.1 Lot. -

4.3.1.1 Lot of completed wire or cable. - A lot of completed wire or cable shall consist of not more than 10,000 feet of completed wire or cable of one type, size and composition, manufactured under the same essential conditions and offered for delivery at one time.

4.3.1.2 Lot of copper wire or conductor. - Unless otherwise specified herein, a lot of wire shall consist of not more than 500 pounds of wire having a diameter of 0.040 inch or less; or of not more than 10 coils, reels, or spools of wire having a diameter of 0.041 inch or over; or of an equivalent quantity of stranded conductor; all of one size and intended for use in the same lot of completed wire or cable.

4.3.2 Sampling procedure. -

4.3.2.1 General. - From each lot offered for delivery the Government inspector shall select samples for the surface inspection (see 4.4) and lot acceptance tests (see 4.5) to be performed at the place of manufacture. Upon satisfactory completion of the inspection at the place of manufacture (see 4.4 and 4.5) the Government inspector shall select samples for the production check tests to be performed at a Government laboratory (see 4.6). At the inspector's discretion, when delivery requirements are urgent, production check tests at the Government laboratory may be conducted simultaneously with lot acceptance tests at the place of manufacture.

4.3.2.2 Selection of samples. -

4.3.2.2.1 Completed wire or cable. - In line with the basic conception of this specification as to minimum acceptable standards, the samples selected for tests at the Government laboratory shall be representative of the poorest of the lot insofar as the visual or dimensional inspection (see 4.4) or the physical and electrical tests (see 4.5.2) disclose any differences in dimensions and/or characteristics. If there is any question as to the satisfactory processing of the completed wire or cable up to the starting and finishing ends, the contractor shall cut off and discard such ends before the lot is submitted for Government inspection to insure that the samples are truly representative of the completed wire or cable offered for delivery.

4.3.2.2.2 Copper wires or conductor. - Sample sections of individual wires or stranded conductor shall be selected in such manner as to be representative of the material to be used in the completed wire or cable.

4.3.2.3 Number of samples. -

4.3.2.3.1 Number of samples for surface inspection (visual examination for defects and finish, and inspection for workmanship and measurement of dimensions). - For the purpose of the inspection specified in 4.4, the Government inspector shall select a sample from each manufactured length of completed wire or cable offered for delivery.

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4.3.2.3.2 Number of samples for lot acceptance tests. -

4.3.2.3.2.1 Completed wire or cable. - For the purpose of the physical tests specified in 4.5.2.1.1, the Government inspector shall select two samples of sufficient length from each lot offered for delivery. Each shipping length of completed wire or cable offered for delivery shall be subjected to the electrical tests specified in 4.5.2.1.2.

4.3.2.3.2.2 Conductors. - For the purpose of the tests specified in 4.5.3, the Government inspector shall select two samples of sufficient length from each lot of conductors submitted for inspection (see 4.3.1.2).

4.3.2.3.3 Number of samples for production check tests.

4.3.2.3.3.1 One or more lots. - When contracts or orders cover one or more full lots of completed cable (see 4.3.1.1), for the purpose of the tests specified in 4.6, the Government inspector shall select two full sample lengths from the first lot of each type, size and composition offered for delivery. Thenceforth, production check samples shall be selected once every two months or from one of every five successive lots of the same type, size, and composition offered for delivery. The method of selection providing the more frequent sampling shall be used.

4.3.2.3.3.2 Less than one lot. - The provisions of 4.3.2.3.3.1 shall not apply to any item when the total quantity of that item is less than one lot of 10,000 feet. When the lot covers 2,500 feet or less of any one type, size, and composition, production check tests are not required. When the lot covers more than 2,500 feet, but less than one full lot of 10,000 feet, of any one type, size and composition of completed cable, one full sample length shall be forwarded to a Government laboratory for the purpose of the tests specified in 4.6.

4.3.2.4 Preparation of samples (for tests at Government laboratory).

4.3.2.4.1 Length of samples. - The length of cable constituting one sample shall be as follows:

(a) Cables for nonflexing service (see 1.2.1).	
All tests except current overload	24 feet
Current overload (see 4.6.11)	11 feet
(b) Cables for repeated flexing service. (see 1.2.2)	24 feet
(c) Types TTRS and TTRSA cables	35 feet
(d) Thin walled cables (see 1.2.3)	30 feet
(e) Watertight cables (see Appendix I)	5 feet
(f) Miscellaneous wires (see 1.2.4)	30 feet

4.3.2.4.2 Identification of samples. - All test samples shall be marked with the following information, as applicable:

(a) Name of manufacturer.
(b) Cable designation (type and size).
(c) The manufacturing "lot" number, or order number either of the bureau or agency concerned or the shipbuilder (if submitted for delivery) and footage represented by sample.
(d) The coil or reel number.
(e) Qualification (see 3.1) designation.

4.3.2.4.3 End seals. - In order to obtain representative end seals, samples for test at the Government laboratory shall be designated after end seals have been prepared on shipping lengths of cable. The length of such sample shall be 2-1/2 feet. The cut end of the shipping length shall be resealed immediately by the manufacturer in accordance with 3.11.22.

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4.3.2.4.4 Packaging of samples for shipment. - To facilitate handling at the Government laboratory, as well as by common carrier, samples of cable for nonflexing service (see 1.2.1) shall be shipped in straight lengths in a suitable wooden box approximately but not more than 16 feet in length. Miscellaneous wires (see 1.2.4) and flexible cables (see 1.2.2) may be shipped in wrapped coils or boxes. When necessary, the total sample length may be cut.

4.4 Lot acceptance inspection (visual and dimensional inspection, at place of manufacture). -

4.4.1 Material and workmanship. - The inspector shall assure himself by careful visual and dimensional inspection of each of the samples of the finished product selected in accordance with 4.3.2 as well as by inspection of the materials and manufacturing processes entering into the finished product, that the completed wire or cable submitted for delivery under the contract or order is in conformance with this specification. The inspector shall observe and record the number of random remnant and scrap lengths and the total footage of such lengths, compared to the number of standard lengths and the total footage in each lot of wire or cable offered for inspection. An excessive number of random, remnant and scrap lengths may have resulted from the cutting out of obvious defects or from breakage during manufacture, and may be taken as indicative of the need for tighter quality control of materials, workmanship, and manufacturing processes. This undesirable condition places an increased burden on the inspector to discover less obvious defects and possible derangement of components in the finished product, and the number of samples that are selected and inspected is determined by the number of shipping lengths. In order that the number of lengths be kept to the minimum consistent with good manufacturing practices, for each type and size of wire or cable on a contract or order, not less than 70 percent of the total footage to be shipped shall be in standard lengths, not more than 30 percent may be in any combination of random and remnant lengths. When the total quantity of any one item is 6 nominal lengths or less, a footage approximating 2 nominal lengths (in lieu of 30 percent) may be in any combination of random and remnant lengths. When the total quantity of any one item is 2 nominal lengths or less, the total footage may be in any combination of random and remnant lengths. In particular, this inspection shall cover such of the following as may be applicable to the individual types and sizes of completed wire or cable involved.

4.4.2 Conductors. - Inspect for number and size of copper wires (see 3.11.1) composing the standard conductors (see 3.10), of specified stranding, total cross-sectional area, etc.

4.4.3 Material components. - Inspect for types of insulation and component materials (see 3.6 and 3.11.2 to 3.11.22, inclusive) and limiting dimensions, etc., of each layer, or stage, as may be applicable (see individual specification sheets, 3.12 to 3.49, inclusive) to the cable under consideration.

4.4.4 Identification and color coding. - Inspect for manufacturer's identification (see 3.8), for circuit identification (see 3.9), and color coding as may be specified for the particular type of cable undergoing inspection.

4.4.5 Outer coverings. - Inspect for braids (see 3.11.16), polychloroprene sheath (see 3.11.17), impervious sheath (see 3.11.18), lead sheath (see 3.11.19), metal armor (see 3.11.20), and paint (see 3.11.21), as applicable.

4.4.6 Dimensions. - Ascertain, by measurement, that the centering, circularity and dimensional tolerances (see 3.7 and 3.12 to 3.49, inclusive, as applicable) have not been exceeded in the completed wire or cable. For insulation nominally less than 1/32 inch in thickness, the minimum thickness shall be determined with a pingage dial micrometer by placing the specimen on the pin and taking the reading immediately after the movable member comes to rest on the specimen.

4.4.7 Patching. - The manufacturer's methods and practices with regard to making patches in the conductor insulation and cable sheath shall be subject to approval by the Government inspector. The materials and methods employed shall be such that the finished product complies with all specification requirements. The frequency of patching shall be held to a minimum consistent with specification lengths and quantity production, and in no case shall any shipping length (see 5.1.1) contain more than one sheath patch. A build-up that is less than half the wall thickness shall not be considered a patch. No build-up or patch shall be made that cannot be contained in a single mold.

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4.4.8 End seals. - Inspect both ends of every shipping length to insure that end seals of the approved design (see 3.11.22) have been applied.

4.4.9 Action in case of failure. - If any one sample representing a lot is found not to be in conformance with this specification, the entire lot represented shall be rejected. The manufacturer shall have the option of inspecting each length of a rejected lot, and, after removing all nonconforming lengths of cable, resubmitting the lot for Government inspection.

4.5 Lot acceptance tests. -

4.5.1 Place of test. - Lot acceptance tests shall be made at the place of manufacture and shall be witnessed by the Government inspector. The manufacturer shall provide the test equipment and facilities necessary for conducting such tests.

4.5.2 Completed wire and cable. -

4.5.2.1 Test procedure. -

4.5.2.1.1 Physical tests. - Specimens of sufficient size from each of the samples selected in accordance with 4.3.2.3.2.1 shall be subject to each of the following tests:

(a) Tensile strength, elongation, set and aging of insulation (see 3.11.2 to 3.11.6, incl., and 4.8.1).
(b) Tensile strength, elongation, set and aging of sheath (see 3.11.17 and 4.8.1).
(c) Flexing endurance (see individual specification sheets and 4.8.14).
(d) Breaking strength (see 3.30 and 4.8.28).
(e) Creepage test (see 3.47, 3.48 and 4.8.20).
(f) Capacitance and characteristic impedance (see 4.8.27).

4.5.2.1.2 Electrical tests. -

4.5.2.1.2.1 Conductor resistance. - The resistance of insulated conductor shall be measured in all completed wires and cables to determine conformance with the individual specification sheets. For stranded conductors, the resistance of individual strand wires need not be measured unless the inspector has reason to doubt the conductivity.

4.5.2.1.2.2 Dielectric tests. - The dielectric tests specified on the individual specification sheets shall be made on each full coil or reel length of completed wire or cable offered for delivery.

4.5.2.1.2.3 Insulation resistance tests. - Insulation resistance tests, when specified on an individual specification sheet shall be made on each full coil or reel length of completed wire or cable offered for delivery.

4.5.2.1.2.4 Induced voltage tests. - Induced voltage tests, when specified on an individual specification sheet, shall be made either on the full-reel length of completed cable or on a specimen not less than 20 feet in length.

4.5.2.2 Action in case of failure. -

4.5.2.2.1 Physical tests. - If any one of the samples representing a lot is found to be not in conformance with this specification, the entire lot shall be rejected. The manufacturer shall have the option of subjecting each length of completed wire or cable in a rejected lot to the test or tests wherein failure occurred, and, after removing all defective lengths, resubmitting the lot for Government inspection.

4.5.2.2.2 Electrical tests. - Each section of completed wire or cable which is found to be not in conformance with this specification shall be rejected.

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4.5.3 Conductors. -

4.5.3.1 Test procedure. - Each of the samples selected in accordance with 4.3.2.3.2.2 shall be subjected to each of the following tests:

(a) Tensile strength and elongation of conductors (see 3.10, 3.11.1, and 4.8.21).
(b) Coating, metallic (see 3.10.3 and 4.8.22).
(c) Resistance, conductors (see individual specification sheets and 4.8.24).

4.5.3.2 Action in case of failure. - If any one of the samples representing a lot of conductors is found to be not in conformance with this specification, the entire lot shall be rejected. The manufacturer shall have the option of testing each coil, reel, or spool of wire in a lot, and, after removing all defective coils, reels, or spools, resubmitting the lot of conductors for Government inspection.

4.6 Production check tests. -

4.6.1 Place of test. - Production check tests shall be conducted at the Material Laboratory, New York Naval Shipyard, Naval Base Station, Brooklyn 1, N.Y., or at the Industrial Laboratory, Mare Island Naval Shipyard, Vallejo, Calif.

4.6.2 Test procedure. - Each of the samples selected by the Government inspector in accordance with 4.3.2.3.3 shall be subjected to such of the following tests as may be applicable to the particular type of cable undergoing inspection:

Synthetic resin, synthetic rubber, butly rubber, polythene, and polychloroprene (see 4.8.1).
Chemical analysis (see 4.8.2)[1].
Varnished cambric as removed from completed cable (see 4.8.3).
Asbestos (see 4.8.4)[1].
Glass (see 4.8.5)[1].
Impervious sheath (see 4.8.6)[1].
Metal armor (see 4.8.7)[1].
Drip (see 4.8.8).
Cold bend (see 4.8.9).
Flammability (see 4.8.10).
Current overload (see 4.8.11).
Water and air leakage (see 4.8.13)[1].
Flexing endurance (see 4.8.14)
Impact resistance (see 4.8.15)[1].
Abrasion resistance (see 4.8.16)[1].
Hot oil test (see 4.8.17).
Oil, water, and chemical resistance (see 4.8.18)[1].
Dielectric tests (see 4.8.19).
Creepage (see 4.8.20).
Flame retardance (see 4.8.26).
Capacitance and characteristic impedance (see 4.8.27).
Breaking strength (see 4.8.28).
Induced voltage (see 4.8.29).
Heat shock (see 4.8.30).
Hydrostatic test (see 4.8.31).
Watertightness (see Appendix I).

[1]These tests shall be conducted only when specifically requested by the Government inspector.

4.6.3 Action in case of failure. - Acceptance of the first lot of completed wire or cable of each type, size and composition offered for delivery under a contract or order shall be withheld pending receipt of a satisfactory test report on the sample or samples representing the lot. Thenceforth, except as hereinafter specified, acceptance and rejection of lots shall normally be on the basis of the sampling, inspection and tests of 4.3.2.3.1, 4.3.2.3.2, 4.4, and 4.5, and acceptance shall not be withheld pending receipt of

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the test report on a production check test sample. In case of receipt of an unsatisfactory test report on a production check sample additional samples shall be selected from every subsequent lot offered for delivery in accordance with 4.3.2. The samples so selected shall be submitted to the Government laboratory, and shall there be subjected to the test or tests wherein failure was observed. Lots shall then be accepted only upon receipt of a satisfactory test report on the samples representing them. This additional testing shall be discontinued and lot acceptance returned to the normal basis of 4.3.2.3.1, 4.3.2.3.2, 4.4 and 4.5, when 4 successive lots have been accepted.

4.7 Qualification tests. -

4.7.1 Place of test. - Tests to determine suitability for Government use shall be made at the Material Laboratory, New York Naval Shipyard, Naval Base Station, Brooklyn 1, N.Y. or the Industrial Laboratory, Mare Island Naval Shipyard, Vallejo, Calif.

4.7.2 List of tests. - The following tests, as applicable to the particular types of cable, shall be conducted at the Government laboratory:

General inspection (see 4.4).
Synthetic resin, synthetic rubber, butyl rubber, polythene, and polychloroprene (see 4.8.1).
Chemical analysis (see 4.8.2).
Varnished cambric (see 4.8.3).
Asbestos (see 4.8.4).
Glass (see 4.8.5).
Impervious sheath (see 4.8.6).
Metal armor (see 4.8.7).
Drip (see 4.8.8).
Cold bend (see 4.8.9).
Flammability (see 4.8.10).
Current overload (see 4.8.11).
Heat endurance (see 4.8.12).
Water and air leakage (see 4.8.13).
Flexing endurance (see 4.8.14)
Impact resistance (see 4.8.15).
Abrasion resistance (see 4.8.16).
Hot oil test (see 4.8.17).
Oil, water and chemical resistance (see 4.8.18).
Dielectric tests (see 4.8.19).
Creepage (see 4.8.20).
Tensile strength and elongation, conductors (see 4.8.21).
Coating, lead or alloy (see 4.8.22).
Flame retardance (see 4.8.26).
Capacitance and characteristic impedance (see 4.8.27).
Breaking strength (see 4.8.28).
Induced voltage (see 4.8.29).
Heat shock (see 4.8.30).
Hydrostatic test (see 4.8.31).
Watertightness (see Appendix I).

4.8 Methods of tests. -

4.8.1 Synthetic resin, synthetic rubber, butyl rubber, polythene, and polychloroprene. -

4.8.1.1 General. - Samples of the insulation (see 3.11.2 to 3.11.6, incl.) and sheath (see 3.11.17) taken from the completed wire or cable shall be tested to determine the physical properties in the "as received" condition and after accelerated aging.

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4.8.1.2 Physical properties. - The tensile strength, elongation and set shall be determined on specimens prepared and tested as follows:

4.8.1.2.1 Specimens. - The test specimen shall be the entire section of the insulation in the case of small conductors or, in the case of a large conductor or in obtaining a specimen of the cable sheath, either a segment of a section cut with a sharp knife held tangentially to the conductor or cable core, or a shaped specimen cut out with a standard die. When the whole cross section is used, the specimen shall not be cut longitudinally. The test specimen shall be as free as possible from surface incisions and imperfections.

4.8.1.2.1.1 Size. - The size of specimen shall be of an area not less than 0.010 square inch and not more than 0.030 square inch. When not possible to use specimens of this size, they shall be as large as practicable.

4.8.1.2.1.2 Condition and age. - Specimens shall not be heated, immersed in water, or subjected to any mechanical or chemical treatment not specifically specified herein. No tests shall be made within 48 hours after vulcanization unless agreed to by the manufacturer.

4.8.1.2.1.3 Buffing. - In event of any irregularities on the surface of a fractional-section specimen, it shall be made smooth and of uniform thickness within 5 percent of the original minimum thickness by buffing, except where large strands are used, in which case the samples shall be buffed sufficiently to remove all inner corrugations.

4.8.1.2.1.4 Calculation of area. - Calculation of the area of the test specimen shall be made as follows:

(a) Where the total cross section of the insulation is used, the area shall be taken as the difference between the area of the circle whose diameter is the minimum average outside diameter of the insulation and the area of the conductor. The area of a stranded conductor shall be figured from its maximum diameter. The following formula may be used to advantage:

$$A = 0.7854 (D + d) (D - d)$$

Where: A = Area of cross section in square inches.
D = Outside diameter of insulation in inches.
d = Diameter over copper in inches.

(b) Where a slice cut from the insulation by a knife held tangentially to the wire is used, and the slice so cut has the cross section of a segment of a circle the area shall be calculated as that of the segment of a circle whose diameter is that of the insulation. The height of the segment is the wall of insulation on the side from which the slice is taken. The values may be obtained from a table giving the areas of segments of a unit circle for the ratio of the height of the segment to the diameter of the circle.

(c) Where the cross section of the slice is not a segment of a circle the area shall be calculated from a direct measurement of the volume or from the specific gravity and the weight of a known length of the specimen having a uniform cross section.

(d) Where a portion of a sector of a circle has to be taken as where the conductor is large and the insulation thin, the area shall be calculated as the thickness times the width. This applies either to a straight test piece or one stamped out with a die, and assumes that corrugations have been removed by buffing.

(e) Where a portion of a sector of a circle has to be taken, as where the conductor is large and the insulation thick, the area shall be calculated as the proportional part of the area of the total cross section.

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4.8.1.2.1.5 Number of specimens. - Five specimens shall be tested for tensile strength, elongation and set from each inspection unit, and the physical property of the inspection unit shall be taken as the median of the results obtained from the five specimens.

4.8.1.2.2 Apparatus. - The testing machine shall be power-driven and preferably to the pendulum type. A spring balanced type of apparatus may be used if provided with a device which will indicate the actual maximum load at which rupture takes place and provision is made to prevent recoil of the spring. The testing machine shall be accurate within 1 percent of the breaking load.

4.8.1.2.3 Procedure. -

4.8.1.2.3.1 Temperature. - Physical tests shall be conducted with the specimens at a temperature between 30° and 32°C.

4.8.1.2.3.2 Rate. - The specimens shall be stretched at a rate of 20 inches per minute in a testing machine having jaws 4 inches apart.

4.8.1.2.3.3 Tensile strength and elongation. - A test specimen, not less than 6 inches in length, having bench marks 2 inches apart, shall be placed in a testing machine and stretched until the specimen breaks. A scale shall be used to measure the elongation at instant of rupture. The break shall be between the bench marks, and the tensile strength calculated from the area of the original sample.

4.8.1.2.3.4 Set. - The test specimen, not less than 6 inches in length, having bench marks 2 inches apart, shall be placed in the jaws of a testing machine and stretched until the bench marks are separated by the distance specified in 3.11.2 to 3.11.6, inclusive, and 3.11.17, as applicable. The test specimen shall then be released within 5 seconds and the set determined 1 minute, unless otherwise specified (see 3.11.2) after the beginning of release.

4.8.1.3 Accelerated aging. -

4.8.1.3.1 Specimens. - Specimens for accelerated aging tests shall be duplicates of those prepared for determination of initial physical properties.

4.8.1.3.2 Procedure. - Specimens shall be suspended vertically in a suitable test chamber without touching each other or the walls of the container. The specified temperature shall be maintained within +2°C. for the specified period of time. At the conclusion of the aging period, the pressure within the test chamber shall be released slowly over a period of not less than 5 minutes. The tension tests shall be conducted not less than 16 nor more than 48 hours after removal from the test chamber. The tensile strength shall be based on the breaking strength after aging and the cross sectional area measured before aging. Aged and unaged specimens should be tested under the same conditions, insofar as practical.

4.8.1.3.2.1 Oxygen pressure test. - The internal pressure of oxygen in the bomb shall be not less than 290 nor more than 310 pounds per square inch.

4.8.1.3.2.2 Air pressure heat test. - The internal pressure of air in the bomb shall be not less than 78 nor more than 82 pounds per square inch.

4.8.1.3.2.3 Air oven test. - Heated air at atmospheric pressure shall be circulated so as to maintain a uniform temperature in the oven, using baffles to avoid local overheating or dead spots. The source of heat shall be located in the air supply outside the aging chamber proper, and the supply shall be controlled with upper and lower vents to provide a continuous change of air during test.

4.8.2 Chemical analysis. - Specimens of the conductor insulation and sheath compounds taken from the completed wire or cable shall be subjected to a careful chemical analysis for purposes of establishing the identity of the compounds, and ascertaining the composition of compounds which have been approved on the basis of compliance with the performance requirements of this specification.

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4.8.3 Varnished cambric. -

4.8.3.1 General. - Samples of the varnished cambric tape (see 3.11.7) removed from the completed cable shall be tested for tensile strength, effect of hot petrolatum, dielectric strength and aging.

4.8.3.2 Tensile strength. - The tests shall be made in the same general manner as specified in 4.8.1.2, except that the rate of separation of the jaws of the testing machine shall be approximately 12 inches per minute.

4.8.3.3 Effect of hot petrolatum. - Samples of the varnished cambric shall be placed in hot petrolatum at a temperature of 150°C. for a period of 15 minutes, then removed from the petrolatum and allowed to cool to room temperature.

4.8.3.4 Dielectric strength. -

4.8.3.4.1 General. - The apparatus used and the methods of measuring voltage shall be as specified in 4.8.19.

4.8.3.4.2 Specimens. - Not less than 3 (unless the conductor has less than 3 tapes, in which case all tapes shall be taken) nor more than 5 tapes shall be taken from each conductor of a completed cable. Specimens of tape shall be selected to be as representative as possible of all of the tapes used.

4.8.3.4.3 Electrodes. - The electrodes shall be of brass or copper, 1/4 inch in diameter, rounded to a radius of 1/32 inch, with flat polished contact surfaces. The electrodes shall be self-aligning, and shall be placed exactly opposite one another, with the specimen in a horizontal plane between them. The electrodes shall be kept clean and polished. The electrodes and the specimen shall be immersed in suitable oil to prevent the occurence of flashover.

4.8.3.4.4 Procedure. - Starting at zero the voltage shall be increased uniformly to breakdown at a rate of approximately 500 volts per second. Three tests shall be made on each tape at room temperature. The tape thicknesses shall be measured near the point of puncture with a machinist's micrometer graduated at 0.001 inch and having anvil and spindle surfaces not less than 0.24 inch nor more than 0.26 inch in diameter. The micrometer shall be tested and calibrated periodically for conformance with accepted commercial standards. The number of thickness readings on the specimen shall be equal to the number of punctures made; these readings shall be distributed along the centerline of the specimen in such manner as to represent closely the thickness at the points of the puncture.

4.8.3.4.5 Test average. - The average of all tests on all tested tapes from any one conductor shall be considered the average breakdown stress, and shall be not less than the appropriate value specified in 3.11.7.11.

4.8.3.5 Aging. - Ten specimens shall be heated in an air oven to a temperature of not less than 124°C. nor more than 126°C. for a period of 150 hours, and then allowed to cool to normal testing room temperature. The specimens shall then be tested by bending them through not less than 180 degrees around a 1/8 inch diameter mandrel. Five specimens shall be bent in the length direction and five in the width direction of the tape.

4.8.4 Asbestos. - Specimens of asbestos taken from the completed cable shall be tested to determine compliance with Specification MIL-I-3053 and 3.11.9 and 3.11.11.

4.8.5 Glass. - Specimens of glass fiber taken from the completed cable shall be tested to determine compliance with Specification MIL-G-1140 and 3.11.10 and 3.11.11.

4.8.6 Impervious sheath. - The materials used shall be in conformance with those employed in the approved sample or otherwise specifically approved.

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4.8.7 Metal armor. -

4.8.7.1 Chemical analysis. - Specimens of armoring wire before application of the completed cable shall be analyzed to determine conformance with 3.11.20.

4.8.7.2 Physical properties. - Specimens of the armoring wire before application to the cable shall be subjected to tests of tensile strength, elongation, springiness, and toughness.

4.8.7.3 Springiness. - A length of wire shall be coiled snugly around a mandrel 1/4 inch in diameter at a tension of approximately but not less than 100 gram (gm.) and then released. The outside diameter of the coil shall be measured in several places at or near its middle. The coil so produced shall not exceed 3/8 inch in outside diameter.

4.8.7.4 Toughness. - The toughness of the wire shall be determined by bending a specimen back and forth, in a specially designed fixture, over a 0.030-inch radius support. Each 90-degree movement in either direction shall be taken as one bend. The bends shall be made under a uniform tension of 100 gm., and at a uniform rate not to exceed 50 bends per minute. The wire shall withstand not less than ten 90-degree bends before a rupture.

4.8.8 Drip. -

4.8.8.1 Specimens. - The specimen shall be a 3-foot section of completed cable.

4.8.8.2 Conditioning. - One specimen of each type of cable size 14 and larger and degaussing cable size 19(6) shall be suspended in a vertical position in an oven maintained for a period of not less than 18 hours at 95°C. For conductor sizes smaller than 14, no drip test will be required.

4.8.8.3 Observation. - No material shall drip or exude from the end of the cable during the test.

4.8.9 Cold bend. -

4.8.9.1 Specimens. - Where a cold bend test is specified in section 3, the specimen of cables for nonflexing service (see 1.2.1) shall consist of a 5-foot section of completed cable, the specimen of miscellaneous wires (see 1.2.4) shall consist of a 1-foot section of completed wire, and the specimens from cables for repeated flexing service (see 1.2.2) shall consist of 1-foot sections of not less than 20 percent of the insulated and braided conductors removed from completed cable. For type MHFA (see 3.23) specimens for the cold bend test shall consist of one 5-foot length of completed cable, and a number (not less than 20 percent of the total number) of the individually insulated and braided conductors removed from completed cable.

4.8.9.2 Apparatus. - Straight wooden bars 5 feet in length, wooden mandrels of specified radii of curvature, and a refrigerator sufficiently large to hold cable specimens and mandrels shall be provided.

4.8.9.3 Procedure. - The specimen shall first be straightened if necessary and then kept straight by mounting on a wooden bar. The mounted specimen shall be placed in the refrigerator and conditioned at 0°C. for not less than 6 hours, then removed from the refrigerator, quickly detached from the mounting, and bent around the appropriate wooden mandrel at a rate of approximately 20 degrees per second. The specimen shall be suitably clamped to maintain its bent position, then it shall be dissected and a visual inspection made of all its component parts. When defects or disarrangements are discovered in any component, examination should be made of specimens in the as received condition to determine whether the defect existed prior to the cold bend test. Any changes or derangements, such as torn tapes or broken or cracked components, resulting from the cold bend test, or a nature or to an extent as to affect the electrical properties or performance of the cable in service, shall be reported to the cognizant Government inspector.

4.8.9.3.1 Cables for nonflexing service. - Cables for nonflexing service (see 1.2.1) shall be bent 180 degrees over a mandrel having a diameter as indicated on the individual specification sheets.

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4.8.9.3.2 Miscellaneous wires and braided conductors. - The miscellaneous wires (see 1.2.4) and the insulated and braided conductors removed from types MHFF (see 3.36), and MHFA (see 3.23) shall be wrapped three turns around a mandrel having a diameter as specified on the individual specification sheets, and shall then be subjected to a 60 cycle potential of 2,000 volts r.m.s. for 1 minute.

4.8.10 Flammability. -

4.8.10.1 Specimens. - Specimens shall consist of straight sections of completed wire or cable approximately 18 inches in length.

4.8.10.2 Apparatus. - An enclosure of sufficient size to contain the specimen, supports, heater coil, spark plugs, flame travel gage and their associated accessories shall be arranged to eliminate air drafts and permit clear view of the interior through shatterproof glass windows. Vent holes, distributed around the sides adjacent to the base, shall be provided to admit fresh air when an exhaust fan connected to the top of the enclosure is operated at a minimum suction just sufficient to carry off smoke and gases.

4.8.10.2.1 Supports shall be suitable for holding the specimens in a vertical position with an unsupported span not less than 14 inches in length. The lower end of the specimen shall be wrapped with varnished cambric in such a manner that gases released through this end are diverted toward the spark plug.

4.8.10.2.2 Heater coils shall consist of 7 turns of 0.102 inch diameter resistance wire conforming to grade E of Specification MIL-W-16970, space wound to 0.25 inch between centers. The inside diameter of the coil shall exceed the overall diameter of the specimen by not less than 0.5 inch nor more than 0.6 inch. The lower end of the heater coil shall be located 1.5 inches above the top of the lower specimen support.

4.8.10.2.3 Two spark plugs, with extended electrodes spaced 1/8 inch from the surface of the specimen and the plugs located on diametrically opposite sides of the specimen, shall be placed with their longitudinal centerlines in a horizontal plane 1/2 inch above the top of the heater coil, to ignite the gases emitted from the heated specimen. A suitable electric circuit shall be provided to maintain continuous sparking at the electrodes during the specified time. The spark plugs shall be mounted in such a manner that they may be moved away from the specimen after ignition takes place so as not to impede the travel of the flame and to prevent their electrodes from becoming fouled by soot.

4.8.10.2.4 A suitable flame travel gage shall be positioned near the specimen to judge the distance of flame travel without appreciable impeding progress of the flame.

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4.8.10.3 Procedure. - Flame tests are specified for the various types of wire and cable in section 3, and the required performance is shown in table XIV. The tests shall be so conducted as to measure, as accurately as practicable, the following factors:

Time of ignition.
Time of heating.
Distance of flame travel.
Time required for self-extinction.

Table XIV - Flammability limits.

Cable types	Ignition time	Burning time	Flame travel
	Minimum	Maximum	Maximum
	Seconds	Seconds	Inches
Cables with braided armor	30	60	2
MHFF, TTOP, MCOS, TTRS, DBSP, TBSP, FBSP, FCSF, SDU TRF, TRXF, FCOTP-4, MMOP, MCSC - Sizes 14 and larger of DCOP, TCOP, CVSF, JHOF, DHOF, THOF, FHOF	30	120	2
Sizes 9 and smaller of DCOP, TCOP, DHOF, THOF, FHOF MDGL, MCGC	No test required.		
SRI, SRIB, SHFS, MRI	25	60	2

4.8.10.3.1 Time of ignition. - With the enclosure closed and ventilated, the specimen centered in the heater coil, and the spark plugs, and flame gage properly located, a stop watch shall be started simultaneously with the engerizing of the heater coil and spark plugs. A constant current as specified in table XV shall be supplied from a suitable transformer source to the heater coil. Ignition shall be considered as occurring when the flame transfers from the escaping gases to the surface of the specimen and continues there, disregarding the flashes which may occur in the gassing space prior to the sustained flame. The time required to ignite the specimen shall be recorded.

4.8.10.3.2 Time of heating. - For all armored cable, the heating current shall be turned off 30 seconds after ignition occurs. When armored cable does not ignite in 250 seconds or less, the specimen is considered satisfactory and the test shall be discontinued. For all unarmored cables, where the ignition time is less than 60 seconds, the heating current shall be turned off 60 seconds after being turned on; where the ignition time is greater than 60 seconds the heating current shall be turned off when ignition occurs.

4.8.10.3.3 Distance of flame travel. - Immediately after ignition occurs, as defined in 4.8.10.3.1, the electrical supply to the spark plugs shall be cut off and the plugs shifted away from the flame. The maximum distance to which the flame travels along the surface of the specimen, measured from the top of the heater coil, before extinction, shall be noted and recorded.

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4.8.10.3.4 Time required for self-extinction. - The number of seconds that the specimen continues to burn, until the cessation of all flaming, after the current in the heater coil has been cut off, shall be recorded as the time required for self-extinction.

4.8.10.3.5 Observations. - The amount and color of the smoke and the amount and size of the soot particles shall be observed and recorded.

Table XV - Current in heating coil.

Nominal diameter of cable	Current amperes A.C.	Nominal diameter of cable	Current amperes A.C.
Inch	R.m.s.	Inches	R.m.s.
0.050 to 0.099	45	0.80 to 0.89	53
.100 to .19	46	.90 to .99	54
.20 to .29	47	1.00 to 1.09	55
.30 to .39	48	1.10 to 1.39	56
.40 to .49	49	1.40 to 1.79	57
.50 to .59	50	1.80 to 2.29	58
.60 to .69	51	2.30 to 3.00	59
.70 to .79	52		

4.8.10.4 Number of tests. - If a sample tested fails to comply with any of the limitations specified in table XIV as to flammability upon the first trial, 2 additional specimens shall be tested, and the results of all 3 trials averaged together.

4.8.11 Current overload. -

4.8.11.1 Specimens. - The specimen shall be a completed cable having a length of not less than 10 feet. The center portion of the specimen shall be bent 180 degrees about a mandrel having a diameter of 13 times the overall diameter of the cable, leaving the ends straight.

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4.8.11.2 Procedure. - The bent specimen shall be fastened to a vertical steel panel so that the straight sections of cable are horizontal. With all copper conductors connected in series, a 60-cycle alternating current for each type and size of cable as shown in table XVI shall be circulated for a continuous period of not less than 18 hours nor more than 24 hours. The ambient temperature shall be recorded, and reported when higher than 30°C.

Table XVI - Test currents.

Conductor size	Type SHFA	Type DHFA	Types FHFA and THFA	Type SDGA	Type MDGA
3, 4, 9	No tests required				
	Amperes per conductor				
14	115	100	85	---	---
23	160	130	115	---	---
30	180	160	140	---	---
40	205	185	160	---	---
50	240	215	185	---	---
60	---	245	215	---	---
75	315	285	250	---	---
100	375	335	295	---	---
125	---	385	340	---	---
150	500	455	400	---	---
200	580	525	460	---	---
250	---	610	535	---	---
300	750	700	590	---	---
350	---	---	665	---	---
400	925	835	740	825	---
500	1080	---	---	975	---
650	1260	---	---	1140	---
800	1510	---	---	1335	---
1000	---	---	---	1560	---
1300	---	---	---	1830	---
1600	---	---	---	2100	---
2000	---	---	---	2400	---
19(6)	---	---	---	---	24
19(14)	---	---	---	---	38
19(23)	---	---	---	---	51
19(40)	---	---	---	---	74

4.8.11.3 Observations. - There shall be no exudation of the cable sheath or of the cable impregnants through the armor during this test. The amount of compound, if any, that drips from the open cable ends, as well as the amount and color of any smoke that is evolved, the maximum temperature obtained from the sheath and the temperature at exudation, if any, shall be observed and recorded. (Exudation is defined as an actual flow or expansion of the sheath compound through the armor to the extent that the armor cuts through the sheath. Bulging of the sheath through the armor with no actual flow or cutting is not considered as exudation.)

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4.8.12 Heat endurance. -

4.8.12.1 General. - A heat endurance test shall be made on completed cables, for qualification tests only (see 4.7) when specified in section 3, to determine the effects of prolonged current loading. For purposes of this test, all specimens are designated as power cables, except type MHFF (see 3.36) and type MHFA (see 3.23) which are designated as control cables.

4.8.12.2 Specimens. -

4.8.12.2.1 Power cables. - For power cables there shall be 1 specimen of completed cable approximately 40 feet in length.

4.8.12.2.2 Control cables. - For control cables there shall be 1 specimen of completed cable approximately 40 feet in length.

4.8.12.3 Apparatus. - The testing apparatus shall consist of a test chamber and the necessary temperature-measuring and control equipment. The test chamber shall be essentially a low temperature oven equipped with racks for installing the cable specimens. It may be of any convenient size and constructed of wood and asbestos board or other suitable material. An ambient temperature of 50°C. shall be maintained in the interior of the chamber by utilizing the heat given off by the cable specimens, by additional heat supplied by resistance type heaters installed on the chamber floor, and by the admission of cool air to the chamber through adjustable vents in the top and bottom of the chamber. Convenient windows should be provided in the chamber walls through which observations of mercury thermometers indicating the interior temperature may be made. Suitable control equipment shall be provided to permit the copper temperature of the cables under test to be held to specified values. The temperature of the copper conductors shall be determined by means of thermocouples placed within the strands of one conductor through a small knife puncture 15 feet from one end. Additional thermocouples for determining the armor temperature shall be provided.

4.8.12.4 Procedure. -

4.8.12.4.1 Power cables. - Specimens shall be racked with thermocouples attached and the conductors of each specimen connected in series to an alternating current power supply adjusted to give a copper temperature of 125°C. This test shall continue for 400 hours.

4.8.12.4.2 Control cables. - The specimen shall be racked within the chamber with thermocouples attached in the manner specified for power cables. The thermocouple to be used for measuring copper temperature shall be in contact with one of the conductors in the inner layer. The conductors shall be connected in series to a power supply adjusted to give a copper temperature of 105°C. in the inner layer of conductors. The test shall continue for 400 hours.

4.8.12.5 Observations. - Readings of copper temperature, armor temperature, current per conductor, chamber temperature, and time shall be taken throughout the duration of the test. Gas or fumes developed from the samples and the oozing of compound from the cable ends or sheath material through the armor during the tests shall be observed and recorded for both power and control cables. At the conclusion of the heat run, the dielectric strength and insulation resistance of the specimens shall be determined, followed by examination and tests (see 4.8.1 to 4.8.3, incl.) of the components to determine the extent of changes during the heat run.

4.8.13 Water and air leakage. -

4.8.13.1 Specimen. - The specimen shall be a 30-inch section of completed cable with or without end seal. Armored specimens shall be tested with armor in place.

4.8.13.2 Apparatus. - The apparatus shall consist of a testing tank, of 15-inch length and suitable diameter to accommodate the test specimen, fitted with Navy standard watertight stuffing tubes with approved packing, connected to the pressure system, a pressure control and leakage detection device permitting test by either air or water under pressure, and necessary connections, gages and valves between testing tank and pressure control system.

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4.8.13.3 Procedure. - The specimen shall be sealed in place into or through the testing tank. Air pressure shall be applied to detect leakage through the cable sheath or end seal, and to enable location and correction of leaks through stuffing tubes and packing. The cable sheath or end seal and the gland shall not leak under this air pressure. After the assembly has withstood the air pressure for a period of 30 minutes without leakage, the specimen shall be subjected to water pressure at room temperature for a period of 4 hours. The amount of water leakage, if any, shall be determined to a precision of 1 cubic inch.

4.8.13.4 Pressures. - Except where otherwise specified in section 3, the air and water pressures applied shall be 75 pounds per square inch for cable sheath tests and 25 pounds per square inch for cable end seal tests.

NOTE: The water and air leakage test is intended primarily for armored cables and end seals (see 3.11.22), but is required in section 3 for certain unarmored cables as a means for checking the integrity of the sheath in instances where this feature appears doubtful. This test is required on type MCSC (see 3.30), but shall not be made on other unarmored cables unless visual inspection discloses a patch, thin spot, cut, gouge, pock-mark, blister, porosity or other defect in the sheath.

4.8.14 Flexing endurance. -

4.8.14.1 Specimens. -

4.8.14.1.1 Bending. - There shall be two bending test specimens, each a 30-inch section of completed cable.

4.8.14.1.2 Twisting. - There shall be two twisting test specimens of completed cable, 24 inches in length for cables of 5/8-inch diameter or less, and 10 times the cable diameter plus 18 inches in length for cables of larger than 5/8-inch diameter.

4.8.14.2 Apparatus. - The bending apparatus and the twisting apparatus may be built into a portable, compact assembly, powered by a single drive and equipped with means for counting the number of cycles impressed upon a specimen. A suitable electric circuit should be provided to stop the apparatus in the event the series conductor circuit of a specimen is broken.

4.8.14.3 Procedure. -

4.8.14.3.1 Conditioning. - Specimens shall be conditioned and tested in a controlled temperature room held at the specified temperature. Specimens for the bending test and for the twisting test shall be exposed to the specified temperature until thermal equilibrium is established before testing is started. All conductors in each specimen shall be connected in a single series circuit for the purpose of detecting breakage of the conductors.

4.8.14.3.2 Bending. - A specimen shall be loosely inserted between a pair of rollers and subjected to 90-degree bending in each direction at the rate of 12 to 14 complete 180-degree cycles per minute. The bends shall be made approximately midway along the length of the specimen, and the portion of the specimen not coming in contact with the rollers shall be loosely restricted to prevent appreciable bending at other points. The upper portion of the specimen shall be anchored to the bending apparatus. A weight which stresses the specimen in tension to approximately 40 pounds per square inch of cross section shall be attached to the free lower end. The grips or clamps shall be designed to apply uniform radial pressure

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to the core of the specimen. The diameter of the rollers shall be approximately but not less than 2-1/2 times the maximum overall diameter specified for the particular type and size undergoing test, except that when the maximum overall diameter specified is greater than 0.800 inch the appropriate roller diameter shall be selected from the following:

Cable type and size	Roller diameter Inches
MHFF-14, TTOP-15, DHOF-23	3
MHFF-19, DHOF-30, THOF-23	3-1/2
MHFF-24,	4
MHFF-30, THOF-42	4-1/2
MCSC-5	5
MHFF-37	6
MHFF-44, DHOF-83	7
(No test required for DHOF-250 and 400, SHOF all sizes, THOF-150, 250 and 400, FHOF-133.)	

4.8.14.3.3 Twisting. - Specimens shall be clamped in cable grips and subjected to 180 degrees of twist in each direction (360 degrees total twist) at the rate of 12 to 14 complete cycles per minute. The distance between grips shall be 6 inches plus 10 times the maximum specified diameter of the specimen. The upper grip shall be oscillated by the twisting apparatus. The lower grip shall be free to move vertically but restrained from oscillation, and have a weight attached which stresses the specimen in tension to approximately 40 pounds per square inch of cable section.

4.8.14.4 Duration of tests. -

4.8.14.4.1 Qualification tests. - The suitability of cables for repeated flexing service (see 1.2.2.1) shall be investigated by bending and twisting tests conducted for 6000 cycles at 75°C. and for 2000 cycles at -30°C., unless the specimen fails in some manner prior thereto. During all tests frequent visual inspection of the sheath shall be made for the purpose of detecting rupture. Upon completion of a test, the specimen shall be removed from the apparatus and checked for open or short circuits between conductors. The specimen shall then be opened, its internal condition examined, and record made of any evidence of damage caused by the test. If failure occurs within 2 inches of either grip, the results shall be disregarded and the test repeated.

4.8.14.4.2 Inspection tests at a Government laboratory. - Cables for repeated flexing service (see 1.2.2) having an overall diameter not greater than 1.500 inches shall be tested for the duration of twisting and bending cycles as required on the individual specification sheets. Flexing endurance tests will not be required in the inspection of cables larger than 1.500 inches in diameter. The specimens then shall be examined for weakening or failure of sheath and conductors and the nature of damage, if any, reported with the results. If failure occurs within 2 inches of either grip, the results shall be disregarded and the test repeated.

4.8.15 Impact resistance. -

4.8.15.1 Specimen. - The specimen shall be a 3-foot section of completed cable.

4.8.15.2 Procedure. - The specimen shall be securely fastened to a steel block and subjected to the blows of a 6-pound steel hammer having sides tapered at a 45-degree angle to a 1/16-inch radius striking edge. The steel block and hammer shall be connected to one side of an alternating current high potential test set and grounded, and all conductors in the specimen connected to the other side of this circuit. With 5000 volts impressed, the hammer shall be dropped from a height of 24 times the overall diameter of the specimen to one spot on the specimen at a rate of 12 blows per minute until potential breakdown occurs. For specimens having an overall diameter of 1.500 inches or less, the point of hammer impact shall then be moved at least 6 inches and the test repeated with 2500 volts applied with

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the hammer being dropped from a height of 48 times the overall diameter of the specimen. The tests shall be made at room temperature and the number of hammer falls to cause potential breakdown at each height determined and recorded. Upon completion of the hammer tests, the specimen shall be checked for short circuits between conductors and the cable then opened and examined for any damage caused by the tests.

4.8.16 Abrasion resistance. -

4.8.16.1 Specimen. - The specimen for cables of 1.250-inch diameter and smaller shall be a 2-foot length of completed cable. The specimen for cables larger than 1.250 inch diameter shall be a 2-foot strip of cable sheath wall of 1-inch width.

4.8.16.2 Procedure. - The specimen shall be hung with a 90-degree arc of contact over a 7-inch diameter abrasion wheel made up of twelve 3/8-inch diameter fine grade tungsten carbide rods and two 7-inch diameter phenolic disks arranged in the form of a squirrel cage. A 15-pound weight should be attached to the free end of the specimen. Specimens of the sheath strip from the larger cable sizes shall be provided with a backing of soft sheet metal. The tests shall be made at room temperature with the abrasion wheel rotating at 60 revolutions per minute (r.p.m.) in the direction which tends to lift the 15-pound weight. The number of wheel revolutions required to wear through a measured thickness of the sheath wall shall be determined and recorded.

4.8.17 Hot oil test. -

4.8.17.1 Specimens. - Specimens shall be duplicates of those prepared for the tensile tests (see 4.6.1).

4.8.17.2 Procedure. - Specimens shall be immersed in lubricating oil, conforming to Navy Symbol No. 3100 of Pamphlet N.B.S. 431, or equal, at a temperature of 120°C. After immersion for 18 hours the specimen shall be removed from the hot oil, blotted lightly to remove excess oil, and suspended in air at room temperature for not less than 3-1/2 hours nor more than 4-1/2 hours. After this conditioning, determinations shall be made of tensile strength based on the original cross section before immersion and of the ultimate elongation based upon the length between bench marks as measured before immersion.

4.8.18 Oil, water and chemical resistance. -

4.8.18.1 Specimens. - Specimens shall be strips of cable sheath suitable for subsequent measurement of tensile strength and elongation, and shall be buffed on the inner surface to a uniform thickness of 0.040 inch.

4.8.18.2 Procedure. - Specimens shall be weighed and surface area measured before immersion. Five specimens shall then be immersed at room temperature for a period of 10 days in each of the following liquids: Distilled water, sea water, storage battery electrolyte, lubricating oil, fuel oil and gasoline. After removal from the liquids, each specimen shall be wiped clean and again weighed. The change in weight of the specimens shall be expressed in terms of weight per unit initial area of surface exposed to the liquids. The specimens shall then be tested to determine change in tensile strength and elongation.

4.8.19 Dielectric tests. -

4.8.19.1 General. - Unless otherwise specified herein, the dielectric tests shall be made on full coil or reel lengths of completed wire or cable to be shipped. Where tests are to be made on short lengths, as required in section 3 for the particular type and size of cable under consideration, these tests shall be made at a Government laboratory only as part of the regular production check tests (see 4.6).

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4.8.19.2 Apparatus. -

4.8.19.2.1 High potential test. - Except for spark testing, the dielectric tests shall be made with alternating potential from a source of ample capacity, and in no case less than 5 kilovolt-ampere, having a frequency not greater than 100 cycles per second and a wave shape approximately as closely as practicable to a sine curve under all test conditions. The testing voltage may be measured by any method satisfactory to the bureau or agency concerned which gives a root-mean-square value, preferably by means of voltmeter connected to voltmeter coil in the high tension winding of the testing transformer or to a separate instrument transformer. A voltmeter on the low-tension side of the transformer is satisfactory if the ratio of transformation does not change under any test condition.

4.8.19.2.2 Spark test. - The power capacity of the sparking equipment shall be as high as practicable and sufficient, to maintain the required testing voltage under all normal load conditions. Means shall be provided for regulating the testing voltage to within 5 percent of the specified value at all times. The testing voltage may be measured by a kilovoltmeter or electrostatic voltmeter having a burden that will not affect the value or wave shape of the testing voltage. An indicating voltmeter connected to the low-tension side of the transformer may be used, provided its readings are calibrated to give a true indication of the voltage across the dielectric of the wire being tested. The equipment shall include a fault-signal system that will automatically stop the machine, and give a visible signal until manually reset. The electrode shall be of a suitable bead chain or fine link mesh construction, and the finished cable or insulated conductor shall be passed through the electrode at a speed such that the insulation will be subjected to the specified test voltage for a period of time not less than 0.2 second, and which shall locate all faults.

4.8.19.3 Rate. - Except for spark testing, the initially applied voltage shall be not greater than 600 volts, and the rate of increase shall be approximately uniform and not over 100 percent in 10 seconds nor less than 100 percent in 60 seconds.

4.8.19.4 Temperature. - Dielectric tests shall be made at normal room temperature.

4.8.19.5 Time of application. -

4.8.19.5.1 Shipping lengths. - Except for spark-tested lengths, the time of voltage application measured from the time the specified voltage is attained shall be not less than the following:

Each conductor against all adjacent conductors and ground:
5 minutes for cables having 4 conductors or less (see 1.2.1 and 1.2.2).
1 minute for cables having 5 conductors or more (see 1.2.1 and 1.2.2).
1 minute for miscellaneous wires (see 1.2.4).

4.8.19.5.2 Spark tested lengths. - Where a spark test is required in section 3 for miscellaneous wires (see 1.2.4), the completed wire shall be passed through the spark electrode at a speed such that the insulation will be subjected to the specified test voltage for a period of time not less than 0.2 second, and which will locate all faults.

4.8.19.5.3 Short lengths. - The time of voltage application, measured from the time the specified voltage is attained, shall be approximately but not less than 1 minute for each conductor against all adjacent conductors connected to ground, and all conductors connected together against ground.

4.8.19.6 Ground electrodes. -

4.8.19.6.1 General. - The following methods for grounding shall be employed while testing electrically. The method of grounding is determined by the cable construction and the nature of the requirements specified in section 3.

4.8.19.6.2 Cable coverings. - In the case of lead sheathed cables or cables with braided metal armor, the lead or armor shall constitute the ground electrode.

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4.8.19.6.3 Water bath. - All unarmored cables except miscellaneous wires types SRIB and SHFS, shall be immersed in a water bath for not less than 1 hour and then tested while still immersed. The water bath shall constitute the ground electrode. When testing types MCOS, DBSP, TBSP, FBSP, and TTRS the shield shall be connected to ground while the insulated conductors are being tested, then the insulated conductors shall be connected together to the shield and potential applied from shield to ground or water bath.

4.8.19.6.4 Spark electrodes. - The copper conductor shall be grounded, and potential applied to the outer surface of the completed wire by a suitable bead chain or fine link mesh device that will insure intimate metallic contact over practically the entire surface of the wire under test. A pipe, helical coil spring, and similar electrodes are not considered to be of an intimate type.

4.8.19.6.5 Electrodes for short lengths. - Where dielectric tests are specified in section 3 on a short length specimen in the as received condition after cold bend (see 4.8.9), or after the flame test (see 4.8.10), the tests shall be made with the external surface of the specimen grounded. Specimens having no lead sheath or braided metal armor shall have the external surface grounded by immersion in water or mercury, by wrapping with metal foil, or by a coating of sprayed metal.

4.8.20 Creepage. -

4.8.20.1 General. - The apparatus used for application and measurement of the test voltage shall conform to 4.8.19.

4.8.20.2 Specimen. - The specimen shall be an 8-inch length of completed wire whose cut ends shall be sealed with an insulating wax of melting point not lower than 65°C.

4.8.20.3 Conditioning. - The specimen shall be conditioned at a temperature of 50°C. and a relative humidity of 90 percent for a continuous period of 18 to 24 hours immediately prior to test.

4.8.20.4 Procedure. - A dielectric test of 5,000 volts alternating current shall be applied for a period of 1 minute between 2 ring type electrodes (narrow tapes of thin copper) spaced at a 2-inch separation along the central portion of the specimen. The test shall be conducted with the specimen in the conditioning chamber at 50°C. and 90 percent relative humidity. There shall be no evidence of breakdown along the braid surface (arcing, smoking, burning). After the voltage test, the surface resistance of the 2-inch section between electrodes shall be measured with a d. c. potential of 200 to 300 volts and the observed value in megohms, multiplied by the overall diameter of the specimen in inches, shall be not less than 20.

4.8.21 Tensile strength and elongation, conductors. - While physical tests need not be made on each lot of copper conductors submitted for delivery, the Government inspector shall require tensile tests at such frequent intervals as may be necessary to ascertain that the physical properties of the conductors are in conformance with 3.10 and 3.11.1. All conductors shall, however, be given a careful visual and dimensional inspection as specified in 4.4.

4.8.22 Coating, metallic. -

4.8.22.1 Specimens. - Specimens shall be coated wire of approximately 6 inches, but not less than 4-1/2 inches, in length.

4.8.22.2 Preparation. - Specimens shall be thoroughly cleaned by immersion in benzine, ether, alcohol or carbon tetrachloride for at least 3 minutes and shall then be removed and wiped dry with a clean, soft cloth.

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4.8.22.3 Solutions. -

4.8.22.3.1 Hydrochloric acid. - The hydrochloric acid solution shall have a specific gravity of 1.088 at 15.5°C. During the test, the solution shall be maintained at a temperature between 15.5°C. and 21°C. A portion of the hydrochloric acid solution having a volume of 180 milliliters shall be considered to be exhausted when 2 cycles have been completed on the numbers of specimens given in table XVII.

Table XVII - Number of specimens to be tested for 2 cycles in 180 ml HCL.

Diameter of wire	Maximum number of specimens
Inch	
0.0850 to 0.0501	6
.0500 to .0381	10
.0380 to .0301	12
.0300 to .0030	14

4.8.22.3.2 Sodium polysulphide. - The sodium polysulphide solution shall have a specific gravity of 1.142 at a temperature of 15.5°C. During the test, tne solution shall be maintained a a temperature between 15.5°C. and 21°C. The sodium polysulphide solution shall contain an excess of sulphur and have sufficient strength to thoroughly blacken a piece of clean bare copper wire in 5 seconds.

4.8.22.4 Procedure. - The cleaned specimen shall be immersed in the hydrochloric acid solution for 1 minute. It shall then be rinsed in clear water, wiped and dried and immersed in the sodium polysulphide solution for 30 seconds and again rinsed and dried. These cycles of operations shall be repeated and if the specimen does not become blackened after the second cycle of immersion, the coating shall be regarded as satisfactory.

4.8.23 Tensile strength, elongation, set and aging, insulation and sheath. - The preparation of specimens and methods of conducting tests shall be as specified in 4.8.1.

4.8.24 Conductor resistance. - The resistance of the insulated conductor shall be measured in all completed wires and cables to determine conformance with the requirements of the individual specification sheets. For stranded conductors, the resistance of individual strand wires need not be measured unless the Government inspector has reason to doubt the conductivity.

4.8.25 Insulation resistance. - The apparent insulation resistance shall be measured as soon as practicable after the dielectric test, and shall be not less than the values specified. The leakage current shall be measured after a 1-minute electrification with a direct current potential of not less than 200 nor more than 500 volts. The conductor shall be maintained negative to the sheath, armor, or water. For multiple conductor cables, the insulation resistance of each conductor against all other conductors connected to ground shall be not less than the specified value. Insulation resistance shall be referred to a temperature of 15.5°C. by multiplying the measured value of insulation resistance by a conversion factor corresponding to the temperature of the measurement and the type of cable. These conversion

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factors are shown in table XVIII. For temperature values between those tabulated, the conversion factor shall be obtained by interpolation. Where the values in table XVIII are not applicable to approved cable constructions, other conversion factors may be used when specifically authorized.

Table XVIII - Temperature conversion factors.

Temperatures		Temperature coefficients	
Degrees F.	Degrees C.	Resin insulated cables (see 1.2.1) Flexible cables (see 1.2.2) Thin-walled cables (see 1.2.3) Miscellaneous wires (see 1.2.4)	Varnished cambric insulated cables (see 1.2.1)
32	0.0	0.032	0.085
35	1.7	.047	.11
38	3.3	.071	.15
41	5.0	.098	.18
44	6.7	.14	.26
47	8.3	.25	.32
50	10.0	.29	.42
53	11.7	.45	.54
56	13.3	.78	.71
59	15.0	.91	.89
60	15.5	1.00	1.00
62	16.7	1.26	1.20
65	18.3	1.80	1.50
68	20.0	2.50	2.00
71	21.7	3.50	2.45
74	23.3	5.00	3.10
77	25.0	6.60	4.00
80	26.7	8.30	5.15
83	28.3	11.0	6.20
86	30.0	17.5	8.00
89	31.7	24.5	10.0
92	33.3	34.0	12.0
95	35.0	48.0	16.0

4.8.26 Flame retardant test. - Various component materials employed in shipboard cables are specified in section 3 to require moisture and flame retardant characteristics. The materials used shall be in conformance with those employed in the approved sample, or otherwise specifically approved.

4.8.27 Capacitance and characteristic impedance (shielded pairs). -

4.8.27.1 Specimens of pair or cable for test. -

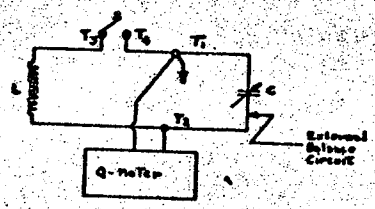
4.8.27.1.1 Length. - Measurements of capacitance and characteristic impedance shall be made upon a specimen 5 feet in length. Measurement shall be checked on a specimen 10 feet in length.

4.8.27.1.2 Preparation of specimen. - The shield shall be removed for a distance of 1 inch from both ends of the pair being tested. Each conductor shall be bared for 1/2 inch from each end. The length of the specimen is considered to be the shielded length.

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4.8.27.1.3 Conditioning. - Measurements shall be made at 25 ± 2.5°C. and prevailing humidity. If a refrigerator, oven or other small chamber is used for conditioning, leads from the specimen to the terminals of the measuring instrument shall be 4000 circular mils or larger wire of the shortest practicable length. Due consideration should be given to the effect of internal capacitances in the instrument and of interlead capacitance and inductance upon the accuracy of the determinations.

4.8.27.2 Test equipment. - The following procedure is specified for a Q-meter and suitable oscillator equipped with the external balance circuit schematically shown below. The general procedure can be adapted to an alternate method of measurement, where it can be demonstrated that such an alternate will yield results of at least equal accuracy and precision.



L: Coil, of inductance suitable for the magnitude of the reactances to be measured.
C: Variable calibrated precision condenser.
T_1, T_2: Terminals of Q-meter, T_1 grounded.
T_3, T_4: Terminals in the inductive branch of the balance circuit (T_4 can be identical with the terminal T_1).
S: Means for shorting T_3 and T_4.

The leads from L and C to the Q-meter terminals shall be short as practicable. The test specimen, or leads from the test specimen, shall be terminated at, or as near as possible to, the terminals of L.

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4.8.27.3 Determination of capacitance. - The shield shall be grounded at both ends for these measurements. The capacitance shall be measured between the inner conductors of each pair. Designate one conductor as No. 1, the other as No. 2 and the shield as No. 3.

(a) Short terminals T_3 and T_4.
(b) Balance the circuit to resonance by adjusting C to a value C_o, at which point balance (resonance) will be indicated by a maximum reading on the Q-meter.
(c) Connect No. 1 to No. 3, terminate 1-3 at T_1, and terminate 2 at T_2. Balance the circuit by adjusting C to a new value C_1. Designate the capacitance $(C_o - C_1)$ between No. 2 and No. 1-3 by the symbol C_a. Disconnect No. 1 from No. 3.
(d) Connect No. 2 to No. 3, terminate No. 2-3 at T_1, and terminate No. 1 at T_2. Balance the circuit by adjusting C to new value C_2. Designate the capacitance $(C_o - C_2)$ between No. 1 and No. 2-3 by the symbol C_b. Disconnect No. 2 from No. 3.
(e) Connect No. 1 to No. 2, terminate No. 3 at T_1 and terminate No. 1-2 at T_2. Balance the circuit by adjusting C to a new value C_3. Designate the capacitance $C_o - C_3$ between No. 1-2 and No. 3 by the symbol C_c. Disconnect No. 1 from No. 2.
(f) The capacitance (C_p) between the two inner conductors is equal to $\frac{2(C_a + C_b) - C_c}{4}$.
(g) The capacitance per foot is the above value of capacitance divided by the length of the sample in feet (see 4.8.27.1.1).

4.8.27.4 Determination of characteristic impedance (surge impedance). - These measurements shall be made upon the same pair or specimen upon which the capacitance measurements were made.

4.8.27.4.1 Measurement of inductance (short circuit impedance Z_{sc}). -

(a) Short T_3 and T_4, and balance the circuit by adjusting C to a value C_o.
(b) Open the circuit at T_3 and T_4, connect the inner conductors of the specimen at the far end, and terminate the conductors at T_3 and T_4. Ground the shield to T_4.
(c) Balance the circuit by adjusting C to value C_1.
(d) Since the inductance of the inductive branch will be increased by an amount dL due to connecting the specimen in series with L, there will be a corresponding change dC (equal to $C_o - C_1$) in the adjustment of the capacitance C necessary to regain balance. The inductance of the specimen (L_p) can be calculated from the formula:

$$L_p = dL = dC/W^2C_o(C_o - dC) \text{ where } W = 2\pi \text{ times the frequency.}$$

If dC is 1 percent or less of C_o, the formula can be reduced to:

$$L_p = dL = \frac{dC}{W^2C_o^2}$$

4.8.27.4.2 Measurement of capacitance (open circuit impedance - Z_{oc}). - Capacitance, C_p shall be measured by the method specified in 4.8.27.3.

4.8.27.4.3 The characteristic impedance, Z_o, can be obtained from either of the formulas:

$$Z_o = \sqrt{\frac{L_p}{C_p}}$$

or

$$Z_o = \sqrt{Z_{oc}\,Z_{sc}}$$

Where: $Z_{oc} = \frac{1}{WC_p}$

$$Z_{sc} = WL_p$$

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4.8.28 Breaking strength.-

4.8.28.1 Completed cable.-

4.8.28.1.1 Specimens.- Specimens shall consist of straight sections of completed type MCSC cable approximately 6 feet in length.

4.8.28.1.2 Apparatus.- The testing machine shall be power-driven. The applied tension shall be indicated on a dial or scale having an accuracy within plus or minus one percent, and the indicator shall remain at the point of maximum load after rupture of the test specimen. A knurled metal mandrel, not less than 3 inches in diameter, shall be fitted into each jaw of the testing machine.

4.8.28.1.3 Procedure.- One end of the specimen shall be wrapped two full turns around the upper mandrel and secured in place so as to provide sufficient snubbing action to prevent slipping, and the other end secured in a similar manner to the lower mandrel. The distance between the jaws with the specimen in place shall be not less than 6 inches. The jaws shall be separated at a uniform rate of approximately one inch per minute until the specimen breaks.

4.8.28.1.4 Number of tests.- If the first specimen representing the lot is rejected, samples from two other reel lengths shall be taken to represent the lot and, if either of these two samples fails to meet the specified breaking strength, the entire lot shall be rejected. Breaks which occur within one inch of either mandrel shall be considered not indicative of the true breaking strength of the specimen.

4.8.28.2 Tie cord.- The apparatus and procedure shall be as specified in 4.8.28.1 except that specimens shall consist of lengths of tie cord removed from the completed cable and the rate of separation between the jaws of the testing machine shall be approximately 12 inches per minute.

4.8.29 Induced voltage, completed cable.-

4.8.29.1 Specimen.- This test applies to completed cable types MCOS-5 and MCOS-6 only. The specimen for production check tests shall be notless than 20 feet in length, and the specimen for lot acceptance tests shall be 20 feet or more in length.

4.8.29.2 Preparation of specimen.- The insulation shall be removed from 1/2 inch of both ends of all conductors. The braided copper shield shall be removed or pushed back to expose 1-1/2 inch of the twisted pair at both ends. The length of braided copper shield is considered to be the length of the specimen. Both ends of each shielded pair shall be terminated in a 150-ohm resistive load. The braided copper shield or shields shall be grounded at one end only.

4.8.29.3 Procedure.-

4.8.29.3.1 Type MCOS-5.- A 60-cycle-per-second current of one ampere r.m.s. shall be circulated; first, in the red and green conductors, then in the red and blue conductors, then in the green and blue conductors. The induced voltage appearing between conductors at the open end of the shielded pair shall be measured to an accuracy of one microvolt while the current is flowing.

4.8.29.3.2 Type MCOS-6.- A 60-cycle-per-second current of one ampere r.m.s. shall be circulated in the yellow and green conductors. The induced voltage appearing between conductors at the open end of each shielded pair shall be measured to an accuracy of one microvolt while the current is flowing.

4.8.29.4 Requirements.- The maximum open-circuit potential induced between the conductors of a shielded pair shall not exceed the value specified in 3.38.

4.8.30 Heat shock.- A specimen of the completed cable shall be wound for six adjacent turns about a mandrel not greater than 3 times the diameter of the specimen. The wound specimen shall be placed in an oven and subjected to air at a temperature of 121°C. for one hour and, after cooling, shall withstand while immersed in water at room temperature the dielectric strength tests shown in the applicable table.

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4.8.31 Hydrostatic test.- Samples shall be selected and tested as specified in Appendix I of this specification, except that special double ended bulkhead stuffing tubes shall be used in lieu of the non-constrictive method of Appendix I, and water pressure shall be applied for 2 hours at 500 pounds per square inch in lieu of 18 hours at 25 pounds per square inch.

4.9 Test reports.- Laboratory reports covering the production check tests (performed at Government laboratory) shall be forwarded directly to the cognizant Government inspector for his action. Two copies of each test report shall be addressed to the bureau or agency concerned.

5. PREPARATION FOR DELIVERY

5.1 Packaging.-

5.1.1 Cable shall be wound in coils or on reels as follows:

5.1.1.1 Coils.- Coils shall each contain one continuous length of cable (see 6.2). The following cable types, up to and including the sizes shown, shall be furnished in coils:

SRI-14	DCOP-2	DHOF-4
SRIB-14	TCOP-2	THOF-4
SHFS-14	FBSP	FHOF-3
MRI-T2-1/2	FCOTP-4	TSP-11
DBSP	MMOP-5	SHOF-3
TBSP		

Both ends of each coil, except for miscellaneous wires as specified in 1.2.4, shall be hermetically sealed as specified in 3.11.22 to prevent the entrance of moisture. Individual coils shall be securely tied, wrapped spirally with one layer of 30-30 waterproof creped paper, a second layer of waterproof string reinforced paper or cloth-backed paper, and securely tied with a minimum of four individual ties. As an alternate wrapping, an inner layer of waterproof-creped paper and a single outer wrap of 30-45 water-resistant, neutral, synthetic resin impregnated and laminated creped paper may be used.

5.1.1.2 Reels.- Reels shall each contain one continuous length of cable (see 6.2). Sizes of cable larger than those specified in 5.1.1.1 shall be wound on reels. Where the gross weight of the reel and cable is more than 1,000 pounds, the reels shall have suitable metal hubs. Reels shall be of suitable design strongly constructed with flanges of sufficient diameter to protect the cable from injury in handling and rolling. The diameter of the drum shall be not less than 14 times the diameter of the cable for non-flexing service and not less than 8 times the diameter of the cable for repeated flexing service. The cable on a reel shall have both ends brought out so that the cable can be readily tested without unreeling. These ends shall be hermetically sealed, in accordance with 3.11.22 to prevent the entrance of moisture and the inner or drum end shall be fully protected from damage during the shipment by a suitable metal shield. Both ends of the cable shall be fastened securely to the reel flange with 1/4-inch rope, applied with two half-hitches, one 6 inches and the other 4 inches approximately back of the end seal, then twisting the loose rope ends together, knotting them, pulling the cable up tight, and stapling the rope to the flange with at least two 1-1/2 inch staples driven into the flange.

5.1.1.2.1 Lagging.- The cable wound on reels shall be covered with flexible waterproof barrier-material secured in a suitable manner to the flange rims. The material used shall conform to Specification JAN-P-125. Reels shall be completely enclosed with wooden lagging with boards touching each other. Lagging boards shall be the same thickness as the outside wooden reel flanges but not thicker than 2-inch commercial lumber and shall be securely nailed to the outside wooden flanges of the reel with cement-coated nails. On reels 36 inches in diameter and larger, lagging shall extend within 1/4 inch of the outside edges of the flanges. On reels smaller than 36 inches in diameter lagging shall be flush with the outside edges of the flanges. Lagging of reels of cable weighing 1000 pounds or more shall be strapped with two 0.1620 ± 0.003 inch (8 gage) high-tensile round wires or two 3/4 by 0.035-inch steel straps. Lagging of reels of cable weighing less than 1000 pounds shall be strapped with two 0.0915 ± 0.003-inch (13 gage) high-tensile round wires or two 5/8 by 0.020-inch steel straps. All straps shall be stapled at intervals of approximately 15 inches. Lagging on metal flanges shall be secured by notching

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lagging of suitable thickness over the flanges and by steel strapping. A minimum clearance of 2 inches, but not less than 1 cable diameter, shall be provided between the inner face of the lagging and the outer layer of cable.

5.1.1.2.2 Plywood casing.- Plywood casing may be used in lieu of lagging but shall be limited to reels smaller than 42 inches in diameter and which contain less than 1000 pounds of cable. Plywood casing shall be applied as follows:

A cushioning layer of asphalt impregnated corrugated fiberboard or its equivalent shall be applied directly over the cable. Two layers of 3-ply water-resistant toxic-treated plywood shall be secured tightly in place between the reel flanges directly over the cushioning material with not less than two 3/8 by 0.020-inch steel straps.

5.2 Packing.-

5.2.1 For domestic shipment.-

5.2.1.1 Cable in coils packaged as specified in 5.1.1.1 weighing up to approximately 50 pounds shall be packed in snug fitting wood cleated fiberboard, cleated plywood, nailed wood, corrugated or solid fiberboard boxes, conforming to Specification NN-B-591, NN-B-601, NN-B-621, LLL-B-631 or LLL-B-636, respectively. Fiberboard boxes shall conform to the special requirements of the applicable box specification. Closure of the fiberboard boxes shall be made with adhesive, metal fastenings, tape, or a combination of these methods, in such a manner that the closures of the boxes shall not break or open when the box is tested in accordance with the applicable container specification. The gross weight of wood boxes shall not exceed 150 pounds unless the weight of the individual item exceeds this amount. Fiberboard boxes shall not exceed the weight limitations of the applicable box specification.

5.2.2 For overseas shipment.-

5.2.2.1 Cable in coils packaged as specified in 5.1.1.1 weighing up to approximately 50 pounds shall be packed in cleated plywood, nailed wood, or wirebound boxes conforming to Specifications JAN-P-105, JAN-P-106, and MIL-B-107, respectively. The gross weight shall not exceed approximately 150 pounds.

5.2.2.2 Cable in coils weighing over approximately 50 pounds, packaged as specified in 5.1.1.1 will require no further packing.

5.2.2.3 Cable wound on reels as specified in 5.1.1.2, will require no further packing.

5.2.3 For domestic or overseas shipment.- Where practicable, shipping containers of similar construction shall be of uniform size, and shall contain the same number of items of one size and type. Containers shall be designed to fit the contents in a compact manner.

5.3 Marking.-

5.3.1 Coils.- Two shipping tags shall be securely attached to each coil, both inside and outside the wrapping, and marked with the following information completed:

Material
Type
Size
Quantity
Contract No.
Contractor
Manufacturer
Gross weight

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5.3.2 Reels.- Each reel shall be plainly marked on both flanges with the following information completed:

Material
Reel No.
Type
Size
Quantity
Contract No.
Contractor
Manufacturer
Gross weight

Where practicable, metal tags, stencils, or paper labels containing the information requested above shall be securely attached to reels. Where paper labels are used, they shall be securely attached and protected by a transparent compound to prevent deterioration of the markings.

5.3.3 Shipping containers.- In addition to any special marking required by the contract or order, shipments shall be marked in accordance with Standard MIL-STD-129.

5.3.4 Special marking.-

5.3.4.1 Sheath material.- All coils, reels, or shipping containers for cables for repeated flexing service (see 1.2.2) shall be marked with the phrase "Thermosetting sheath" or "Thermoplastic sheath", as applicable.

5.3.4.2 Yearly marking.- In addition to the marking specified in 5.3.1 to 5.3.4.1, inclusive, all shipping containers shall be marked with a keyed series of colors to indicate the year of manufacture. This marking shall consist of a stripe approximately 2 inches wide and colored as follows for the particular year of manufacture. This cycle of colors shall be repeated every fifth year.

Year of manufacture	Identifying color
1946	Red
1947	Green
1948	Orange
1949	Blue
1950	White
1951	Red
1952	Green
1953	Orange
1954	Blue

5.3.4.2.1 Coils.- On coils wrapped for shipment the colored stripe shall be applied in at least 2 locations spaced equidistant around the coil. At each location the stripe shall be applied around the cross-section of the coil and each marking shall cover not less than the outer half of the periphery of the cross-section.

5.3.4.2.2 Reels.- On reels the colored stripe shall be applied circumferentially over the lagging and midway between the flanges. The stripe shall consist of one coat, commercial quality, outside paint of appropriate color. In addition to the stripe, both flanges of all reels shall be stenciled with 4-inch high figures to show the year of manufacture.

5.3.4.2.3 Shipping containers.- When the color-identified wrapped coils are packed more than one in a container, such containers also shall be marked with the colored stripe on one side and one end.

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6. NOTES

6.1 Intended use. - Ratings, characteristics and applications for cables covered by this specification are specified in Section S62-2 of General Specifications for Machinery.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and size of cable required (see 1.2 and 1.3).
(c) Whether domestic or overseas shipment is required (see 5.2).
(d) "In order to compensate for handling and probable loss on issue and for reasons specified in 4.4.1 remnant lengths shall be subject to the price reductions given in table XX. Scrap lengths are not acceptable.
(e) "Time required. - The length of time normally required for conducting inspection tests on completed cable will be approximately 21 days from the time the necessary samples are forwarded, by the Government inspector to the Government laboratory, to the date the results will be reported to the inspector. All bidders shall take this time interval into account and due allowance should be made for it in the computation of the bidder's delivery dates. If a longer time than 21 days is actually required for conducting such laboratory inspection tests, the time taken in excess of 21 days will not be charged against the contractor. It should be noted, however, that the foregoing estimate is based upon the assumption that only one series of tests will be required for a given set of samples (i.e., that the results of the first series of tests are satisfactory); where rejections are found necessary, the Government shall be considered as responsible for only that time, in excess of 21 days, required for the first series of tests."

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6.2.1 Quantity. - The quantity of each type and size of wire or cable on a contract or order should be specified as an integral multiple of the unit ordering length given in table XIX for reels; the unit ordering length for wire or cable coils (see 5.1.1.1) is 500 feet.

Table XIX - Unit ordering lengths; reels.

Designation		Feet	Designation		Feet	Designation		Feet
Types	Sizes	Nominal	Types	Sizes	Nominal	Types	Sizes	Nominal
	3	2500	TTRS	2	1500		6	2000
	4	2500		4	1500		9	1500
	9	2500	TTRSA	6	1500		14	1500
	14	2500		8	1500		23	1000
	23	2500		10	1500		30	1000
	30	2000		12	1500	DHOF	83	1000
SHFA	40	2000		16	1500		250	500
SHFP	50	1500					400	500
SHFR	75	1500				DSS	2	1000
	100	1500					3	1000
	150	1000				TSS	4	1000
	200	1000				FSS	2	1000
	300	1000				MSS	6	1000
	400	1000		2	2000			
	500	1000		4	2000		6	2000
	650	1000		7	1500		9	1500
	800	1000	MHFA	10	1500		14	1500
	3	2500	MHFF	14	1000		23	1000
	4	2000		19	1000		42	1000
	9	2000		24	1000	THOF	150	1000
	14	2000		30	1000		250	500
	23	2000		37	1000		400	500
	30	1500		44	1000			
	40	1500		1-1/2	2500		23	2000
DHFA	50	1500		3	2500		60	1000
DHFR	60	1000		5	2500		150	1000
	75	1000		10	2500		200	1000
	100	1000	TTHFWA	15	1500	SHOF	250	1000
	125	1000	TTOP	20	1500		800	500
	150	1000		30	1000			
	200	800		40	1000		400	1000
	250	800		50	1000		500	1000
	300	800		60	1000		650	1000
	400	800		26	1000	SDGA	800	1000
	3	2500	SHFS	42	1000		1000	1000
	4	2000		66	1000		1300	1000
	9	2000		100	1000		1600	500
THFA	14	2000					2000	500
THFR	23	2000		26	1500			
FHFA	30	1500	SRI	42	1500		19(6)	1000
	40	1500	SRIB	66	1500	MDGA	19(14)	1000
	50	1500		100	1500	MDGI	19(23)	1000
	60	1000	FHOF	4	2000		19(40)	1000
	75	1000		9	1500			
	100	1000		133	1000		84	1000
	125	1000	MDGB	12(1-1/2)	1500	TRF	105	1000
	150	1000				TRXF	133	1000
	200	800	MCOS	2	2000		168	1000
	250	800		4	2000			
	300	800		5	2000			
	350	800		6	2000			
	400	800		7	2000			
CVSF	8	1000						
MCSC	5	[1] 1800						
PBJX	4	2000						
PBTX	4	2000						
FCSF	220	[2] 1600						
FCSF	66	[2] 1600						
MCGC	8	[3] 1000						
SDU	500	1000						
SDU	800	700						

[1] See 3.30
[2] See 3.32
[3] See 3.33

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6.2.2 Tabulation.- The range of standard, random, remnant and scrap lengths for each nominal length is shown in table XX.

Table XX - Cable lengths.

Type of package	Nominal length	Standard lengths	Random lengths	Remnant lengths		Scrap lengths
		No price reduction	No price reduction	Price reduction		Not acceptable
				5 percent	10 percent	
	Feet	Feet	Feet	Feet	Feet	Feet
Coil	500	550 to 450	449 to 150	149 to 100	99 to 50	49 to 0
Reel	500	550 to 450	449 to 150	149 to 100	159 to 80	49 to 0
Reel	800	880 to 720	719 to 240	239 to 160	159 to 80	79 to 0
Reel	1000	1100 to 900	899 to 300	299 to 200	199 to 100	99 to 0
Reel	1500	1650 to 1350	1349 to 450	449 to 300	299 to 150	149 to 0
Reel	2000	2200 to 1800	1799 to 600	599 to 400	399 to 200	199 to 0
Reel	2500	2750 to 2250	2249 to 750	749 to 500	499 to 250	249 to 0
Reel	3000	3300 to 2700	2699 to 900	899 to 600	599 to 300	299 to 0

6.3 In the procurement of products requiring qualification, the right is reserved to reject bids on products that have not been subjected to the required tests and found satisfactory for inclusion on the Military Qualified Products List. The attention of suppliers is called to this requirement, and manufacturers are urged to communicate with the Bureau of Ships, Navy Department, Washington 25, D.C., and arrange to have the products that they propose to offer to the Army, the Navy or the Air Force, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products covered by this specification may be obtained from the Chief of the Bureau of Ships, Navy Department, Washington 25, D.C.

6.4 Length of samples.- The samples required for production check tests at a Government laboratory (see 4.6) are in addition to the lengths stated in the contract or order, and should be included in the contract price. Except where otherwise specified, the length of cable required for these tests should be as specified in 4.3.2.4. When a reel or coil of cable becomes a remnant length by reason of the removal therefrom of samples selected by the Government inspector, no price reduction will be applicable to such remnant length (see 6.2.2).

6.5 Superseding documents.- This specification supersedes MIL-C-915. The latter superseded Navy Specification 15C1j.

Notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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FHOF	3.40	55
FSS	3.26	40
MCSC	3.30	45
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MCGC	3.33	48
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THFA	3.14	28
THFR	3.19	33
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APPENDIX I

10. The manufacturer shall endeavor to fill all watertight cable, as specified in 3.6, so that no water can be forced into the cable core, and so that the electrical characteristics of the cable are not affected or are affected to a minimum extent by exposure of the open ends to water under pressure. The quality of the watertight feature shall be judged, for inspection purposes, by the total amount of water which leaks from the open free end of a 5-foot specimen, when tested as specified herein.

20. One 5-foot specimen from each manufactured length of watertight cable shall be tested at a Government laboratory, and the leakage found shall be considered as representing the watertight quality of the entire length from which the specimen is cut. A manufactured length under this test may be a remnant length, random length, or standard length (see 6.2.2), or any greater length which is cabled and sheathed in one operation. In the event, that a specimen from the manufactured length is tested, and the manufactured length is cut subsequently in two or more pieces for delivery to different purchasers or to different destinations, specimens need not be cut from the several pieces or shipping lengths for further watertightness tests, unless additional tests are specifically requested by the purchaser.

30. In general, one test should be sufficient to determine the watertight quality of a length of cable. However, if the manufacturer feels that the leakage reported on the first specimen tested from any reel does not represent the quality of the length on that reel, the manufacturer shall have the option of making additional tests at the plant on the questionable length, and after removing the non-representative ends in accordance with 4.3.2.2.1, of resubmitting the remaining length for Government inspection. In this event, the results of prior tests shall be disregarded and the last test only shall determine the relative acceptability of any retested length. The cost or value of specimens consumed in retesting shall be borne by the manufacturer and shall not be included in the contract or order.

40. One end of the 5-foot specimen shall be subjected to tap water at a pressure of 25 pounds per square inch applied continuously for 18 hours. The method employed for applying the water pressure shall not constrict nor permit expansion of the specimen within the gland. Since various contractors may employ different methods for packing the gland, and since it has been found that variation in packing methods will affect the leakage results, it is necessary that all specimens be tested at a Government laboratory under uniform procedure. Specimens will be tested in the "as received" condition; that is, specimens will not be subjected to prior flexing, heating or electrical tests, although the braided metal armor may be removed without disturbing the watertight core, when this is necessary, to facilitate making a watertight gland without constriction of the specimen.

50. The following tabulation shows several degrees of water leakage for each type and size of watertight cable. These values are based on an analysis of inspection test data, and the values in column 1 are set at a low level which may be difficult to meet, unless manufacturers continue to cooperate in this development by discovering and introducing new refinements to improve watertight quality. These values have and will be reduced from time to time, as warranted by current inspection test data. Watertight quality is improved continuously in this manner. Eventually, zero leakage will be specified. Failure of a reasonable number of specimens to conform to the column 1 values is expected at the present stage of the development and is not considered as evidence of substandard material, but as evidence of the need for tighter quality control and care in processing.

60. The results of all tests and retests made at the Government laboratory shall be reported to the bureau or agency concerned. and shall show the footage represented by each specimen.

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70. The following tabulation, and the explanatory notes thereto, constitute a part of this specification.

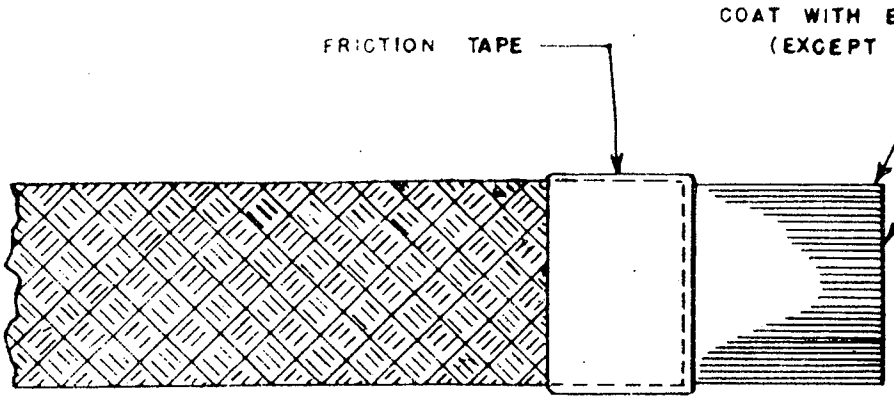
All cable types with conductor sizes	Total water leakage during 18 hr. test at 25 p.s.i.	
	Column 1	Column 2
	No price reduction	10% price reduction
	Cubic inches	Cubic inches
9 and smaller	2.0 or less	2.1 or more
14 to 40 incl.	10.0 or less	10.1 or more
50 to 100 incl.	20.0 or less	20.1 or more
125 to 300 incl.	40.0 or less	40.1 or more
350 and larger	60.0 or less	60.1 or more

Note 1. - The water leakage reported for any specimen will determine whether the cable footage represented by that specimen comes within the limits of column 1 or column 2.

Note 2. - Cable footage represented by specimens which do not exceed the leakage permitted in column 1 is acceptable at the contract price.

Note 3. - Cable footage represented by specimens which exceed the leakage permitted in column 1 is acceptable under column 2 at a 10 percent reduction in the contract price for that footage, provided the watertight sealing compounds are found to be present as specified in 3.6.

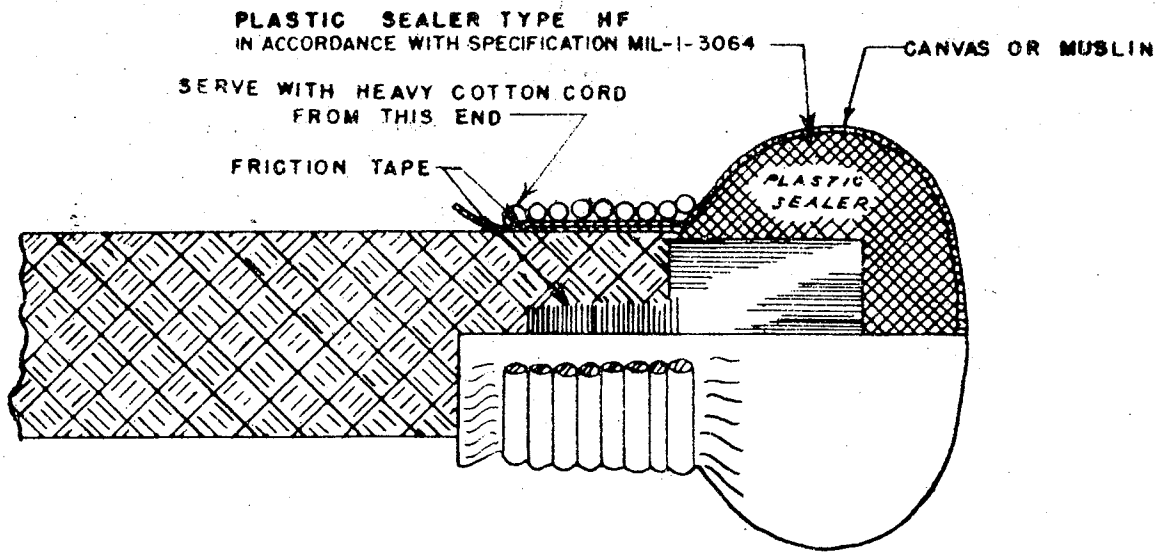
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(1) REMOVE ABOUT ONE INCH OF CABLE ARMOR

(2) SEIZE ARMOR WITH ONE LAYER OF FRICTION TAPE AS SHOWN

(3) APPLY BONDING AGENT IF NECESSARY SUCH AS A SOLUTION OF BATTERY SEALING COMPOUND AND CARBON TETRACHLORIDE TO CABLE END AND SHEATH



(4) PREPARE A PAD OF PLASTIC SEALER ABOUT 1/2 INCH THICK. KNEAD AND SHAPE AS SHOWN.

(5) COVER WITH CANVAS OR MUSLIN.

(6) SERVE WITH HEAVY COTTON CORD SO AS TO FORCE PLASTIC SEALER TIGHTLY AGAINST CABLE SIDE AND END.

Figure 1. - Method of treating cable ends on reels to prevent entrance of water.

E14650

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